DevOrSysAdminScripts

Laurent Frédéric Bernard François Lyaudet
https://lyaudet.eu/laurent/
Laurent.Lyaudet@gmail.com

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Abstract

Some "small" useful scripts for enhancing development or system-administrators tasks

Current version: 2026-03-26
Current number of commits: 780
Current git SHA1: 294d640a725292077b8458025030b87547614361
Code lines: 11872 total lines, 10910 not empty lines, 962 empty lines.

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1 Files tree

```
.
├── build_and_checks_dependencies
│   ├── build_and_checks.exec.sh.tpl
│   ├── build_dependencies_notes.exec.php
│   ├── build_dependencies_notes.libr.php
│   ├── build_md_from_printable_md.exec.sh
│   ├── check_shell_scripts_beginnings.libr.sh
│   ├── check_shell_scripts_indentation.libr.sh
│   ├── check_URLs.libr.sh
│   ├── common_build_and_checks.exec.sh
│   ├── common_build_and_checks.exec.sh.tpl
│   ├── common_options.libr.sh
│   ├── comparisons.libr.sh
│   ├── create_PDF.exec.sh
│   ├── generate_from_template.libr.sh
│   ├── get_common_text_glob_patterns.libr.sh
│   ├── grammar_and_spelling_check.libr.sh
│   ├── licenses_templates
│   │   ├── build_licenses_templates.exec.sh
│   │   ├── license_file_header_AGPLv3+.tpl
│   │   ├── license_file_header_GPLv3+.tpl
│   │   └── license_file_header_LGPLv3+.tpl
│   ├── lines_counts.libr.sh
│   ├── lines_filters.libr.sh
│   ├── lines_maps.libr.sh
│   ├── overwrite_if_not_equal.libr.sh
│   ├── python_black_complement.libr.sh
│   ├── python_isort_complement.libr.sh
│   ├── repository_data.libr.sh
│   ├── shell_checks_complement.libr.sh
│   ├── split_score.exec.php
│   ├── split_score.libr.php
│   ├── strings_functions.libr.sh
│   ├── too_long_code_lines.libr.sh
│   ├── update_common_build_and_checks.exec.sh
│   └── update_or_check_files_names_listing.exec.sh
├── build_and_checks.exec.sh
├── build_and_checks_variables
│   ├── bandit_baseline.json
│   ├── bandit.ini
│   ├── black.toml
│   ├── dependencies_data.json
│   ├── dependencies_notes.txt
│   ├── DevOrSysAdminScripts.html.tpl
```

```
│   ├── DevOrSysAdminScripts.tex
│   ├── DevOrSysAdminScripts.tex.tpl
│   ├── files_names_listing.txt
│   ├── .gitignore
│   ├── licenses_notes.txt
│   ├── phpcs_ruleset.xml
│   ├── phpmd_baseline.xml
│   ├── pylintrc.toml
│   ├── repository_data.txt
│   ├── repository_data.txt.tpl
│   ├── ruff.toml
│   ├── shellcheck.ini
│   └── shellcheck_notes.txt
├── COPYING
├── COPYING.LESSER
├── DevOrSysAdminScripts.html
├── DevOrSysAdminScripts.pdf
├── git_helpers
│   ├── git_checks.libr.sh
│   ├── post-commit
│   ├── post_commit_renaming_comment.libr.sh
│   └── pre-commit
├── .gitignore
├── IMAP_transfert.py
├── LiensUtiles.txt
├── miscellaneous
│   ├──
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 0123\
│   ├──
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 01234\
│   ├──
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 012345
│   ├──
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 012345\
│   ├──
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 0123456
│   ├──
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 0123456\
│   ├──
│   │ a_long_long_long_long_long_long_long_long_long_filename_so_that_you_c
```

```
│   │ an_see_that_it_works.txt
│   ├── funny.sh
│   ├── spampty_data.txt
│   ├── string_escaping.html
│   ├── string_escaping.xhtml
│   └──
│     yet_another_long_long_long_long_long_long_long_long_long_filename_so_
│     that_you_can_see_that_it_works.txt
├── README.html
├── README.md
├── README.md.tpl
├── string_escaping.libr.php
├── unicode.libr.php
├── upgrade_build_and_checks.exec.sh
└── wget_sha512.libr.sh

6 directories, 83 files

[ 12K Mar 26 23:51]  ./
├── [4.0K Mar 26 23:50]  build_and_checks_dependencies/
│   ├── [4.4K Mar 25 14:39]  build_and_checks.exec.sh.tpl*
│   ├── [1.2K Mar 26 12:30]  build_dependencies_notes.exec.php*
│   ├── [6.9K Mar 26 12:37]  build_dependencies_notes.libr.php
│   ├── [4.2K Mar 25 14:54]  build_md_from_printable_md.exec.sh*
│   ├── [3.3K Mar 25 15:01]  check_shell_scripts_beginnings.libr.sh
│   ├── [5.4K Mar 25 15:35]  check_shell_scripts_indentation.libr.sh
│   ├── [3.6K Mar 23 10:11]  check_URLs.libr.sh
│   ├── [ 32K Mar 26 23:49]  common_build_and_checks.exec.sh*
│   ├── [ 27K Mar 26 23:49]  common_build_and_checks.exec.sh.tpl
│   ├── [ 16K Mar 23 15:17]  common_options.libr.sh
│   ├── [6.1K Mar 22 16:03]  comparisons.libr.sh
│   ├── [ 21K Mar 24 21:57]  create_PDF.exec.sh*
│   ├── [3.7K Mar 22 16:36]  generate_from_template.libr.sh
│   ├── [2.0K Mar 24 20:52]  get_common_text_glob_patterns.libr.sh
│   ├── [3.3K Mar 23 10:18]  grammar_and_spelling_check.libr.sh
│   ├── [4.0K Mar 26 12:19]  licenses_templates/
│   │   ├── [ 14K Mar 26 12:19]  build_licenses_templates.exec.sh*
│   │   ├── [ 728 Feb 28 20:38]  license_file_header_AGPLv3+.tpl
│   │   ├── [ 707 Feb 28 20:38]  license_file_header_GPLv3+.tpl
│   │   └── [ 728 Feb 28 20:38]  license_file_header_LGPLv3+.tpl
│   ├── [6.2K Mar 23 15:40]  lines_counts.libr.sh
│   ├── [ 15K Mar 26 11:10]  lines_filters.libr.sh
│   ├── [2.7K Feb 19 16:16]  lines_maps.libr.sh
│   ├── [2.7K Mar 22 18:33]  overwrite_if_not_equal.libr.sh
│   ├── [2.5K Mar 23 15:29]  python_black_complement.libr.sh
│   ├── [3.0K Mar 23 15:30]  python_isort_complement.libr.sh
```

```
│   ├── [2.1K Mar 23 10:14]  repository_data.libr.sh
│   ├── [4.2K Mar 24 21:29]  shell_checks_complement.libr.sh
│   ├── [2.0K Mar 26 12:37]  split_score.exec.php
│   ├── [3.2K Mar 26 23:11]  split_score.libr.php
│   ├── [ 48K Mar 23 10:16]  strings_functions.libr.sh
│   ├── [4.5K Mar 24 21:31]  too_long_code_lines.libr.sh
│   ├── [4.2K Mar 26 23:29]  update_common_build_and_checks.exec.sh*
│   └── [6.1K Mar 23 17:56]  update_or_check_files_names_listing.exec.sh*
├── [4.6K Mar 26 23:49]  build_and_checks.exec.sh*
├── [4.0K Mar 26 23:51]  build_and_checks_variables/
│   ├── [  15 Jan  5 20:00]  bandit_baseline.json
│   ├── [  25 Jan  5 18:49]  bandit.ini
│   ├── [  30 Feb 28 20:30]  black.toml
│   ├── [4.9K Mar 26 23:19]  dependencies_data.json
│   ├── [4.1K Mar 26 23:20]  dependencies_notes.txt
│   ├── [5.7K Mar 26 23:16]  DevOrSysAdminScripts.html.tpl
│   ├── [ 35K Mar 26 23:50]  DevOrSysAdminScripts.tex
│   ├── [4.2K Mar 26 23:16]  DevOrSysAdminScripts.tex.tpl
│   ├── [5.5K Mar 26 22:08]  files_names_listing.txt
│   ├── [  23 Mar 24 19:33]  .gitignore
│   ├── [1.4K Mar 26 23:16]  licenses_notes.txt
│   ├── [ 13K Mar 26 22:59]  phpcs_ruleset.xml
│   ├── [2.2K Mar 26 12:12]  phpmd_baseline.xml
│   ├── [ 678 Feb 24 22:38]  pylintrc.toml
│   ├── [4.5K Mar 22 15:51]  repository_data.txt
│   ├── [ 786 Mar 22 15:39]  repository_data.txt.tpl
│   ├── [1.4K Feb 24 23:25]  ruff.toml
│   ├── [ 166 Mar 24 19:33]  shellcheck.ini
│   └── [3.5K Mar 26 23:50]  shellcheck_notes.txt
├── [ 34K May  6  2024]  COPYING
├── [7.5K Dec 27 21:48]  COPYING.LESSER
├── [472K Mar 26 23:50]  DevOrSysAdminScripts.html
├── [408K Mar 26 23:50]  DevOrSysAdminScripts.pdf
├── [4.0K Mar 24 21:26]  git_helpers/
│   ├── [3.8K Mar 22 21:06]  git_checks.libr.sh
│   ├── [2.1K Mar 22 21:09]  post-commit*
│   ├── [ 15K Mar 24 21:26]  post_commit_renaming_comment.libr.sh
│   └── [1.8K Mar 22 21:11]  pre-commit*
├── [   8 Mar 26 10:37]  .gitignore
├── [8.3K Feb 28 20:31]  IMAP_transfert.py
├── [ 830 Dec 29 19:10]  LiensUtiles.txt
├── [4.0K Mar 18 17:27]  miscellaneous/
│   ├── [   0 Mar  4 10:37]
│   1234567890123456789012345678901234567890123456789012345678901234567890
│   0123\
│   ├── [   0 Mar  4 10:03]
```

```
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
    │ 01234\
    │── [   0 Mar  4 10:01]
    │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
    │ 012345
    │── [   0 Mar  4 10:37]
    │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
    │ 012345\
    │── [   0 Mar  4 10:37]
    │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
    │ 0123456
    │── [   0 Mar  4 10:37]
    │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
    │ 0123456\
    │── [   0 Mar  4 09:13]
    │ a_long_long_long_long_long_long_long_long_long_filename_so_that_you_c
    │ an_see_that_it_works.txt
    │── [1.3K Mar  9 15:15]  funny.sh
    │── [  94 Mar  3 19:56]  spampty_data.txt
    │── [ 442 Mar 18 16:50]  string_escaping.html
    │── [ 654 Mar 18 17:27]  string_escaping.xhtml
    └── [   0 Mar  4 09:13]
      yet_another_long_long_long_long_long_long_long_long_long_filename_so_
      that_you_can_see_that_it_works.txt
├── [8.1K Mar 22 21:29]  README.html
├── [4.3K Mar 22 21:29]  README.md
├── [4.3K Mar 22 21:29]  README.md.tpl
├── [7.0K Mar 26 22:16]  string_escaping.libr.php
├── [ 16K Mar 26 20:11]  unicode.libr.php
├── [1.8K Mar 23 18:45]  upgrade_build_and_checks.exec.sh*
└── [2.5K Mar 23 18:45]  wget_sha512.libr.sh

6 directories, 83 files
```

2 Notes on licenses and dependencies

This repository is dual licensed. All the Bash scripts source code is covered by GPLv3+. All the PHP scripts source code is covered by LGPLv3+. All the Python scripts source code is covered by LGPLv3+.

The text of the licenses is available at: `https://www.gnu.org/licenses/`. The git repository of this source code is also available at: `https://github.com/LLyaudet/DevOrSysAdminScripts/`. Clone it with this URL: `git@github.com:LLyaudet/DevOrSysAdminScripts.git`.

Here is a listing of the dependencies of the code in this repository:

- The following dependencies are GPLv3+ licensed:

basename, bash (and its builtins), cat, chmod, cp, cut, date, diff, dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed, sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;

- The following dependencies are GPLv2+ licensed:
  more, pandoc, pdftex, pdflatex, tree;
- The following dependencies are GPLv2 licensed:
  git, pylint;
- The following dependencies are LGPLv2.1+ licensed:
  notify-send;
- The following dependencies are MIT licensed:
  black, isort, mypy, phpdoc, pip, ruff;
- The following dependencies are Apache-2.0 licensed:
  bandit;
- The following dependencies are BSD-3-Clause licensed:
  phpcs, phpmd;
- php is licensed under PHP License v3.01;
- tex is licensed under Knuth license;
- pcregrep, pcre2grep has a custom license from Cambridge University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception,
- python3 is licensed under PSF License Version 2 and the Zero-Clause BSD license.

Remark: Some commands listed under GPLv3+ are POSIX and can have different implementations under another license; for example, uutils coreutils is MIT licensed, whilst GNU coreutils is GPLv3+ licensed.

Bash can be found at the following URLs:
`https://www.gnu.org/software/bash/`,
`https://tiswww.case.edu/php/chet/bash/bashtop.html`,
`https://savannah.gnu.org/projects/bash/`.

GNU coreutils can be found at the following URLs:
`https://www.gnu.org/software/coreutils/`,
`https://savannah.gnu.org/projects/coreutils/`.

uutils coreutils can be found at the following URLs:
`https://uutils.github.io/coreutils/`,
`https://github.com/uutils/coreutils`.

diff (GNU diffutils) can be found at the following URLs:
`https://www.gnu.org/software/diffutils/`,
`https://savannah.gnu.org/projects/diffutils/`.

find, xargs (GNU findutils) can be found at the following URLs:
https://www.gnu.org/software/findutils/,
https://savannah.gnu.org/projects/findutils/.

grep (GNU grep) can be found at the following URLs:
https://www.gnu.org/software/grep/,
https://savannah.gnu.org/projects/grep/.

sed (GNU sed) can be found at the following URLs:
https://www.gnu.org/software/sed/,
https://savannah.gnu.org/projects/sed/.

ShellCheck can be found at the following URLs:
https://www.shellcheck.net/,
https://github.com/koalaman/shellcheck.

GNU Wget can be found at the following URLs:
https://www.gnu.org/software/wget/,
https://savannah.gnu.org/projects/wget/.

more of util-linux can be found at the following URLs:
https://www.kernel.org/pub/linux/utils/util-linux/,
https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/,
https://github.com/util-linux/util-linux official?.

pandoc can be found at the following URLs:
https://pandoc.org/,
https://github.com/jgm/pandoc.

pdftex/pdflatex can be found at the following URLs:
https://www.tug.org/applications/pdftex/,
https://tug.org/texlive/svn/.

tree can be found at the following URLs:
https://oldmanprogrammer.net/source.php?dir=projects/tree,
https://gitlab.com/OldManProgrammer/unix-tree,
https://github.com/Old-Man-Programmer/tree.

git can be found at the following URLs:
https://git-scm.com/about,
https://github.com/git/git.

pylint can be found at the following URLs:
https://www.pylint.org/,
https://pylint.readthedocs.io/en/stable/,
https://github.com/pylint-dev/pylint,
https://pypi.org/project/pylint/.

notify-send can be found at the following URLs:
https://gitlab.gnome.org/GNOME/libnotify/,
https://gnome.pages.gitlab.gnome.org/libnotify/.

black can be found at the following URLs:
https://github.com/psf/black,
https://black.readthedocs.io/en/stable/,
https://pypi.org/project/black/.

isort can be found at the following URLs:
https://pycqa.github.io/isort/,

https://github.com/pycqa/isort/,
https://pypi.org/project/isort/.

mypy can be found at the following URLs:
https://mypy-lang.org/,
https://mypy.readthedocs.io/en/stable/,
https://github.com/python/mypy,
https://pypi.org/project/mypy/.

phpdoc (phpDocumentor) can be found at the following URLs:
https://phpdoc.org/,
https://docs.phpdoc.org/,
https://github.com/phpdocumentor/phpdocumentor.

pip can be found at the following URLs:
https://pip.pypa.io/en/stable/,
https://github.com/pypa/pip,
https://pypi.org/project/pip/ (egg and chicken ;)).

ruff can be found at the following URLs:
https://docs.astral.sh/ruff/,
https://github.com/astral-sh/ruff,
https://pypi.org/project/ruff/.

bandit can be found at the following URLs:
https://bandit.readthedocs.io/en/latest/,
https://github.com/PyCQA/bandit,
https://pypi.org/project/bandit/.

phpcs (PHP_CodeSniffer) can be found at the following URLs:
https://github.com/PHPCSStandards/PHP_CodeSniffer,
https://packagist.org/packages/squizlabs/php_codesniffer.

phpmd can be found at the following URLs:
https://phpmd.org/,
https://github.com/phpmd/phpmd,
https://packagist.org/packages/phpmd/phpmd.

php can be found at the following URLs:
https://www.php.net/,
https://github.com/php/php-src.

tex can be found at the following URL:
https://ctan.org/tex-archive/systems/knuth/dist/tex.

pcregrep, pcre2grep can be found at the following URLs:
https://www.pcre.org/,
https://pcre2project.github.io/pcre2/,
https://github.com/PCRE2Project/pcre1,
https://github.com/PCRE2Project/pcre2,
https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c.

python3 can be found at the following URLs:
https://www.python.org/,
https://docs.python.org/3/.

If you see something that I missed regarding my dependencies, please tell me/email me :).

3 Listing of files

3.1 build_and_checks_dependencies/build_and_checks.exec.sh.tpl

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
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# The file "build_and_checks.exec.sh.tpl" was renamed from
# "build_and_checks.sh.tpl" to "build_and_checks.exec.sh.tpl".
# The file "build_and_checks.exec.sh" generated from the file
# "build_and_checks.sh.tpl" or "build_and_checks.exec.sh.tpl"
# was renamed from
# "build_and_checks.sh" to "build_and_checks.exec.sh".

build_and_checks(){
  # $1=work_directory
  # Options:
  #   --check-download (=keep to keep the downloaded files)
  #   --verbose
  local LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  readonly LFBFL_work_directory

  local LFBFL_verbose=""
```

```
if [[ "$*" == *--verbose* ]]; then
  printf "%s %s\n" "$0" "$*"
  LFBFL_verbose="--verbose"
fi
readonly LFBFL_verbose

local LFBFL_check_download=""
if [[ "$*" == *--check-download=keep* ]]; then
  LFBFL_check_download="--check-download=keep"
elif [[ "$*" == *--check-download* ]]; then
  LFBFL_check_download="--check-download"
fi
readonly LFBFL_check_download

source ./wget_sha512.libr.sh

LFBFL_subdir="build_and_checks_dependencies"
mkdir --parents "${LFBFL_subdir}/licenses_templates"
local LFBFL_variables_directory
LFBFL_variables_directory="${LFBFL_work_directory}/"
LFBFL_variables_directory+="build_and_checks_variables"
readonly LFBFL_variables_directory
mkdir --parents "${LFBFL_variables_directory}/temp"
if [[ ! -f "${LFBFL_variables_directory}/.gitignore" ]]; then
  printf "temp/\nupgrade_venvs_ts\n"\
    > "${LFBFL_variables_directory}/.gitignore"
fi

local LFBFL_dependencies_URL
LFBFL_dependencies_URL="https://raw.githubusercontent.com/LLyaudet/"
LFBFL_dependencies_URL+="DevOrSysAdminScripts/main/${LFBFL_subdir}"
readonly LFBFL_dependencies_URL

declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
local LFBFL_script_download_URL
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/"
LFBFL_script_download_URL+="${LFBFL_common_file_name}"
readonly LFBFL_script_download_URL
declare -r LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_common_file_name}"
@sha512_common_build_and_checks.exec.sh@
wget_sha512 "${LFBFL_file_path}"\
  "${LFBFL_script_download_URL}"\
  "${LFBFL_correct_sha512}"\
  "${LFBFL_verbose}"\
  "${LFBFL_check_download}"
chmod +x "${LFBFL_file_path}"
```

```

  "${LFBFL_file_path}" "${LFBFL_work_directory}"\
    "${LFBFL_dependencies_URL}"\
    "${LFBFL_verbose}"\
    "${LFBFL_check_download}"
}

if [[ "$*" == *--fixed-point-build* ]]; then
  printf -- "--fixed-point-build\n"
  source "./build_and_checks_dependencies/lines_filters.libr.sh"
  LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  LFBFL_data_file_name="${LFBFL_variables_directory}/"
  LFBFL_data_file_name+="repository_data.txt"
  LFBFL_repository_name=""
  grep_variable "${LFBFL_data_file_name}" repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  # Touching the 3 following files first let us iterate
  # upgrade_build_and_checks 2 times instead of 3 to reach
  # a fixed point.
  touch "${LFBFL_variables_directory}/${LFBFL_repository_name}.tex"
  touch "${LFBFL_work_directory}/${LFBFL_repository_name}.pdf"
  touch "${LFBFL_work_directory}/${LFBFL_repository_name}.html"
  # The two iterations of build_and_checks needs to complete
  # during the same minute to have a fixed point.
  build_and_checks "$@"
  # build_and_checks "$@" second iteration below
  # Then calling this script without --fixed-point-build should not
  # yield any new superficial modifications.
fi

build_and_checks "$@"
```

3.2 build_and_checks_dependencies/build_dependencies_notes.exec.php

```
#!/usr/bin/php
<?php

/**
This file is part of DevOrSysAdminScripts library.

```

```
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@category Executable
@package DevOrSysAdminScripts
@author Laurent Lyaudet <laurent.lyaudet@gmail.com>
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*/

declare(strict_types=1);
declare(encoding='UTF-8');

require_once 'string_escaping.libr.php';
require_once 'build_dependencies_notes.libr.php';

build_dependencies_notes($argv[1]);
?>
```

3.3 build_and_checks_dependencies/build_dependencies_notes.libr.php

```
<?php

/**
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2026-03-26T11:59+01:00: This file was renamed from
"build_dependencies_notes.exec.php"
to
"build_dependencies_notes.libr.php".

@category Library
@package DevOrSysAdminScripts
@author Laurent Lyaudet <laurent.lyaudet@gmail.com>
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*/

declare(strict_types=1);
declare(encoding='UTF-8');

// require_once 'string_escaping.libr.php';



/**
This function looks at the dependencies data found in a work directory,
and generate three files: a txt file, a TeX file (.sub),
and an HTML file (.sub) with the data in a nicer format.

@param string $s_work_directory The directory where the work must be done.

@return void
*/
```

```
function build_dependencies_notes(string $s_work_directory) : void {
  $s_subdir1 = $s_work_directory.'/build_and_checks_variables';
  $s_subdir2 = $s_subdir1.'/temp';
  $s_data_file_path = $s_subdir1.'/dependencies_data.json';
  $s_data = file_get_contents($s_data_file_path);
  $arr_data = json_decode($s_data, true);
  $s_result_content1 = '';
  $s_result_content2 = '';
  $s_result_content3 = '';

  foreach($arr_data as $arr_dependency){
    $s_URL_word = 'URL';
    if(count($arr_dependency['URLs']) > 1){
      $s_URL_word .= 's';
    }

    $s_result_content3 .= "<p>\n";

    $s_dependency1 = $arr_dependency['dependency_name'];
    $s_dependency2 = string_escaping\TeX\escape_text($s_dependency1);
    $s_dependency3 = string_escaping\HTML\escape_text($s_dependency1);

    $s_sentence_end = ' can be found at the following '.$s_URL_word.':';
    $s_result_content1 .= $s_dependency1.$s_sentence_end."\n";
    $s_result_content2 .= $s_dependency2.$s_sentence_end."\\newline\n";
    $s_result_content3 .= $s_dependency3.$s_sentence_end."<br>\n";

    for(
      $i = 0, $i_max = count($arr_dependency['URLs']) - 1;
      $i <= $i_max;
      ++$i
    ){
      $mixed_URL = $arr_dependency['URLs'][$i];
      $s_comment = '';
      $s_URL = '';
      if(is_array($mixed_URL)){
        $s_URL = $mixed_URL['URL'];
        $s_comment = ' '.$mixed_URL['comment'];
      }
      else{
        $s_URL = $mixed_URL;
      }

      $s_result_content1 .= $s_URL.$s_comment."\n";

      if(strlen($s_URL) > 60){
```

```
        $s_result_content2 .= (
          "{\\small\n"
          .'  \url{'.$s_URL."}\n"
          .'}'
        );
      }
      else{
        $s_result_content2 .= '\url{'.$s_URL.'}';
      }
      $s_result_content2 .= string_escaping\TeX\escape_text($s_comment);
      if($i === $i_max){
        $s_result_content2 .= ".\n";
      }
      else{
        $s_result_content2 .= ",\\newline\n";
      }

      $s_result_content3 .= (
        "<a\n"
        .'href="'.string_escaping\HTML\escape_attribute($s_URL)."\"\n"
        ."><span style=\"font-family:monospace\"\n"
        .'>'.string_escaping\HTML\escape_text($s_URL)."</span\n"
        .'></a>'
      );
      $s_result_content3 .= string_escaping\HTML\escape_text($s_comment);
      if($i === $i_max){
        $s_result_content3 .= ".\n";
      }
      else{
        $s_result_content3 .= ",<br>\n";
      }
    }//end for urls

    $s_result_content1 .= "\n";
    $s_result_content2 .= "\n";
    $s_result_content3 .= "</p>\n\n";
  }//end foreach($arr_data as $arr_dependency)

  // Add a "footer".
  $s_closing_sentence = (
    "If you see something that I missed regarding my dependencies,\n"
    ."please tell me/email me :).\n"
  );
  $s_result_content1 .= $s_closing_sentence;
  $s_result_content2 .= $s_closing_sentence;
  $s_result_content3 .= "<p>\n".$s_closing_sentence."</p>\n";
```

```
// We overwrite results if needed.
$s_result_path1 = $s_subdir1.'/dependencies_notes.txt';
$s_existing_result_content1 = '';
if(file_exists($s_result_path1)){
  $s_existing_result_content1 = file_get_contents($s_result_path1);
}
if($s_result_content1 !== $s_existing_result_content1){
  file_put_contents($s_result_path1, $s_result_content1);
}
/*
Example of generated content of dependencies_notes.txt:
------------------------------------------------------------------------
more of util-linux can be found at the following URLs:
https://www.kernel.org/pub/linux/utils/util-linux/
https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/
https://github.com/util-linux/util-linux official?
------------------------------------------------------------------------
*/
$s_result_path2 = $s_subdir2.'/dependencies_notes.tex.sub';
$s_existing_result_content2 = '';
if(file_exists($s_result_path2)){
  $s_existing_result_content2 = file_get_contents($s_result_path2);
}
if($s_result_content2 !== $s_existing_result_content2){
  file_put_contents($s_result_path2, $s_result_content2);
}
/*
Example of generated content of dependencies_notes.tex.sub:
------------------------------------------------------------------------
more of util-linux can be found at the following URLs:\newline
\url{https://www.kernel.org/pub/linux/utils/util-linux/},\newline
{\small
  \url{https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/}
},\newline
\url{https://github.com/util-linux/util-linux} official?.
------------------------------------------------------------------------
*/
$s_result_path3 = $s_subdir2.'/dependencies_notes.html.sub';
$s_existing_result_content3 = '';
if(file_exists($s_result_path3)){
  $s_existing_result_content3 = file_get_contents($s_result_path3);
}
if($s_result_content3 !== $s_existing_result_content3){
  file_put_contents($s_result_path3, $s_result_content3);
}
```

```
  /*
  Example of generated content of dependencies_notes.html.sub:
  <p>
  more of util-linux can be found at the following URLs:<br>
  <a
  href="https://www.kernel.org/pub/linux/utils/util-linux/"
  ><span style="font-family:monospace"
  >https://www.kernel.org/pub/linux/utils/util-linux/</span
  ></a>,<br>
  <a
  href="https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/"
  ><span style="font-family:monospace"
  >https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/</span
  ></a>,<br>
  <a
  href="https://github.com/util-linux/util-linux"
  ><span style="font-family:monospace"
  >https://github.com/util-linux/util-linux</span
  ></a> official?.
  </p>
  */
}//end build_dependencies_notes()
?>
```

3.4 build_and_checks_dependencies/build_md_from_printable_md.exec.sh

```
#!/usr/bin/env bash
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# This file was renamed from "build_md_from_printable_md.sh"
# to "build_md_from_printable_md.exec.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

build_md_from_printable_md(){
  # Options:
  #   --verbose
  #   --work-directory=""
  #   --base-name="README"
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  local LFBFL_base_name="README"
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --base-name=* ]]; then
      LFBFL_base_name=${LFBFL_arg#--base-name=}
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        printf "Searching md file: %s.\n" "${LFBFL_base_name}"
      fi
      break
    fi
  done
  readonly LFBFL_base_name

  # Remove line returns here to keep lines short.
  local LFBFL_sed_expression
  LFBFL_sed_expression='s/(\n[ ]*\[[a-zA-Z0-9:-]*\]: [^\n\\]*)\\\n/\1/Mg'
  LFBFL_sed_expression+=';s/(\n[ ]*<http[^\n\\]*)\\\n/\1/Mg'
  LFBFL_sed_expression+=';s/(\n[ ]*- <http[^\n\\]*)\\\n/\1/Mg'
  readonly LFBFL_sed_expression
```

```
  # Since Markdown can contain code, it is limited to Markdown URLs.
  # Hence it is way more complicated than 's/\\\n//Mg' expression
  # in other files.
  # Another downside is that more than one pass is needed.

  if [[ -f "${LFBFL_base_name}.md.tpl" ]]; then
    sed -Ez -e "${LFBFL_sed_expression}"\
            -e "${LFBFL_sed_expression}"\
            -e "${LFBFL_sed_expression}"\
            -e "${LFBFL_sed_expression}"\
      "${LFBFL_base_name}.md.tpl"\
      > "${LFBFL_base_name}.md.temp"
    overwrite_if_not_equal "${LFBFL_base_name}.md"\
      "${LFBFL_base_name}.md.temp"
  else
    printf "No file %s.md.tpl\n" "${LFBFL_base_name}"
    # Unfortunately, we cannot return early, since we want to propose
    # HTML file generation with pandoc from .md file,
    # if the file .md.tpl doesn't exist.
  fi

  if [[ ! -f "${LFBFL_base_name}.md" ]]; then
    printf "No file %s.md\n" "${LFBFL_base_name}"
    return 1
  fi

  pandoc --from=markdown --to=html --standalone\
    --shift-heading-level-by=-1 "${LFBFL_base_name}.md"\
    --output="${LFBFL_base_name}.html.temp"
  # Correcting pandoc HTML "by hand".
  # https://github.com/jgm/pandoc/issues/10957
  # With the most open-minded answer found on Internet ;) XD.
  # This second sed expression is what I need,
  # but it may be unsuitable for you.
  # Other problems:
  # https://github.com/jgm/pandoc/issues/11484
  local LFBFL_prewrap="s~(code \{(\n|[^}])*)(    \})"
  LFBFL_prewrap+="~\1      white-space: pre-wrap;\n\3~M"
  readonly LFBFL_prewrap
  sed -i -E -e 's~(  <meta .*) />~\1>~g'\
            -e 's~<html .*>~<html lang="en">~'\
            -e '/    code\{white-space: pre-wrap;\}/d'\
    "${LFBFL_base_name}.html.temp"
  sed -i -Ez -e 's~(<img(\n|[^a-z0-9])(\n|[^>])*) />~\1>~Mg'\
             -e "${LFBFL_prewrap}"\
    "${LFBFL_base_name}.html.temp"
```

```
  overwrite_if_not_equal "${LFBFL_base_name}.html"\
    "${LFBFL_base_name}.html.temp"

  # java -Xss128M -jar "../jar_venv/vnu.jar" --exit-zero-always\
  #   --verbose ${LFBFL_base_name}.html"
}

build_md_from_printable_md "$@"
```

3.5 build_and_checks_dependencies/check_shell_scripts_beginnings.libr.sh

```
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# This file was renamed from "check_shell_scripts_beginnings.sh"
# to "check_shell_scripts_beginnings.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

declare -gr LFBFL_SHELL_SCRIPT_BEGINNING="#!/usr/bin/env bash"

check_one_shell_script_beginning(){
```

```
  # $1=file_path
  #
  # declare -r LFBFL_file_path=$(basename "$1")
  # if [[ "${LFBFL_file_path}" =~ license_file_header_.*\.sh ]]; then
  #   return 0
  # fi
  # diff <(head --lines=1 "$1") <(printf '#!/usr/bin/env bash')
  # ------------------------------------------------------------------
  # is of course slower than
  # ------------------------------------------------------------------
  # diff <(head --lines=1 "$1") <(printf '#!/usr/bin/env bash')
  # ------------------------------------------------------------------
  # is of course slower than
  # ------------------------------------------------------------------
  declare -r LFBFL_head=$(head --lines=1 "$1")
  [[ "${LFBFL_head}" == "${LFBFL_SHELL_SCRIPT_BEGINNING}" ]]\
    || printf "%s:File has wrong shell script beginning.\n" "$1"
}

check_shell_scripts_beginnings(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  # shopt -s globstar
  # for LFBFL_file_path in **/*.sh; do
  local LFBFL_file_path
  local LFBFL_head
  declare -r LFBFL_s_files_paths=$(
    find . -type f -name "*.sh" -printf '%P\n'\
    | relevant_find
  )
  if [[ -z "${LFBFL_s_files_paths}" ]]; then
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "check_shell_scripts_beginnings: No file found.\n"
    return
  fi
```

```
  declare -a LFBFL_arr_files_paths
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "%s:Checking shell script beginning.\n"\
          "${LFBFL_file_path}"
    LFBFL_head=$(head --lines=1 "${LFBFL_file_path}")
    [[ "${LFBFL_head}" == "${LFBFL_SHELL_SCRIPT_BEGINNING}" ]]\
      || printf "%s:File has wrong shell script beginning.\n"\
          "${LFBFL_file_path}"
  done
}
```

3.6 build_and_checks_dependencies/check_shell_scripts_indentation.libr.sh

```
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LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

check_one_shell_script_indentation(){
```

```
32    # $1=file_path
33    declare -i LFBFL_i_file_with_error=0
34    local LFBFL_some_line
35    local LFBFL_previous_line="workeduntilsomeasshole"
36    # created a command named "workeduntilsomeasshole"
37    # and called it without neither indentation, nor parameters
38    # in a .sh script XD.
39    # Manual justification of the lines L-4 and L-3 was a serendipity.
40    # I just tried a joke and was lucky ;) :).
41    local LFBFL_line_end
42    local LFBFL_previous_line_end
43    declare -i LFBFL_i_line_length
44    declare -i LFBFL_i_line_end_length
45    declare -i LFBFL_i_line_start_length
46    local LFBFL_spacing
47    declare -i LFBFL_i_offset
48    local LFBFL_substring1
49    local LFBFL_substring2
50    declare -r LFBFL_pattern='^ *-e '
51
52    # This grep would have been enough without some sed commands.
53    declare -r LFBFL_s_lines=$(
54      grep --before-context=1 --extended-regexp --line-number\
55        --regexp='^(  )* ([^ ]|$)' --with-filename "$1"
56    )
57    if [[ -z "${LFBFL_s_lines}" ]]; then
58      return
59    fi
60
61    declare -a LFBFL_arr_lines
62    mapfile -t LFBFL_arr_lines <<< "${LFBFL_s_lines}"
63    readonly LFBFL_arr_lines
64
65    for LFBFL_some_line in "${LFBFL_arr_lines[@]}"; do
66      if [[ "${LFBFL_some_line}" == "--" ]]; then
67        # When switching between code fragments, we reinit.
68        LFBFL_previous_line="workeduntilsomeasshole"
69        continue
70      fi
71      # Note how it's funny that this loop always work because we check
72      # shell scripts beginning in another script and thus we always
73      # have a line before.
74      if [[ "${LFBFL_previous_line}" == "workeduntilsomeasshole" ]]; then
75        LFBFL_previous_line="${LFBFL_some_line}"
76        continue
77      fi
```

```
    # Remove the name of the file and then the line number
    LFBFL_line_end=${LFBFL_some_line#*:}
    LFBFL_line_end=${LFBFL_line_end#*:}
    # But for the previous line '-' or ':' is used instead of ':'
    # and filenames with '-' are much more frequent than with ':'...
    LFBFL_i_line_length=${#LFBFL_some_line}
    LFBFL_i_line_end_length=${#LFBFL_line_end}
    LFBFL_i_line_start_length=$((
      LFBFL_i_line_length - LFBFL_i_line_end_length
    ))
    LFBFL_i_offset=$((LFBFL_i_line_start_length - 2))
    # Checking if increasing number of digits in the line numbers
    # (99 -> 100).
    LFBFL_substring1=${LFBFL_previous_line:${LFBFL_i_offset}:1}
    # Nice puzzle below :) Look for comments above :)
    case ${LFBFL_substring1} in
      -) ((LFBFL_i_offset += 1));;
      :) ((LFBFL_i_offset += 1));;
      [0-9]) ((LFBFL_i_offset += 2));;
      *) printf ":-(\n";;
    esac
    LFBFL_previous_line_end=${LFBFL_previous_line:${LFBFL_i_offset}}
    if [[ "${LFBFL_line_end}" =~ ${LFBFL_pattern} ]]; then
      # Line starts with spacing and -e
      # Keep max spaces
      LFBFL_spacing=${LFBFL_line_end%%-*}
      LFBFL_i_offset=${#LFBFL_spacing}
      LFBFL_substring1=${LFBFL_line_end:${LFBFL_i_offset}:3}
      LFBFL_substring2=${LFBFL_previous_line_end:${LFBFL_i_offset}:3}
      if [[ "${LFBFL_substring1}" != "${LFBFL_substring2}" ]]; then
        printf "%s\n" "${LFBFL_some_line}"
        LFBFL_i_file_with_error=1
      fi
    else
      printf "%s\n" "${LFBFL_some_line}"
      LFBFL_i_file_with_error=1
    fi
    LFBFL_previous_line="${LFBFL_some_line}"
  done

  if [[ LFBFL_i_file_with_error -eq 1 ]]; then
    printf "%s:File has some lines with odd number of spaces.\n" "$1"
  fi
}

check_shell_scripts_indentation(){
```

```
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  # shopt -s globstar
  # for LFBFL_file_path in **/*.sh; do
  local LFBFL_file_path
  declare -r LFBFL_s_files_paths=$(
    find . -type f -name "*.sh" -printf '%P\n'\
    | relevant_find
  )
  if [[ -z "${LFBFL_s_files_paths}" ]]; then
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "check_shell_scripts_indentation: No file found.\n"
    return
  fi
  declare -a LFBFL_arr_files_paths
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  readonly LFBFL_arr_files_paths
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "%s:Checking shell script indentation.\n"\
          "${LFBFL_file_path}"
    check_one_shell_script_indentation "${LFBFL_file_path}"
  done
}
```

3.7 build_and_checks_dependencies/check_URLs.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
```

```
# or (at your option) any later version.
#
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#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "check_URLs.sh" to "check_URLs.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

check_URLs(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  get_COMMON_TEXT_FILES_GLOB_PATTERNS

  declare -Ar LFBFL_substitutions=(
    ["http://www.gnu.org/licenses/"]="https://www.gnu.org/licenses/"
  )
```

```

  local LFBFL_pattern
  local LFBFL_base_file_name
  local LFBFL_substitution
  local LFBFL_substitution2
  local LFBFL_file_path
  local LFBFL_s_files_paths
  declare -a LFBFL_arr_files_paths
  for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "check_URLs: Iterating on pattern: %s.\n" "${LFBFL_pattern}"
    fi
    find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | xargs grep -H 'http:'\
      | grep -v "| xargs grep -H 'htt"\
      | grep -vP "['\"]http(:[^'\"]*)['\"].*['\"]https\\1['\"]"\
      | grep -v 'http://www.w3.org/1999/xhtml'\
      | grep -v 'http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd'
    # Second grep after xargs grep just above will remove false positives
    # from substitutions that fit on one line.

    LFBFL_s_files_paths=$(
      find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | relevant_find\
      | not_main_html_find
    )
    if [[ -z "${LFBFL_s_files_paths}" ]]; then
      [[ LFBFL_i_verbose -eq 1 ]] && printf "check_URLs: No file found.\n"
      continue
    fi
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      LFBFL_base_file_name=$(basename "${LFBFL_file_path}")
      [[ "${LFBFL_base_file_name}" != "check_URLs.libr.sh" ]]\
        || continue
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        printf "check_URLs: Handling the file: %s.\n" "${LFBFL_file_path}"
      fi
      for LFBFL_substitution in "${!LFBFL_substitutions[@]}"; do
        LFBFL_substitution2=${LFBFL_substitutions[${LFBFL_substitution}]}
        if grep -q "${LFBFL_substitution}" "${LFBFL_file_path}"; then
          sed -i "s|${LFBFL_substitution}|${LFBFL_substitution2}|g"\
            "${LFBFL_file_path}"
        fi
      done
    done
```

```
  done
}
```

3.8 build_and_checks_dependencies/common_build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# The file "common_build_and_checks.exec.sh.tpl" was renamed from
# "common_build_and_checks.sh.tpl" to
# "common_build_and_checks.exec.sh.tpl".
# The file "common_build_and_checks.exec.sh" generated from the file
# "common_build_and_checks.sh.tpl"
# or "common_build_and_checks.exec.sh.tpl"
# was renamed from
# "common_build_and_checks.sh" to "common_build_and_checks.exec.sh".

common_build_and_checks(){
  # $1 LFBFL_work_directory
  # $2 LFBFL_dependencies_URL
  # Options:
  #   --check-download (=keep to keep the downloaded files)
  #   --verbose
  local LFBFL_work_directory="${1:-.}"
  LFBFL_work_directory=$(realpath "${LFBFL_work_directory}")
  local LFBFL_work_directory_option="--work-directory="
```

```
LFBFL_work_directory_option+="${LFBFL_work_directory}"
declare -r LFBFL_dependencies_URL="$2"
local LFBFL_verbose=""
declare -i LFBFL_i_verbose=0
if [[ "$*" == *--verbose* ]]; then
  printf "%s %s\n" "$0" "$*"
  LFBFL_verbose="--verbose"
  LFBFL_i_verbose=1
fi
readonly LFBFL_verbose
readonly LFBFL_i_verbose
local LFBFL_check_download=""
if [[ "$*" == *--check-download=keep* ]]; then
  LFBFL_check_download="--check-download=keep"
elif [[ "$*" == *--check-download* ]]; then
  LFBFL_check_download="--check-download"
fi
readonly LFBFL_check_download

declare -a LFBFL_some_common_options=(
  "${LFBFL_verbose}"
  # "$--work-directory={LFBFL_work_directory}" doesn't work
  "${LFBFL_work_directory_option}"
)

source ./wget_sha512.libr.sh

local LFBFL_subdir="build_and_checks_dependencies"
declare -r LFBFL_subdir2="${LFBFL_subdir}/licenses_templates"
declare -r LFBFL_subdir3="build_and_checks_variables"
local LFBFL_file_name
local LFBFL_script_download_URL
local LFBFL_file_path
local LFBFL_correct_sha512

wrapped_wget_sha512(){
  wget_sha512 "${LFBFL_file_path}"\
    "${LFBFL_script_download_URL}"\
    "${LFBFL_correct_sha512}"\
    "${LFBFL_verbose}"\
    "${LFBFL_check_download}"
}

LFBFL_file_name="build_dependencies_notes.exec.php"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
```

```
LFBFL_correct_sha512='624368d161351d7a261fbe3165d51907f46394c1b0e79'
LFBFL_correct_sha512+='b980ec11173c324b2c5531cf2a2b60b2dc62edcaa9c8'
LFBFL_correct_sha512+='71dca55065f8802df4275420836252786f9606b'
wrapped_wget_sha512
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="build_dependencies_notes.libr.php"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='2152594899ac7ebd1bf56f8cb84b8b68f2e45d56d2942'
LFBFL_correct_sha512+='e6c6ed520f84c130012009be712e510f61c06f135937'
LFBFL_correct_sha512+='dc908b0db213071bae736cc3e18bca22b10bd02'
wrapped_wget_sha512

LFBFL_file_name="build_md_from_printable_md.exec.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='ceae57a25ad40c149a7a020ddbf5e85941f4212e2ffd0'
LFBFL_correct_sha512+='1f2b00b14c895aeb628cca53565b0afacb8f1e24dbbc'
LFBFL_correct_sha512+='39b270cdc0c61e4cfe42bafd0d9b6bec4215405'
wrapped_wget_sha512
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="check_shell_scripts_beginnings.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='6a57c0f30a0f6f15e9375384bb0ff214bdcbc3377c238'
LFBFL_correct_sha512+='e33febdb3a905d96358c7392b7b66c71aad022435527'
LFBFL_correct_sha512+='00bf10f30dfeb0db9fc84d06eba7343c1e06184'
wrapped_wget_sha512

LFBFL_file_name="check_shell_scripts_indentation.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='86bccfa7181772622db577cbfe0d2be7c4ea8a8c3550e'
LFBFL_correct_sha512+='5edcd21b20628dd5f5539c287a2af9b7a27933a7c812'
LFBFL_correct_sha512+='3a26c057bbd54175b89917336830529eb3831bd'
wrapped_wget_sha512

LFBFL_file_name="check_URLs.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='2027e3f9cdb847cc71b4b69149486779df666a1b18c30'
LFBFL_correct_sha512+='ab161b7c96ee2c48fed4ef78edb489d566d568fb7b1c'
LFBFL_correct_sha512+='8390a680a5212f5a04c9bd6dd05744374bbc5b3'
wrapped_wget_sha512
```

```
LFBFL_file_name="common_options.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='58f125f10f9c10cd38f04cdeec997bb7dc376ba6f0c09'
LFBFL_correct_sha512+='dfc1d2c0927855d709fbf002c027e67370cc0f8864b7'
LFBFL_correct_sha512+='8b23ab7685fc192413cc657df46475e28b989e2'
wrapped_wget_sha512

LFBFL_file_name="comparisons.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='eb2bfabf11d219a4f7a105c95ea5fc2b1f6ddd668fad7'
LFBFL_correct_sha512+='344963c62d574eb3ecb93b35aad8b5ee4c2558792478'
LFBFL_correct_sha512+='f7ef9da058fba1cfe2195b61c87e1ed8686f200'
wrapped_wget_sha512

LFBFL_file_name="create_PDF.exec.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='e57d7603f18596a0e259dcf25f6f922cbdbad9ae34b8d'
LFBFL_correct_sha512+='63a3085179fcc8eb1236658a7dc81bbb72122f5e8091'
LFBFL_correct_sha512+='253dc270f72337ef638d83cb2490cfb187da5c1'
wrapped_wget_sha512
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="generate_from_template.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='6369d2f6f28986d56398b2a9533fddbd14251fd8c8853'
LFBFL_correct_sha512+='203f3d8561065b2109abc8c328f0045b4dfb47895690'
LFBFL_correct_sha512+='a275d560ac684aed15e41307c1531c2e88a3643'
wrapped_wget_sha512

LFBFL_file_name="get_common_text_glob_patterns.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='d38b894f63ac0db1eab4447a765b723642e12e3443aee'
LFBFL_correct_sha512+='01bd08e44bd2971bf11d3f0b98d2932bcaaeab4f30b7'
LFBFL_correct_sha512+='9f7da666e02a0fb1dada8a348ad866a104e4d53'
wrapped_wget_sha512

LFBFL_file_name="grammar_and_spelling_check.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='48f0edc5841f59f92b8126240df13e651a887bb399640'
```

```
LFBFL_correct_sha512+='589fb4bdc56dfd9006e6908ad952f1ce037e6dd22293'
LFBFL_correct_sha512+='5667301e528a767d0ef00ad43d733990b1ae89f'
wrapped_wget_sha512

# /licenses_templates/ ---------------------------------------------
local LFBFL_dependencies_URL2
LFBFL_dependencies_URL2="${LFBFL_dependencies_URL}/licenses_templates"
readonly LFBFL_dependencies_URL2
LFBFL_file_name="build_licenses_templates.exec.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
LFBFL_correct_sha512='7af3968a0e765d5da1d15a14b3e40ba622e07ba6fa758'
LFBFL_correct_sha512+='7d3ee875e67c59da8a84d6b302affdbc8407fce7fbbc'
LFBFL_correct_sha512+='d65c52eb2965c4f1cf021f42de671f1a0739536'
wrapped_wget_sha512
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="license_file_header_AGPLv3+.tpl"
LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
LFBFL_correct_sha512='9f9aa56d9b90d5003a12c5ec3370ac31aab74fc4a2524'
LFBFL_correct_sha512+='2c5fbcb0197b0abbb8242f62dafc76eaf60faa5da173'
LFBFL_correct_sha512+='05924b31b64b037083532fced9f98445000e2d0'
wrapped_wget_sha512

LFBFL_file_name="license_file_header_GPLv3+.tpl"
LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
LFBFL_correct_sha512='2bdef9a729a5a9ca5ed87ef8a518877c5ee6b31b45336'
LFBFL_correct_sha512+='08660316a4e595fb31465a4847f4d53a2f7264118bd4'
LFBFL_correct_sha512+='eab7c1d1892e00bc1861455ed02898f194db067'
wrapped_wget_sha512

LFBFL_file_name="license_file_header_LGPLv3+.tpl"
LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
LFBFL_correct_sha512='265fd0d086b48ab798f35a072335a4976e4eda8c256fe'
LFBFL_correct_sha512+='bba6c77c98ca86cdd844c6ac150c3445c80debbb3fc2'
LFBFL_correct_sha512+='574b5fceeeaf79ec734a759bdef5fe798067c62'
wrapped_wget_sha512
# ----------------------------------------------------------------

LFBFL_file_name="lines_counts.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='ec36e818b0aab4e386e4b01e643b07ad6bb1ef2ec5ccc'
```

```
LFBFL_correct_sha512+='4cfa697bac206f9a65d3cd710994a18db7397597a403'
LFBFL_correct_sha512+='463ec1353bc951ec4d79f54ccee61261dd279d8'
wrapped_wget_sha512

LFBFL_file_name="lines_filters.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='4f0aaa7e72721289d97f53e77cc735142352cb4a80bf5'
LFBFL_correct_sha512+='bc668905013a079cbca8b01d3f5bbfb7455f2c257b86'
LFBFL_correct_sha512+='6a9bf0a976c2de141995fc7871b98efaa6be217'
wrapped_wget_sha512

LFBFL_file_name="lines_maps.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='25c3e5082951260295b5f0e87ba1f2c978efdea1e8926'
LFBFL_correct_sha512+='072ce4d2062dc2a47bb92b9098f081e553eeae5d9154'
LFBFL_correct_sha512+='fdc50c9d88ae5365b776a23eb6e026b884a8374'
wrapped_wget_sha512

LFBFL_file_name="overwrite_if_not_equal.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='2e2e9c804ec8e1879fc96df595f30994a0685934483ce'
LFBFL_correct_sha512+='c9deed4c2fb98f79d4a393e13b6511b3b740ce9f073a'
LFBFL_correct_sha512+='e7ed124f08a60b35c896d60dd710223d2592d16'
wrapped_wget_sha512

LFBFL_file_name="python_black_complement.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='abe0e5701ad6a56937b1a1e3687366e87e8cb2578a179'
LFBFL_correct_sha512+='8e15ce81e0aaafc208c1b325058287c331092765ef73'
LFBFL_correct_sha512+='2c584cadf33e3acbcc195f25b0b632a0ce229d3'
wrapped_wget_sha512

LFBFL_file_name="python_isort_complement.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='70542f00cb59ed52d6576e6c066c5dc701ac90274ab68'
LFBFL_correct_sha512+='475086182b13b8cdf75f6b0263ac813e0010dd6cdb86'
LFBFL_correct_sha512+='9239899c2a6161f912bf1f789ad30965b8c1fe5'
wrapped_wget_sha512

LFBFL_file_name="repository_data.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
```

```
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='e3e8b487c45bf3960c5eb739fd1d29229ca358bfecb23'
LFBFL_correct_sha512+='e585fa7abf65c9cdaa263e799e96a83a22c660323e31'
LFBFL_correct_sha512+='bf2cdbc2f857b43957ef9405d2b8bed95dac7c0'
wrapped_wget_sha512

LFBFL_file_name="shell_checks_complement.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='c41e25b9d06078f524ce429d92f9fb436f0e40acca954'
LFBFL_correct_sha512+='0d0b534b5bc2941f6c81bffca7ad2faaa02bf75e4ba2'
LFBFL_correct_sha512+='031bffea4e6254056474c09f18824ce6e4090cf'
wrapped_wget_sha512

LFBFL_file_name="split_score.exec.php"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='0eaca0166e71a1992f897c7f87e77da3babd297cd85c6'
LFBFL_correct_sha512+='e4251ae920f15b4cbd9c3d4a7e2ae21eaf74b366a0e7'
LFBFL_correct_sha512+='681977dff58901c6d40b3e46f38e0cbec86be54'
wrapped_wget_sha512

LFBFL_file_name="split_score.libr.php"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='47b667d83125fa17babd46201ae8a8b7f7abeef123276'
LFBFL_correct_sha512+='40e7ed87faeacac997455207c4ab4df2734649676960'
LFBFL_correct_sha512+='1efcadaf35338ba2bec2410e01065ff7ed871f7'
wrapped_wget_sha512

LFBFL_file_name="strings_functions.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='1c5cb59a9ad306ca8cd49b398146dc6621de0e69cb510'
LFBFL_correct_sha512+='b9f41d4412d5b853f26eae7128410419be7dfb0d36cf'
LFBFL_correct_sha512+='1a6272f70f270e551406866b1e1fd209bdcce44'
wrapped_wget_sha512

LFBFL_file_name="too_long_code_lines.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='c8bf3f61560a1f41371a913bec2d18aaa1d6c80a24b98'
LFBFL_correct_sha512+='c2580c524c3483b9a109274bb7240dca857031f686fb'
LFBFL_correct_sha512+='0b261b41b221825cc0138e2f44ed5c54acbfd20'
wrapped_wget_sha512
```

```
LFBFL_file_name="update_or_check_files_names_listing.exec.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='9d9100a8527ea9717aff99f483c81b7921fd079ace4f9'
LFBFL_correct_sha512+='ead5cecd3c89da90f43959297ff78184eadf55bff79e'
LFBFL_correct_sha512+='b716060befe5eff9f00cc3b2f4dc69522e8bf8f'
wrapped_wget_sha512
chmod +x "./${LFBFL_file_path}"

# shellcheck source=check_shell_scripts_beginnings.libr.sh
source "./${LFBFL_subdir}/check_shell_scripts_beginnings.libr.sh"
# shellcheck source=check_shell_scripts_indentation.libr.sh
source "./${LFBFL_subdir}/check_shell_scripts_indentation.libr.sh"
# shellcheck source=check_URLs.libr.sh
source "./${LFBFL_subdir}/check_URLs.libr.sh"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=comparisons.libr.sh
source "./${LFBFL_subdir}/comparisons.libr.sh"
# shellcheck source=generate_from_template.libr.sh
source "./${LFBFL_subdir}/generate_from_template.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=grammar_and_spelling_check.libr.sh
source "./${LFBFL_subdir}/grammar_and_spelling_check.libr.sh"
# shellcheck source=lines_counts.libr.sh
source "./${LFBFL_subdir}/lines_counts.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=lines_maps.libr.sh
source "./${LFBFL_subdir}/lines_maps.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
# shellcheck source=python_black_complement.libr.sh
source "./${LFBFL_subdir}/python_black_complement.libr.sh"
# shellcheck source=python_isort_complement.libr.sh
source "./${LFBFL_subdir}/python_isort_complement.libr.sh"
# shellcheck source=shell_checks_complement.libr.sh
source "./${LFBFL_subdir}/shell_checks_complement.libr.sh"
# shellcheck source=strings_functions.libr.sh
source "./${LFBFL_subdir}/strings_functions.libr.sh"
# shellcheck source=too_long_code_lines.libr.sh
source "./${LFBFL_subdir}/too_long_code_lines.libr.sh"

enhanced_set_bash_option extglob
# shellcheck source=repository_data.libr.sh
```

```
source "./${LFBFL_subdir}/repository_data.libr.sh"
# shellcheck disable=SC2154,SC2309
[[ i_enhanced_set_bash_option_extglob_result -eq 0 ]]\
  && enhanced_unset_bash_option extglob

declare -a LFBFL_return_traps_stack
local LFBFL_previous_return_trap
init_return_trap

enhanced_set_shell_option pipefail --trap-unset

declare -a LFBFL_some_common_options2=("${LFBFL_some_common_options[@]}")
keep_options LFBFL_some_common_options2 --verbose

printf "Building license headers\n"
"./${LFBFL_subdir2}/build_licenses_templates.exec.sh"\
  "${LFBFL_some_common_options[@]}"

printf "Building README.md\n"
"./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
  --work-directory="${LFBFL_work_directory}"\
  --base-name=README\
  "${LFBFL_verbose}"

printf "Building other MarkDown files\n"
local LFBFL_some_directory
declare -r LFBFL_readme="${LFBFL_work_directory}/README.md.tpl"
local LFBFL_s_files_paths
LFBFL_s_files_paths=$(
  find "${LFBFL_work_directory}" -type f -name "*.md.tpl"\
  | relevant_find
)
declare -a LFBFL_arr_files_paths
if [[ -n "${LFBFL_s_files_paths}" ]]; then
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    [[ "${LFBFL_file_path}" == "${LFBFL_readme}" ]] && continue
    printf "Found template %s.\n" "${LFBFL_file_path}"
    LFBFL_some_directory=$(dirname "${LFBFL_file_path}")
    LFBFL_file_name=$(basename "${LFBFL_file_path}")
    LFBFL_file_name=${LFBFL_file_name%.md.tpl}
    "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
      --work-directory="${LFBFL_some_directory}"\
      --base-name="${LFBFL_file_name}"\
      "${LFBFL_verbose}"
  done
```

```
410     fi
411
412     printf "Building dependencies_notes\n"
413     # https://github.com/php/php-src/issues/21538
414     local LFBFL_annoying_warning
415     LFBFL_annoying_warning="PHP Warning:  declare(encoding=...) ignored"
416     LFBFL_annoying_warning=" because Zend multibyte feature is turned off"
417     LFBFL_annoying_warning=" by settings in"
418     readonly LFBFL_annoying_warning
419     php "./${LFBFL_subdir}/build_dependencies_notes.exec.php"\
420       "${LFBFL_work_directory}"\
421       2>&1\
422       | grep -v "${LFBFL_annoying_warning}"
423
424     declare -i LFBFL_i_directory_changed
425     pushd_to_work_directory
426     LFBFL_i_directory_changed=$?
427     can_continue_after_enhanced_pushd || return 1
428
429     local LFBFL_data_file_name="${LFBFL_subdir3}/repository_data.txt"
430
431     declare -i LFBFL_i_max_line_length
432     grep_variable "${LFBFL_data_file_name}" max_line_length\
433       --result-variable-prefix=LFBFL_i_
434
435     LFBFL_some_common_options+=(
436       "--max-line-length=${LFBFL_i_max_line_length}"
437     )
438     LFBFL_some_common_options2+=(
439       "--max-line-length=${LFBFL_i_max_line_length}"
440     )
441
442     declare -i LFBFL_i_upgrade_venvs=0
443     local LFBFL_upgrade_venvs_ts_file="${LFBFL_subdir3}/upgrade_venvs_ts"
444     readonly LFBFL_upgrade_venvs_ts_file
445     declare -i LFBFL_i_upgrade_venvs_ts
446     declare -i LFBFL_i_current_ts
447     local LFBFL_upgrade_venvs_answer
448
449     local LFBFL_upgrade_venvs_time_interval_in_seconds=""
450     get_upgrade_venvs_time_interval_in_seconds "${LFBFL_data_file_name}"\
451       "${LFBFL_some_common_options2[@]}"
452
453     if [[ -f "${LFBFL_upgrade_venvs_ts_file}" ]]; then
454       LFBFL_i_upgrade_venvs_ts=$(
455         stat --format %Y "${LFBFL_upgrade_venvs_ts_file}"
```

```
    )
    LFBFL_i_current_ts=$(date +%s)
    ((LFBFL_i_current_ts -= LFBFL_i_upgrade_venvs_ts))
    ((LFBFL_i_current_ts -= LFBFL_upgrade_venvs_time_interval_in_seconds))
    if [[ LFBFL_i_current_ts -gt 0 ]]; then
      LFBFL_i_upgrade_venvs=1
    fi
  else
    LFBFL_i_upgrade_venvs=1
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    local LFBFL_upgrade_venvs=""
    grep_variable "${LFBFL_data_file_name}" upgrade_venvs\
      --result-variable-prefix=LFBFL_
    if [[ "${LFBFL_upgrade_venvs}" != "auto" ]]; then
      read -r -n 1 -t 10\
        -p "Upgrade venvs and composer global? [Y/n]"\
        LFBFL_upgrade_venvs_answer
      if [[ "${LFBFL_upgrade_venvs_answer}" == "n" ]]; then
        LFBFL_i_upgrade_venvs=0
      fi
    fi
  fi

  printf "Running shellcheck\n"
  LFBFL_s_files_paths=$(
    find . -type f -name "*.sh"\
    | relevant_find
  )
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      shellcheck --rcfile="${LFBFL_subdir3}/shellcheck.ini"\
        "${LFBFL_file_path}"
    done
  fi
  printf "Running shell_checks_complement\n"
  shell_checks_complement "${LFBFL_some_common_options2[@]}"

  printf -- "---Python---\n"

  maybe_deactivate(){
    if command -v deactivate
    then
      deactivate
    fi
```

```
502     }
503
504     source_without_return_trap(){
505       declare -i LFBFL_i_no_return_traps=1
506       # shellcheck disable=SC1090,SC1091
507       source "$1"
508     }
509
510     printf "Running isort\n"
511     local LFBFL_isort_venv=""
512     grep_variable "${LFBFL_data_file_name}" isort_venv\
513       --result-variable-prefix=LFBFL_\
514       --replace-line-returns-by=""
515     if [[ -n "${LFBFL_isort_venv}" ]]; then
516       maybe_deactivate
517       source_without_return_trap "${LFBFL_isort_venv}/bin/activate"
518     fi
519     if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
520       pip install --upgrade isort
521     fi
522     isort --profile=black .
523     if [[ -n "${LFBFL_isort_venv}" ]]; then
524       deactivate
525     fi
526     printf "Running python_isort_complement\n"
527     python_isort_complement "${LFBFL_some_common_options2[@]}"
528
529     printf "Running black\n"
530     # First, we update the configuration file with max_line_length.
531     local LFBFL_update_max_length="s/^line-length = [0-9]*$"
532     LFBFL_update_max_length+="/line-length = ${LFBFL_i_max_line_length}/"
533     sed -E -e "${LFBFL_update_max_length}"\
534       "${LFBFL_subdir3}/black.toml"\
535       > "${LFBFL_subdir3}/black.toml.temp"
536     overwrite_if_not_equal "${LFBFL_subdir3}/black.toml"\
537       "${LFBFL_subdir3}/black.toml.temp"
538
539     local LFBFL_black_venv=""
540     grep_variable "${LFBFL_data_file_name}" black_venv\
541       --result-variable-prefix=LFBFL_\
542       --replace-line-returns-by=""
543     if [[ -n "${LFBFL_black_venv}" ]]; then
544       maybe_deactivate
545       source_without_return_trap "${LFBFL_black_venv}/bin/activate"
546     fi
547     if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
```

```
    pip install --upgrade black
  fi
  black --config="${LFBFL_subdir3}/black.toml" .
  if [[ -n "${LFBFL_black_venv}" ]]; then
    deactivate
  fi
  printf "Running python_black_complement\n"
  python_black_complement "${LFBFL_some_common_options2[@]}"

  printf "Probing if mypy should be runned\n"
  local LFBFL_mypy_venv=""
  grep_variable "${LFBFL_data_file_name}" mypy_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_mypy_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade mypy
  fi
  local LFBFL_directory_path
  declare -i LFBFL_i_no_toml=1
  LFBFL_s_files_paths=$(
    find . -type f -name "pyproject.toml"\
    | relevant_find
  )
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      if grep --quiet "Typing :: Typed" "${LFBFL_file_path}"; then
        printf "Running mypy\n"
        LFBFL_directory_path=$(dirname "${LFBFL_file_path}")
        mypy "${LFBFL_directory_path}"
      fi
      LFBFL_i_no_toml=0
    done
  fi
  if [[ LFBFL_i_no_toml -eq 1 ]]; then
    printf "Running mypy\n"
    mypy .
  fi
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    deactivate
  fi

```

```
  printf "Running bandit\n"
  LFBFL_bandit_venv=""
  grep_variable "${LFBFL_data_file_name}" bandit_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_bandit_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_bandit_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade bandit
  fi
  bandit --ini="${LFBFL_subdir3}/bandit.ini"\
    --baseline="${LFBFL_subdir3}/bandit_baseline.json"\
    --recursive .
  # Saving new baseline in temp if necessary.
  bandit --ini="${LFBFL_subdir3}/bandit.ini"\
    --format=json\
    --output="${LFBFL_subdir3}/temp/bandit_baseline.json"\
    --recursive .
  if [[ -n "${LFBFL_bandit_venv}" ]]; then
    deactivate
  fi

  printf "Running pylint\n"
  LFBFL_pylint_venv=""
  grep_variable "${LFBFL_data_file_name}" pylint_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_pylint_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_pylint_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade pylint
  fi
  pylint --rcfile="${LFBFL_subdir3}/pylintrc.toml" --recursive=y .
  if [[ -n "${LFBFL_pylint_venv}" ]]; then
    deactivate
  fi

  printf "Running ruff\n"
  LFBFL_ruff_venv=""
  grep_variable "${LFBFL_data_file_name}" ruff_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
```

```
640   if [[ -n "${LFBFL_ruff_venv}" ]]; then
641     maybe_deactivate
642     source_without_return_trap "${LFBFL_ruff_venv}/bin/activate"
643   fi
644   if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
645     pip install --upgrade ruff
646   fi
647   ruff check --config="${LFBFL_subdir3}/ruff.toml"
648   if [[ -n "${LFBFL_ruff_venv}" ]]; then
649     deactivate
650   fi
651   printf -- "---Python end---\n"
652
653   printf -- "---PHP---\n"
654   printf "Running PHP native linter\n"
655   LFBFL_s_files_paths=$(
656     find . -type f -name "*.php"\
657     | relevant_find
658   )
659   if [[ -n "${LFBFL_s_files_paths}" ]]; then
660     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
661     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
662       php --syntax-check "${LFBFL_file_path}"
663     done
664   fi
665
666   printf "Running PHP_CodeSniffer\n"
667   if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
668     composer global require squizlabs/php_codesniffer
669   fi
670   phpcs --report=code\
671     --standard=build_and_checks_variables/phpcs_ruleset.xml
672
673   printf "Running PHPMD\n"
674   if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
675     composer global require phpmd/phpmd
676   fi
677
678   declare -r LFBFL_phpmd_baseline="${LFBFL_subdir3}/phpmd_baseline.xml"
679   local LFBFL_temp_phpmd_baseline="${LFBFL_subdir3}/temp/"
680   LFBFL_temp_phpmd_baseline+="phpmd_baseline.xml"
681   readonly LFBFL_temp_phpmd_baseline
682
683   local LFBFL_phpmd_rulesets="cleancode,codesize,controversial,"
684   LFBFL_phpmd_rulesets+="design,naming,unusedcode"
685   readonly LFBFL_phpmd_rulesets
```

```
686
687   phpmd --color --baseline-file="${LFBFL_phpmd_baseline}"\
688     . text "${LFBFL_phpmd_rulesets}"
689   # Saving new baseline in temp if necessary.
690   rm "${LFBFL_temp_phpmd_baseline}"
691   phpmd --color --generate-baseline\
692     --baseline-file="${LFBFL_temp_phpmd_baseline}"\
693     . text "${LFBFL_phpmd_rulesets}"
694   sed -i -e 's/" file="/"\n    file="/g' "${LFBFL_temp_phpmd_baseline}"
695   sed -i -E -e 's~(file=".*)/>~\1\n  />~g' "${LFBFL_temp_phpmd_baseline}"
696   diff "${LFBFL_phpmd_baseline}" "${LFBFL_temp_phpmd_baseline}"
697
698   printf "Running phpDocumentor\n"
699   # shellcheck disable=SC2312
700   docker run --rm -v "$(pwd):/data" phpdoc/phpdoc
701   printf -- "---PHP end---\n"
702
703   printf -- "---JS---\n"
704   LFBFL_npm_lint_directories=""
705   grep_variable "${LFBFL_data_file_name}" npm_lint_directories\
706     --result-variable-prefix=LFBFL_
707   if [[ -n "${LFBFL_npm_lint_directories}" ]]; then
708     printf "Running ESLint\n"
709     local LFBFL_JS_directory
710     declare -a LFBFL_arr_paths
711     mapfile -t LFBFL_arr_paths <<< "${LFBFL_npm_lint_directories}"
712     for LFBFL_JS_directory in "${LFBFL_arr_paths[@]}"; do
713       (cd "${LFBFL_JS_directory}" && npm run lint)
714     done
715   fi
716   printf -- "---JS end---\n"
717
718   printf "Analyzing too long lines\n"
719   # shellcheck disable=SC2248
720   too_long_code_lines "${LFBFL_some_common_options2[@]}"
721
722   printf "Analyzing shell scripts beginnings\n"
723   check_shell_scripts_beginnings "${LFBFL_some_common_options2[@]}" \
724     | relevant_grep
725
726   printf "Analyzing shell scripts indentation\n"
727   check_shell_scripts_indentation "${LFBFL_some_common_options2[@]}" \
728     | relevant_grep
729
730   printf "Analyzing URLs\n"
731   check_URLs "${LFBFL_some_common_options2[@]}" \
```

```
    | relevant_grep

  printf "Analyzing strange characters: hover over in doubt\n"
  # shellcheck disable=SC1111
  LFBFL_usual_characters="\x00-\x7Fàâéèêëîïôûç©‘”└├│«»..."
  grep --binary-files=without-match --color=always --exclude-dir .git\
    --line-number --perl-regexp --recursive --\
    "[^${LFBFL_usual_characters}]" .\
    | relevant_grep

  printf "Grammar and spelling check\n"
  grammar_and_spelling_check "${LFBFL_data_file_name}"\
    "${LFBFL_some_common_options2[@]}"

  [[ LFBFL_i_directory_changed -eq 0 ]] && popd_from_work_directory

  printf "Checking listed files\n"
  "./${LFBFL_subdir}/update_or_check_files_names_listing.exec.sh"\
    "${LFBFL_some_common_options[@]}"

  printf "Creating the PDF file of the listing of the source code\n"
  "./${LFBFL_subdir}/create_PDF.exec.sh" "${LFBFL_some_common_options[@]}"

  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  if [[ -f "${LFBFL_subdir3}/post_build.sh" ]]; then
    "./${LFBFL_subdir3}/post_build.sh" "${LFBFL_verbose}"
  fi

  # ------------------------------------------------------------------
  # It is at the end that all the HTML files are generated and that
  # we can check there are no mistakes.
  printf "Running Nu W3C HTML CSS and SVG validator\n"
  local LFBFL_upgrade_vnu_jar_time_interval_in_seconds=""
  grep_variable "${LFBFL_data_file_name}"\
    upgrade_vnu_jar_time_interval_in_seconds\
    --result-variable-prefix=LFBFL_

  if [[ "${LFBFL_upgrade_vnu_jar_time_interval_in_seconds}" == "wat" ]];
  then
    LFBFL_upgrade_vnu_jar_time_interval_in_seconds=${RANDOM}
  fi

  declare -i LFBFL_i_vnu_jar_ts
  local LFBFL_vnu_jar_path=""
```

```
  grep_variable "${LFBFL_data_file_name}" vnu_jar_path\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -f "${LFBFL_vnu_jar_path}" ]]; then
    LFBFL_i_vnu_jar_ts=$(stat --format %Y "${LFBFL_vnu_jar_path}")
    LFBFL_i_current_ts=$(date +%s)
    ((LFBFL_i_current_ts -= LFBFL_i_vnu_jar_ts))
    ((
      LFBFL_i_current_ts -= LFBFL_upgrade_vnu_jar_time_interval_in_seconds
    ))
    if [[ LFBFL_i_current_ts -gt 0 ]]; then
      printf $'/!\\ATTENTION: Nu W3C validator is too old./!\\\n'
      printf "Go get the latest version here:\n"
      printf "https://github.com/validator/validator/releases\n"
    fi
    # vnu has false positives for xhtml
    local LFBFL_use_vnu_for_xhtml=""
    grep_variable "${LFBFL_data_file_name}" use_vnu_for_xhtml\
      --result-variable-prefix=LFBFL_
    local LFBFL_find_regexp
    if [[ "${LFBFL_use_vnu_for_xhtml}" == "true" ]]; then
      LFBFL_find_regexp=".*\.\(css\|html\|svg\|xhtml\)"
    else
      LFBFL_find_regexp=".*\.\(css\|html\|svg\)"
    fi
    declare -r LFBFL_files_for_Nu=$(
      find . -iregex "${LFBFL_find_regexp}"\
      | relevant_find
    )
    if [[ -n "${LFBFL_files_for_Nu}" ]]; then
      declare -a LFBFL_files_for_Nu2
      mapfile -t LFBFL_files_for_Nu2 <<< "${LFBFL_files_for_Nu}"
      readonly LFBFL_files_for_Nu2
      java -Xss128M -jar "${LFBFL_vnu_jar_path}" --exit-zero-always\
        --verbose "${LFBFL_files_for_Nu2[@]}"
    fi
  fi
  # ------------------------------------------------------------------

  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    touch "${LFBFL_upgrade_venvs_ts_file}"
  fi
}

common_build_and_checks "$@"
```

3.9 build_and_checks_dependencies/common_build_and_checks.exec.sh.tpl

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# The file "common_build_and_checks.exec.sh.tpl" was renamed from
# "common_build_and_checks.sh.tpl" to
# "common_build_and_checks.exec.sh.tpl".
# The file "common_build_and_checks.exec.sh" generated from the file
# "common_build_and_checks.sh.tpl"
# or "common_build_and_checks.exec.sh.tpl"
# was renamed from
# "common_build_and_checks.sh" to "common_build_and_checks.exec.sh".

common_build_and_checks(){
  # $1 LFBFL_work_directory
  # $2 LFBFL_dependencies_URL
  # Options:
  #   --check-download (=keep to keep the downloaded files)
  #   --verbose
  local LFBFL_work_directory="${1:-.}"
  LFBFL_work_directory=$(realpath "${LFBFL_work_directory}")
  local LFBFL_work_directory_option="--work-directory="
  LFBFL_work_directory_option+="${LFBFL_work_directory}"
  declare -r LFBFL_dependencies_URL="$2"
  local LFBFL_verbose=""
```

```
45    declare -i LFBFL_i_verbose=0
46    if [[ "$*" == *--verbose* ]]; then
47      printf "%s %s\n" "$0" "$*"
48      LFBFL_verbose="--verbose"
49      LFBFL_i_verbose=1
50    fi
51    readonly LFBFL_verbose
52    readonly LFBFL_i_verbose
53    local LFBFL_check_download=""
54    if [[ "$*" == *--check-download=keep* ]]; then
55      LFBFL_check_download="--check-download=keep"
56    elif [[ "$*" == *--check-download* ]]; then
57      LFBFL_check_download="--check-download"
58    fi
59    readonly LFBFL_check_download
60
61    declare -a LFBFL_some_common_options=(
62      "${LFBFL_verbose}"
63      # "$--work-directory={LFBFL_work_directory}" doesn't work
64      "${LFBFL_work_directory_option}"
65    )
66
67    source ./wget_sha512.libr.sh
68
69    local LFBFL_subdir="build_and_checks_dependencies"
70    declare -r LFBFL_subdir2="${LFBFL_subdir}/licenses_templates"
71    declare -r LFBFL_subdir3="build_and_checks_variables"
72    local LFBFL_file_name
73    local LFBFL_script_download_URL
74    local LFBFL_file_path
75    local LFBFL_correct_sha512
76
77    wrapped_wget_sha512(){
78      wget_sha512 "${LFBFL_file_path}"\
79        "${LFBFL_script_download_URL}"\
80        "${LFBFL_correct_sha512}"\
81        "${LFBFL_verbose}"\
82        "${LFBFL_check_download}"
83    }
84
85    LFBFL_file_name="build_dependencies_notes.exec.php"
86    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
87    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
88    @sha512_build_dependencies_notes.exec.php@
89    wrapped_wget_sha512
90    chmod +x "./${LFBFL_file_path}"
```

```
91
92    LFBFL_file_name="build_dependencies_notes.libr.php"
93    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
94    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
95    @sha512_build_dependencies_notes.libr.php@
96    wrapped_wget_sha512
97
98    LFBFL_file_name="build_md_from_printable_md.exec.sh"
99    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
100   LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
101   @sha512_build_md_from_printable_md.exec.sh@
102   wrapped_wget_sha512
103   chmod +x "./${LFBFL_file_path}"
104
105   LFBFL_file_name="check_shell_scripts_beginnings.libr.sh"
106   LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
107   LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
108   @sha512_check_shell_scripts_beginnings.libr.sh@
109   wrapped_wget_sha512
110
111   LFBFL_file_name="check_shell_scripts_indentation.libr.sh"
112   LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
113   LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
114   @sha512_check_shell_scripts_indentation.libr.sh@
115   wrapped_wget_sha512
116
117   LFBFL_file_name="check_URLs.libr.sh"
118   LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
119   LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
120   @sha512_check_URLs.libr.sh@
121   wrapped_wget_sha512
122
123   LFBFL_file_name="common_options.libr.sh"
124   LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
125   LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
126   @sha512_common_options.libr.sh@
127   wrapped_wget_sha512
128
129   LFBFL_file_name="comparisons.libr.sh"
130   LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
131   LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
132   @sha512_comparisons.libr.sh@
133   wrapped_wget_sha512
134
135   LFBFL_file_name="create_PDF.exec.sh"
136   LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
```

```
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_create_PDF.exec.sh@
    wrapped_wget_sha512
    chmod +x "./${LFBFL_file_path}"

    LFBFL_file_name="generate_from_template.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_generate_from_template.libr.sh@
    wrapped_wget_sha512

    LFBFL_file_name="get_common_text_glob_patterns.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_get_common_text_glob_patterns.libr.sh@
    wrapped_wget_sha512

    LFBFL_file_name="grammar_and_spelling_check.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_grammar_and_spelling_check.libr.sh@
    wrapped_wget_sha512

    # /licenses_templates/ ---------------------------------------------
    local LFBFL_dependencies_URL2
    LFBFL_dependencies_URL2="${LFBFL_dependencies_URL}/licenses_templates"
    readonly LFBFL_dependencies_URL2
    LFBFL_file_name="build_licenses_templates.exec.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
    @sha512_build_licenses_templates.exec.sh@
    wrapped_wget_sha512
    chmod +x "./${LFBFL_file_path}"

    LFBFL_file_name="license_file_header_AGPLv3+.tpl"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
    @sha512_license_file_header_AGPLv3+.tpl@
    wrapped_wget_sha512

    LFBFL_file_name="license_file_header_GPLv3+.tpl"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
    @sha512_license_file_header_GPLv3+.tpl@
    wrapped_wget_sha512

```

```
LFBFL_file_name="license_file_header_LGPLv3+.tpl"
LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
@sha512_license_file_header_LGPLv3+.tpl@
wrapped_wget_sha512
# ------------------------------------------------------------------

LFBFL_file_name="lines_counts.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_lines_counts.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="lines_filters.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_lines_filters.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="lines_maps.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_lines_maps.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="overwrite_if_not_equal.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_overwrite_if_not_equal.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="python_black_complement.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_python_black_complement.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="python_isort_complement.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_python_isort_complement.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="repository_data.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
```

```
229     @sha512_repository_data.libr.sh@
230     wrapped_wget_sha512
231
232     LFBFL_file_name="shell_checks_complement.libr.sh"
233     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
234     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
235     @sha512_shell_checks_complement.libr.sh@
236     wrapped_wget_sha512
237
238     LFBFL_file_name="split_score.exec.php"
239     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
240     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
241     @sha512_split_score.exec.php@
242     wrapped_wget_sha512
243
244     LFBFL_file_name="split_score.libr.php"
245     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
246     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
247     @sha512_split_score.libr.php@
248     wrapped_wget_sha512
249
250     LFBFL_file_name="strings_functions.libr.sh"
251     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
252     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
253     @sha512_strings_functions.libr.sh@
254     wrapped_wget_sha512
255
256     LFBFL_file_name="too_long_code_lines.libr.sh"
257     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
258     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
259     @sha512_too_long_code_lines.libr.sh@
260     wrapped_wget_sha512
261
262     LFBFL_file_name="update_or_check_files_names_listing.exec.sh"
263     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
264     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
265     @sha512_update_or_check_files_names_listing.exec.sh@
266     wrapped_wget_sha512
267     chmod +x "./${LFBFL_file_path}"
268
269     # shellcheck source=check_shell_scripts_beginnings.libr.sh
270     source "./${LFBFL_subdir}/check_shell_scripts_beginnings.libr.sh"
271     # shellcheck source=check_shell_scripts_indentation.libr.sh
272     source "./${LFBFL_subdir}/check_shell_scripts_indentation.libr.sh"
273     # shellcheck source=check_URLs.libr.sh
274     source "./${LFBFL_subdir}/check_URLs.libr.sh"
```

```
275     # shellcheck source=common_options.libr.sh
276     source "./${LFBFL_subdir}/common_options.libr.sh"
277     # shellcheck source=comparisons.libr.sh
278     source "./${LFBFL_subdir}/comparisons.libr.sh"
279     # shellcheck source=generate_from_template.libr.sh
280     source "./${LFBFL_subdir}/generate_from_template.libr.sh"
281     # shellcheck source=get_common_text_glob_patterns.libr.sh
282     source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
283     # shellcheck source=grammar_and_spelling_check.libr.sh
284     source "./${LFBFL_subdir}/grammar_and_spelling_check.libr.sh"
285     # shellcheck source=lines_counts.libr.sh
286     source "./${LFBFL_subdir}/lines_counts.libr.sh"
287     # shellcheck source=lines_filters.libr.sh
288     source "./${LFBFL_subdir}/lines_filters.libr.sh"
289     # shellcheck source=lines_maps.libr.sh
290     source "./${LFBFL_subdir}/lines_maps.libr.sh"
291     # shellcheck source=overwrite_if_not_equal.libr.sh
292     source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
293     # shellcheck source=python_black_complement.libr.sh
294     source "./${LFBFL_subdir}/python_black_complement.libr.sh"
295     # shellcheck source=python_isort_complement.libr.sh
296     source "./${LFBFL_subdir}/python_isort_complement.libr.sh"
297     # shellcheck source=shell_checks_complement.libr.sh
298     source "./${LFBFL_subdir}/shell_checks_complement.libr.sh"
299     # shellcheck source=strings_functions.libr.sh
300     source "./${LFBFL_subdir}/strings_functions.libr.sh"
301     # shellcheck source=too_long_code_lines.libr.sh
302     source "./${LFBFL_subdir}/too_long_code_lines.libr.sh"
303
304     enhanced_set_bash_option extglob
305     # shellcheck source=repository_data.libr.sh
306     source "./${LFBFL_subdir}/repository_data.libr.sh"
307     # shellcheck disable=SC2154,SC2309
308     [[ i_enhanced_set_bash_option_extglob_result -eq 0 ]]\
309       && enhanced_unset_bash_option extglob
310
311     declare -a LFBFL_return_traps_stack
312     local LFBFL_previous_return_trap
313     init_return_trap
314
315     enhanced_set_shell_option pipefail --trap-unset
316
317     declare -a LFBFL_some_common_options2=("${LFBFL_some_common_options[@]}")
318     keep_options LFBFL_some_common_options2 --verbose
319
320     printf "Building license headers\n"
```

```
  "./${LFBFL_subdir2}/build_licenses_templates.exec.sh"\
    "${LFBFL_some_common_options[@]}"

  printf "Building README.md\n"
  "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
    --work-directory="${LFBFL_work_directory}"\
    --base-name=README\
    "${LFBFL_verbose}"

  printf "Building other MarkDown files\n"
  local LFBFL_some_directory
  declare -r LFBFL_readme="${LFBFL_work_directory}/README.md.tpl"
  local LFBFL_s_files_paths
  LFBFL_s_files_paths=$(
    find "${LFBFL_work_directory}" -type f -name "*.md.tpl"\
    | relevant_find
  )
  declare -a LFBFL_arr_files_paths
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      [[ "${LFBFL_file_path}" == "${LFBFL_readme}" ]] && continue
      printf "Found template %s.\n" "${LFBFL_file_path}"
      LFBFL_some_directory=$(dirname "${LFBFL_file_path}")
      LFBFL_file_name=$(basename "${LFBFL_file_path}")
      LFBFL_file_name=${LFBFL_file_name%.md.tpl}
      "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
        --work-directory="${LFBFL_some_directory}"\
        --base-name="${LFBFL_file_name}"\
        "${LFBFL_verbose}"
    done
  fi

  printf "Building dependencies_notes\n"
  # https://github.com/php/php-src/issues/21538
  local LFBFL_annoying_warning
  LFBFL_annoying_warning="PHP Warning:  declare(encoding=...) ignored"
  LFBFL_annoying_warning=" because Zend multibyte feature is turned off"
  LFBFL_annoying_warning=" by settings in"
  readonly LFBFL_annoying_warning
  php "./${LFBFL_subdir}/build_dependencies_notes.exec.php"\
    "${LFBFL_work_directory}"\
    2>&1\
    | grep -v "${LFBFL_annoying_warning}"

  declare -i LFBFL_i_directory_changed
```

```
367     pushd_to_work_directory
368     LFBFL_i_directory_changed=$?
369     can_continue_after_enhanced_pushd || return 1
370
371     local LFBFL_data_file_name="${LFBFL_subdir3}/repository_data.txt"
372
373     declare -i LFBFL_i_max_line_length
374     grep_variable "${LFBFL_data_file_name}" max_line_length\
375       --result-variable-prefix=LFBFL_i_
376
377     LFBFL_some_common_options+=(
378       "--max-line-length=${LFBFL_i_max_line_length}"
379     )
380     LFBFL_some_common_options2+=(
381       "--max-line-length=${LFBFL_i_max_line_length}"
382     )
383
384     declare -i LFBFL_i_upgrade_venvs=0
385     local LFBFL_upgrade_venvs_ts_file="${LFBFL_subdir3}/upgrade_venvs_ts"
386     readonly LFBFL_upgrade_venvs_ts_file
387     declare -i LFBFL_i_upgrade_venvs_ts
388     declare -i LFBFL_i_current_ts
389     local LFBFL_upgrade_venvs_answer
390
391     local LFBFL_upgrade_venvs_time_interval_in_seconds=""
392     get_upgrade_venvs_time_interval_in_seconds "${LFBFL_data_file_name}"\
393       "${LFBFL_some_common_options2[@]}"
394
395     if [[ -f "${LFBFL_upgrade_venvs_ts_file}" ]]; then
396       LFBFL_i_upgrade_venvs_ts=$(
397         stat --format %Y "${LFBFL_upgrade_venvs_ts_file}"
398       )
399       LFBFL_i_current_ts=$(date +%s)
400       ((LFBFL_i_current_ts -= LFBFL_i_upgrade_venvs_ts))
401       ((LFBFL_i_current_ts -= LFBFL_upgrade_venvs_time_interval_in_seconds))
402       if [[ LFBFL_i_current_ts -gt 0 ]]; then
403         LFBFL_i_upgrade_venvs=1
404       fi
405     else
406       LFBFL_i_upgrade_venvs=1
407     fi
408     if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
409       local LFBFL_upgrade_venvs=""
410       grep_variable "${LFBFL_data_file_name}" upgrade_venvs\
411         --result-variable-prefix=LFBFL_
412       if [[ "${LFBFL_upgrade_venvs}" != "auto" ]]; then
```

```
      read -r -n 1 -t 10\
        -p "Upgrade venvs and composer global? [Y/n]"\
        LFBFL_upgrade_venvs_answer
      if [[ "${LFBFL_upgrade_venvs_answer}" == "n" ]]; then
        LFBFL_i_upgrade_venvs=0
      fi
    fi
  fi

  printf "Running shellcheck\n"
  LFBFL_s_files_paths=$(
    find . -type f -name "*.sh"\
    | relevant_find
  )
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      shellcheck --rcfile="${LFBFL_subdir3}/shellcheck.ini"\
        "${LFBFL_file_path}"
    done
  fi
  printf "Running shell_checks_complement\n"
  shell_checks_complement "${LFBFL_some_common_options2[@]}"

  printf -- "---Python---\n"

  maybe_deactivate(){
    if command -v deactivate
    then
      deactivate
    fi
  }

  source_without_return_trap(){
    declare -i LFBFL_i_no_return_traps=1
    # shellcheck disable=SC1090,SC1091
    source "$1"
  }

  printf "Running isort\n"
  local LFBFL_isort_venv=""
  grep_variable "${LFBFL_data_file_name}" isort_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_isort_venv}" ]]; then
    maybe_deactivate
```

```
    source_without_return_trap "${LFBFL_isort_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade isort
  fi
  isort --profile=black .
  if [[ -n "${LFBFL_isort_venv}" ]]; then
    deactivate
  fi
  printf "Running python_isort_complement\n"
  python_isort_complement "${LFBFL_some_common_options2[@]}"

  printf "Running black\n"
  # First, we update the configuration file with max_line_length.
  local LFBFL_update_max_length="s/^line-length = [0-9]*$"
  LFBFL_update_max_length+="/line-length = ${LFBFL_i_max_line_length}/"
  sed -E -e "${LFBFL_update_max_length}"\
    "${LFBFL_subdir3}/black.toml"\
    > "${LFBFL_subdir3}/black.toml.temp"
  overwrite_if_not_equal "${LFBFL_subdir3}/black.toml"\
    "${LFBFL_subdir3}/black.toml.temp"

  local LFBFL_black_venv=""
  grep_variable "${LFBFL_data_file_name}" black_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_black_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_black_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade black
  fi
  black --config="${LFBFL_subdir3}/black.toml" .
  if [[ -n "${LFBFL_black_venv}" ]]; then
    deactivate
  fi
  printf "Running python_black_complement\n"
  python_black_complement "${LFBFL_some_common_options2[@]}"

  printf "Probing if mypy should be runned\n"
  local LFBFL_mypy_venv=""
  grep_variable "${LFBFL_data_file_name}" mypy_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
```

```
    maybe_deactivate
    source_without_return_trap "${LFBFL_mypy_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade mypy
  fi
  local LFBFL_directory_path
  declare -i LFBFL_i_no_toml=1
  LFBFL_s_files_paths=$(
    find . -type f -name "pyproject.toml"\
    | relevant_find
  )
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      if grep --quiet "Typing :: Typed" "${LFBFL_file_path}"; then
        printf "Running mypy\n"
        LFBFL_directory_path=$(dirname "${LFBFL_file_path}")
        mypy "${LFBFL_directory_path}"
      fi
      LFBFL_i_no_toml=0
    done
  fi
  if [[ LFBFL_i_no_toml -eq 1 ]]; then
    printf "Running mypy\n"
    mypy .
  fi
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    deactivate
  fi

  printf "Running bandit\n"
  LFBFL_bandit_venv=""
  grep_variable "${LFBFL_data_file_name}" bandit_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_bandit_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_bandit_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade bandit
  fi
  bandit --ini="${LFBFL_subdir3}/bandit.ini"\
    --baseline="${LFBFL_subdir3}/bandit_baseline.json"\
    --recursive .
```

```
551   # Saving new baseline in temp if necessary.
552   bandit --ini="${LFBFL_subdir3}/bandit.ini"\
553     --format=json\
554     --output="${LFBFL_subdir3}/temp/bandit_baseline.json"\
555     --recursive .
556   if [[ -n "${LFBFL_bandit_venv}" ]]; then
557     deactivate
558   fi
559
560   printf "Running pylint\n"
561   LFBFL_pylint_venv=""
562   grep_variable "${LFBFL_data_file_name}" pylint_venv\
563     --result-variable-prefix=LFBFL_\
564     --replace-line-returns-by=""
565   if [[ -n "${LFBFL_pylint_venv}" ]]; then
566     maybe_deactivate
567     source_without_return_trap "${LFBFL_pylint_venv}/bin/activate"
568   fi
569   if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
570     pip install --upgrade pylint
571   fi
572   pylint --rcfile="${LFBFL_subdir3}/pylintrc.toml" --recursive=y .
573   if [[ -n "${LFBFL_pylint_venv}" ]]; then
574     deactivate
575   fi
576
577   printf "Running ruff\n"
578   LFBFL_ruff_venv=""
579   grep_variable "${LFBFL_data_file_name}" ruff_venv\
580     --result-variable-prefix=LFBFL_\
581     --replace-line-returns-by=""
582   if [[ -n "${LFBFL_ruff_venv}" ]]; then
583     maybe_deactivate
584     source_without_return_trap "${LFBFL_ruff_venv}/bin/activate"
585   fi
586   if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
587     pip install --upgrade ruff
588   fi
589   ruff check --config="${LFBFL_subdir3}/ruff.toml"
590   if [[ -n "${LFBFL_ruff_venv}" ]]; then
591     deactivate
592   fi
593   printf -- "---Python end---\n"
594
595   printf -- "---PHP---\n"
596   printf "Running PHP native linter\n"
```

```
LFBFL_s_files_paths=$(
  find . -type f -name "*.php"\
  | relevant_find
)
if [[ -n "${LFBFL_s_files_paths}" ]]; then
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    php --syntax-check "${LFBFL_file_path}"
  done
fi

printf "Running PHP_CodeSniffer\n"
if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
  composer global require squizlabs/php_codesniffer
fi
phpcs --report=code\
  --standard=build_and_checks_variables/phpcs_ruleset.xml

printf "Running PHPMD\n"
if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
  composer global require phpmd/phpmd
fi

declare -r LFBFL_phpmd_baseline="${LFBFL_subdir3}/phpmd_baseline.xml"
local LFBFL_temp_phpmd_baseline="${LFBFL_subdir3}/temp/"
LFBFL_temp_phpmd_baseline+="phpmd_baseline.xml"
readonly LFBFL_temp_phpmd_baseline

local LFBFL_phpmd_rulesets="cleancode,codesize,controversial,"
LFBFL_phpmd_rulesets+="design,naming,unusedcode"
readonly LFBFL_phpmd_rulesets

phpmd --color --baseline-file="${LFBFL_phpmd_baseline}"\
  . text "${LFBFL_phpmd_rulesets}"
# Saving new baseline in temp if necessary.
rm "${LFBFL_temp_phpmd_baseline}"
phpmd --color --generate-baseline\
  --baseline-file="${LFBFL_temp_phpmd_baseline}"\
  . text "${LFBFL_phpmd_rulesets}"
sed -i -e 's/" file="/"\n    file="/g' "${LFBFL_temp_phpmd_baseline}"
sed -i -E -e 's~(file=".*)/>~\1\n  />~g' "${LFBFL_temp_phpmd_baseline}"
diff "${LFBFL_phpmd_baseline}" "${LFBFL_temp_phpmd_baseline}"

printf "Running phpDocumentor\n"
# shellcheck disable=SC2312
docker run --rm -v "$(pwd):/data" phpdoc/phpdoc
```

```
  printf -- "---PHP end---\n"

  printf -- "---JS---\n"
  LFBFL_npm_lint_directories=""
  grep_variable "${LFBFL_data_file_name}" npm_lint_directories\
    --result-variable-prefix=LFBFL_
  if [[ -n "${LFBFL_npm_lint_directories}" ]]; then
    printf "Running ESLint\n"
    local LFBFL_JS_directory
    declare -a LFBFL_arr_paths
    mapfile -t LFBFL_arr_paths <<< "${LFBFL_npm_lint_directories}"
    for LFBFL_JS_directory in "${LFBFL_arr_paths[@]}"; do
      (cd "${LFBFL_JS_directory}" && npm run lint)
    done
  fi
  printf -- "---JS end---\n"

  printf "Analyzing too long lines\n"
  # shellcheck disable=SC2248
  too_long_code_lines "${LFBFL_some_common_options2[@]}"

  printf "Analyzing shell scripts beginnings\n"
  check_shell_scripts_beginnings "${LFBFL_some_common_options2[@]}" \
    | relevant_grep

  printf "Analyzing shell scripts indentation\n"
  check_shell_scripts_indentation "${LFBFL_some_common_options2[@]}" \
    | relevant_grep

  printf "Analyzing URLs\n"
  check_URLs "${LFBFL_some_common_options2[@]}" \
    | relevant_grep

  printf "Analyzing strange characters: hover over in doubt\n"
  # shellcheck disable=SC1111
  LFBFL_usual_characters="\x00-\x7Fàâéèêëîïôûç©‘”└├│«»..."
  grep --binary-files=without-match --color=always --exclude-dir .git\
    --line-number --perl-regexp --recursive --\
    "[^${LFBFL_usual_characters}]" .\
    | relevant_grep

  printf "Grammar and spelling check\n"
  grammar_and_spelling_check "${LFBFL_data_file_name}"\
    "${LFBFL_some_common_options2[@]}"

  [[ LFBFL_i_directory_changed -eq 0 ]] && popd_from_work_directory
```

```
689
690   printf "Checking listed files\n"
691   "./${LFBFL_subdir}/update_or_check_files_names_listing.exec.sh"\
692     "${LFBFL_some_common_options[@]}"
693
694   printf "Creating the PDF file of the listing of the source code\n"
695   "./${LFBFL_subdir}/create_PDF.exec.sh" "${LFBFL_some_common_options[@]}"
696
697   pushd_to_work_directory --trap-popd
698   can_continue_after_enhanced_pushd || return 1
699
700   if [[ -f "${LFBFL_subdir3}/post_build.sh" ]]; then
701     "./${LFBFL_subdir3}/post_build.sh" "${LFBFL_verbose}"
702   fi
703
704   # ------------------------------------------------------------------
705   # It is at the end that all the HTML files are generated and that
706   # we can check there are no mistakes.
707   printf "Running Nu W3C HTML CSS and SVG validator\n"
708   local LFBFL_upgrade_vnu_jar_time_interval_in_seconds=""
709   grep_variable "${LFBFL_data_file_name}"\
710     upgrade_vnu_jar_time_interval_in_seconds\
711     --result-variable-prefix=LFBFL_
712
713   if [[ "${LFBFL_upgrade_vnu_jar_time_interval_in_seconds}" == "wat" ]];
714   then
715     LFBFL_upgrade_vnu_jar_time_interval_in_seconds=${RANDOM}
716   fi
717
718   declare -i LFBFL_i_vnu_jar_ts
719   local LFBFL_vnu_jar_path=""
720   grep_variable "${LFBFL_data_file_name}" vnu_jar_path\
721     --result-variable-prefix=LFBFL_\
722     --replace-line-returns-by=""
723   if [[ -f "${LFBFL_vnu_jar_path}" ]]; then
724     LFBFL_i_vnu_jar_ts=$(stat --format %Y "${LFBFL_vnu_jar_path}")
725     LFBFL_i_current_ts=$(date +%s)
726     ((LFBFL_i_current_ts -= LFBFL_i_vnu_jar_ts))
727     ((
728       LFBFL_i_current_ts -= LFBFL_upgrade_vnu_jar_time_interval_in_seconds
729     ))
730     if [[ LFBFL_i_current_ts -gt 0 ]]; then
731       printf $'/!\\ATTENTION: Nu W3C validator is too old./!\\\n'
732       printf "Go get the latest version here:\n"
733       printf "https://github.com/validator/validator/releases\n"
734     fi
```

```
    # vnu has false positives for xhtml
    local LFBFL_use_vnu_for_xhtml=""
    grep_variable "${LFBFL_data_file_name}" use_vnu_for_xhtml\
      --result-variable-prefix=LFBFL_
    local LFBFL_find_regexp
    if [[ "${LFBFL_use_vnu_for_xhtml}" == "true" ]]; then
      LFBFL_find_regexp=".*\.\(css\|html\|svg\|xhtml\)"
    else
      LFBFL_find_regexp=".*\.\(css\|html\|svg\)"
    fi
    declare -r LFBFL_files_for_Nu=$(
      find . -iregex "${LFBFL_find_regexp}"\
      | relevant_find
    )
    if [[ -n "${LFBFL_files_for_Nu}" ]]; then
      declare -a LFBFL_files_for_Nu2
      mapfile -t LFBFL_files_for_Nu2 <<< "${LFBFL_files_for_Nu}"
      readonly LFBFL_files_for_Nu2
      java -Xss128M -jar "${LFBFL_vnu_jar_path}" --exit-zero-always\
        --verbose "${LFBFL_files_for_Nu2[@]}"
    fi
  fi
  # ------------------------------------------------------------------

  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    touch "${LFBFL_upgrade_venvs_ts_file}"
  fi
}

common_build_and_checks "$@"
```

3.10 build_and_checks_dependencies/common_options.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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get_where_was_i(){
  # Auxiliary function needed below.
  # See usages below if you want to reuse it.
  # $1=offset
  LFBFL_where_was_i="$0"
  if [[ "${BASH_SOURCE[$1]}" != "" && "${BASH_SOURCE[$1]}" != "$0" ]]; then
    LFBFL_where_was_i+=" ${BASH_SOURCE[$1]}"
  fi
  if [[ "${FUNCNAME[$1]}" != "" ]]; then
    LFBFL_where_was_i+=" ${FUNCNAME[$1]}()"
  fi
  readonly LFBFL_where_was_i
  # Comment out line above in case someone would use distinct
  # offsets in the same function.
}

# RETURN traps handling
# Add the following lines at the start of a function.
#   declare -a LFBFL_return_traps_stack
#   local LFBFL_previous_return_trap
#   init_return_trap

init_return_trap(){
  local LFBFL_previous_return_trap2
  LFBFL_previous_return_trap2=$(trap -p RETURN)
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    get_where_was_i 2
    printf "%s previous_return_trap is: %s.\n"\
      "${LFBFL_where_was_i}"\
      "${LFBFL_previous_return_trap2}"
  fi
  if [[ -z "${LFBFL_previous_return_trap2}" ]]; then
    LFBFL_previous_return_trap2="trap - RETURN"
  fi
```

```
  declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
  LFBFL_previous_return_trap="${LFBFL_function_depth}:"
  LFBFL_previous_return_trap+="${LFBFL_previous_return_trap2}"
  trap 'execute_return_traps' RETURN
}

execute_return_traps(){
  if [[ "${FUNCNAME[1]}" == "init_return_trap" ]]; then
    return
  fi
  # ----------------------------------------------------------------------
  # See https://lists.gnu.org/archive/html/help-bash/2026-03/msg00052.html
  if [[ "${FUNCNAME[1]}" == "source" ]]; then
    # Not Working
    return
  fi
  if [[ ${LFBFL_i_no_return_traps:-0} -eq 1 ]]; then
    return
  fi
  # ----------------------------------------------------------------------
  local LFBFL_some_return_trap
  local LFBFL_where_was_i
  get_where_was_i 2
  declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
  while [[ -n "${LFBFL_return_traps_stack}" ]]; do
    LFBFL_some_return_trap="${LFBFL_return_traps_stack[-1]}"
    if [[ "${LFBFL_some_return_trap}" != ${LFBFL_function_depth}:* ]]; then
      # We only treat the top of the stack that should correspond to the
      # current function.
      break
    fi
    LFBFL_some_return_trap=${LFBFL_some_return_trap#*:}
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "%s executing some return trap: %s.\n"\
        "${LFBFL_where_was_i}"\
        "${LFBFL_some_return_trap}"
    fi
    eval "${LFBFL_some_return_trap}"
    unset 'LFBFL_return_traps_stack[-1]'
  done
  if [[ "${LFBFL_previous_return_trap}" != ${LFBFL_function_depth}:* ]];
  then
    # We only set the previous return trap if such a trap was replaced
    # by an init_return_trap.
    return
  fi
```

```
  LFBFL_previous_return_trap=${LFBFL_previous_return_trap#*:}
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    printf "%s setting back to previous_return_trap: %s.\n"\
      "${LFBFL_where_was_i}"\
      "${LFBFL_previous_return_trap}"
  fi
  eval "${LFBFL_previous_return_trap}"
}


is_top_dirstack_directory(){
  # This function should not be needed since when testing "${DIRSTACK[0]}"
  # is already a realpath. But I keep it, in case it is needed.
  # https://lists.gnu.org/archive/html/help-bash/2026-03/msg00005.html
  # $1=to_directory needs to be "realpathed" already
  if [[ -z "$1" ]]; then
    return 0
  fi
  local LFBFL_real_path
  LFBFL_real_path=$(realpath "${DIRSTACK[0]}")
  readonly LFBFL_real_path
  [[ "${LFBFL_real_path}" == "$1" ]]
}

enhanced_pushd(){
  # Auxiliary function needed below.
  # See usages below if you want to reuse it.
  # $1=to_directory must be sanitized with realpath before call
  # $2=offset_for_where_was_i
  # $3=error_message_intermediate_complement
  # returns an error code whenever no pushd happened
  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "enhanced_pushd requested: %s\n" "$1"
  declare -gi i_enhanced_pushd_result=0
  if [[ -z "$1" ]]; then
    i_enhanced_pushd_result=110
    # shellcheck disable=SC2248
    return ${i_enhanced_pushd_result}
  fi
  [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_pushd validated1\n"
  is_top_dirstack_directory "$1" && {
    i_enhanced_pushd_result=111
    # shellcheck disable=SC2248
    return ${i_enhanced_pushd_result}
  }
  [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_pushd validated2\n"
```

```

  pushd "$1" || {
    i_enhanced_pushd_result=$?
    local LFBFL_where_was_i
    get_where_was_i "$2"
    printf "%s %s%s no such directory.\n" "${LFBFL_where_was_i}" "$3" "$1"
    # shellcheck disable=SC2248
    return ${i_enhanced_pushd_result}
  }
  [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_pushd executed\n"
  # shellcheck disable=SC2248
  return ${i_enhanced_pushd_result}
}

enhanced_popd(){
  # Auxiliary function needed below.
  # See usages below if you want to reuse it.
  # $1=to_directory
  # $2=offset_for_where_was_i
  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "enhanced_popd requested: %s\n" "$1"
  if [[ -z "$1" ]]; then
    return 0
  fi

  declare -i LFBFL_i_popd_result
  popd || {
    LFBFL_i_popd_result=$?
    readonly LFBFL_i_popd_result
    local LFBFL_where_was_i
    get_where_was_i "$2"
    printf "%s popd failed.\n" "${LFBFL_where_was_i}"
    # shellcheck disable=SC2248
    return ${LFBFL_i_popd_result}
  }
  [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_popd executed\n"
  return 0
}

can_continue_after_enhanced_pushd(){
  [[ i_enhanced_pushd_result -eq 0
  || i_enhanced_pushd_result -eq 110
  || i_enhanced_pushd_result -eq 111
  ]]
}

```

```
# Main usage for the next functions relative to verbose mode:
# Add the two following lines at the start of a function.
#   declare -i LFBFL_i_verbose=0
#   get_verbose_option "$@"
# /!\ You really need all of the two lines. /!\
# Otherwise, the call to get_verbose_option may impact a function above;
# in case you didn't copy all of "$@" in nested function calls...
# /!\ The same warning applies for other functions below. /!\
# /!\ I will not repeat it below. /!\
# Use get_verbose_options_array, if needed.

get_verbose_options_array(){
  # Use this constant with code like:
  # "${LFBFL_VERBOSE_OPTIONS[LFBFL_i_verbose]}"
  # It could be extended if there is more than one level of verboseness.
  if [[ -n "${LFBFL_VERBOSE_OPTIONS}" ]]; then
    return
  fi

  declare -agr LFBFL_VERBOSE_OPTIONS=(
    ""
    "--verbose"
  )
}

get_verbose_option(){
  # This command is to be called in another one with same arguments.
  # Options:
  #   --verbose
  # This test is faster than:
  #   for LFBFL_arg in "$@"; do
  #     if [[ "${LFBFL_arg}" == "--verbose" ]]; then
  # But it is not bug free.
  if [[ "$*" == *--verbose* ]]; then
    local LFBFL_where_was_i
    get_where_was_i 2
    printf "%s %s\n" "${LFBFL_where_was_i}" "$*"
    LFBFL_i_verbose=1
  fi
  # shellcheck disable=SC2034
  readonly LFBFL_i_verbose
}


get_some_option(){
  # $1=variable_name
```

```
  # $2=some_option_long_name
  # $3=option_default_value
  # $4=read_only 0 or 1 to make the variable readonly
  # This command is to be called in another one with same arguments on top
  # of the three previous.
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == "$2"=* ]]; then
      printf -v "$1" "%s" "${LFBFL_arg#"$2"=}"
      printf -v "$1" "%s" "${!1:-$3}"
      break
    fi
  done
  if [[ $4 -eq 1 ]]; then
    readonly "$1"
  fi
}

keep_options(){
  # $1=variable_name of array of options
  # Other arguments: the options to keep.
  declare -a LFBFL_array_of_filtered_options
  declare -n LFBFL_array_of_options=$1
  local LFBFL_some_argument
  local LFBFL_some_argument2
  for LFBFL_some_argument in "${LFBFL_array_of_options[@]}"; do
    for LFBFL_some_argument2 in "$@"; do
      if [[
        "${LFBFL_some_argument}" == "${LFBFL_some_argument2}"
        || "${LFBFL_some_argument}" == "${LFBFL_some_argument2}"=*
      ]]; then
        LFBFL_array_of_filtered_options+=("${LFBFL_some_argument}")
      fi
    done
  done
  LFBFL_array_of_options=("${LFBFL_array_of_filtered_options[@]}")
}


# Main usage for the next functions relative to work_directory:
# Add the five following lines at the start of a function.
#   local LFBFL_work_directory=""
#   get_work_directory_option "$@"
#   pushd_to_work_directory\
#     && trap 'popd_from_work_directory' RETURN
#   can_continue_after_enhanced_pushd || return 1
```

```
290 # Or:
291 #   local LFBFL_work_directory=""
292 #   get_work_directory_option "$@"
293 #   pushd_to_work_directory --trap-popd
294 #   can_continue_after_enhanced_pushd || return 1
295
296 get_work_directory_option(){
297   # This command is to be called in another one with same arguments.
298   # Options:
299   #   --work-directory=""
300   get_some_option LFBFL_work_directory --work-directory "." 0 "$@"
301   # LFBFL_work_directory=${LFBFL_work_directory/#~/${HOME}} line below
302   if [[ -n "${LFBFL_work_directory}" ]]; then
303     LFBFL_work_directory=$(realpath "${LFBFL_work_directory}")
304   fi
305   readonly LFBFL_work_directory
306 }
307
308 pushd_to_work_directory(){
309   # Options:
310   #   --trap-popd
311   enhanced_pushd "${LFBFL_work_directory}" 3 " --work-directory="
312   if [[ i_enhanced_pushd_result -eq 0 ]]; then
313     local LFBFL_arg
314     for LFBFL_arg in "$@"; do
315       if [[ "${LFBFL_arg}" == --trap-popd ]]; then
316         declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
317         local LFBFL_some_return_trap
318         LFBFL_some_return_trap="${LFBFL_function_depth}:"
319         LFBFL_some_return_trap+="popd_from_work_directory 4"
320         readonly LFBFL_some_return_trap
321         LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
322         break
323       fi
324     done
325   fi
326   # shellcheck disable=SC2248
327   return ${i_enhanced_pushd_result}
328 }
329
330 popd_from_work_directory(){
331   # $1=offset for where_was_i
332   enhanced_popd "${LFBFL_work_directory}" "${1:-3}"
333 }
334
335 work_directory_is_top_dirstack_directory(){
```

```
  is_top_dirstack_directory "${LFBFL_work_directory}"
}


get_offset_option(){
  # This command is to be called in another one with same arguments on top
  # of the default value.
  # $1=option_default_value
  # Options:
  #   --offset=2
  get_some_option LFBFL_i_offset --offset "$1" 1 "$@"
}


# Functions relative to shell and bash options
# (See SHELLOPTS and BASHOPTS, set and shopt):
# - shell options usage, example with pipefail:
#     enhanced_set_shell_option pipefail\
#       && trap 'enhanced_unset_shell_option pipefail' RETURN
#   or maybe:
#     enhanced_unset_shell_option pipefail\
#       && trap 'enhanced_set_shell_option pipefail' RETURN
#   if you need to unset it in some function.
#   or better if you use my RETURN traps handler:
#     enhanced_set_shell_option pipefail --trap-unset
#     enhanced_unset_shell_option pipefail --trap-set
# - bash options usage, example with globstar:
#     enhanced_set_bash_option globstar\
#       && trap 'enhanced_unset_bash_option globstar' RETURN
#   or maybe:
#     enhanced_unset_bash_option globstar\
#       && trap 'enhanced_set_bash_option globstar' RETURN
#   or better if you use my RETURN traps handler:
#     enhanced_set_bash_option globstar --trap-unset
#     enhanced_unset_bash_option globstar --trap-set

enhanced_set_shell_option(){
  # $1=optname some shell option for set
  # Options:
  #   --offset=2
  #   --trap-unset
  declare -r LFBFL_result_name="i_enhanced_set_shell_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  if [[ -o "$1" ]]; then
    printf -v "${LFBFL_result_name}" "%d" "1"
    # shellcheck disable=SC2086
```

```
    return ${!LFBFL_result_name}
  fi
  set -o "$1"
  declare -i LFBFL_i_offset=2
  get_offset_option 2 "$@"
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    # shellcheck disable=SC2248
    get_where_was_i ${LFBFL_i_offset}
    printf "%s %s shell option activated.\n" "${LFBFL_where_was_i}" "$1"
  fi
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --trap-unset ]]; then
      declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
      local LFBFL_some_return_trap
      LFBFL_some_return_trap="${LFBFL_function_depth}:"
      LFBFL_some_return_trap+="enhanced_unset_shell_option '$1' --offset=3"
      readonly LFBFL_some_return_trap
      LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
      break
    fi
  done
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
}

enhanced_unset_shell_option(){
  # $1=optname some shell option for set
  # Options:
  #   --offset=2
  #   --trap-set
  declare -r LFBFL_result_name="i_enhanced_unset_shell_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  if [[ ! -o "$1" ]]; then
    printf -v "${LFBFL_result_name}" "%d" "1"
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
  set +o "$1"
  declare -i LFBFL_i_offset=2
  get_offset_option 2 "$@"
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    # shellcheck disable=SC2248
    get_where_was_i ${LFBFL_i_offset}
```

```
    printf "%s %s shell option unactivated.\n" "${LFBFL_where_was_i}" "$1"
  fi
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --trap-set ]]; then
      declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
      local LFBFL_some_return_trap
      LFBFL_some_return_trap="${LFBFL_function_depth}:"
      LFBFL_some_return_trap+="enhanced_set_shell_option '$1' --offset=3"
      readonly LFBFL_some_return_trap
      LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
      break
    fi
  done
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
}

enhanced_set_bash_option(){
  # $1=optname some bash option for shopt
  # Options:
  #   --offset=2
  #   --trap-unset
  declare -r LFBFL_result_name="i_enhanced_set_bash_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  # declare -r LFBFL_regexp="^(.*:)?$1(:.*)?$" funny but brainfucked
  # if [[ "${BASHOPTS}" =~ ${LFBFL_regexp} ]]; then
  if shopt -q "$1"; then
    printf -v "${LFBFL_result_name}" "%d" "1"
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
  shopt -s "$1"
  declare -i LFBFL_i_offset=2
  get_offset_option 2 "$@"
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    # shellcheck disable=SC2248
    get_where_was_i ${LFBFL_i_offset}
    printf "%s %s bash option activated.\n" "${LFBFL_where_was_i}" "$1"
  fi
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --trap-unset ]]; then
      declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
      local LFBFL_some_return_trap
```

```
      LFBFL_some_return_trap="${LFBFL_function_depth}:"
      LFBFL_some_return_trap+="enhanced_unset_bash_option '$1' --offset=3"
      readonly LFBFL_some_return_trap
      LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
      break
    fi
  done
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
}

enhanced_unset_bash_option(){
  # $1=optname some bash option for shopt
  # Options:
  #   --offset=2
  #   --trap-set
  declare -r LFBFL_result_name="i_enhanced_unset_bash_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  if ! shopt -q "$1"; then
    printf -v "${LFBFL_result_name}" "%d" "1"
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
  shopt -u "$1"
  declare -i LFBFL_i_offset=2
  get_offset_option 2 "$@"
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    # shellcheck disable=SC2248
    get_where_was_i ${LFBFL_i_offset}
    printf "%s %s bash option unactivated.\n" "${LFBFL_where_was_i}" "$1"
  fi
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --trap-set ]]; then
      declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
      local LFBFL_some_return_trap
      LFBFL_some_return_trap="${LFBFL_function_depth}:"
      LFBFL_some_return_trap+="enhanced_set_bash_option '$1' --offset=3"
      readonly LFBFL_some_return_trap
      LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
      break
    fi
  done
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
```

```
}
```

3.11 build_and_checks_dependencies/comparisons.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# of the GNU General Public License
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# the GNU General Public License
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# The file "comparisons.sh" was renamed to "comparisons.libr.sh".

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

equal(){
  # Returns 1 if all the arguments strings are equal.
  # Renvoie 1 dans $? si toutes les chaînes en arguments sont égales.
  # J'ai rerepensé au 3-way-merge
  # et à mon article sur le principe de première différence
  # appliqué à la largeur modulaire, de clique ou de rang avec
  # l'idée d'un principe de première différence ternaire.
  declare -i LFBFL_i_first=1
  local LFBFL_current
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    if [[ LFBFL_i_first -eq 1 ]]; then
      LFBFL_current="${LFBFL_arg}"
      LFBFL_i_first=0
```

```
43        continue
44      fi
45      if [[ "${LFBFL_current}" != "${LFBFL_arg}" ]]; then
46        return 0
47      fi
48    done
49    return 1
50  }
51
52  some_distinct(){
53    ! equal "$@"
54  }
55
56  all_distinct(){
57    # Returns 1 if all the arguments strings are distinct.
58    declare -ir LFBFL_ij_max=$#
59    declare -i LFBFL_i
60    declare -i LFBFL_j
61    for ((LFBFL_i=1; LFBFL_i<LFBFL_ij_max; ++LFBFL_i)); do
62      for ((LFBFL_j = LFBFL_i + 1; LFBFL_j<=LFBFL_ij_max; ++LFBFL_j)); do
63        [[ "${!LFBFL_i}" == "${!LFBFL_j}" ]] && return 0
64      done
65    done
66    return 1
67  }
68
69  all_distinct2(){
70    # Returns 1 if all the arguments strings are distinct.
71    local LFBFL_element
72    local LFBFL_previous_element
73    if [[ $# -le 1 ]]; then
74      return 1
75    fi
76    declare -r LFBFL_s_sorted_arguments=$(
77      printf "%s\n" "$@" \
78      | sort
79    )
80    declare -a LFBFL_arr_sorted_arguments
81    mapfile -t LFBFL_arr_sorted_arguments <<< "${LFBFL_s_sorted_arguments}"
82    declare -i LFBFL_i
83    declare -ir LFBFL_i_max=${#LFBFL_arr_sorted_arguments[@]}
84    LFBFL_previous_element=${LFBFL_arr_sorted_arguments[0]}
85    for ((LFBFL_i=1; LFBFL_i<LFBFL_i_max; ++LFBFL_i)); do
86      LFBFL_element="${LFBFL_arr_sorted_arguments[${LFBFL_i}]}"
87      if [[ "${LFBFL_previous_element}" == "${LFBFL_element}" ]]; then
88        return 0
```

```
    fi
    LFBFL_previous_element="${LFBFL_element}"
  done
  return 1
}

all_distinct3(){
  # Returns 1 if all the arguments strings are distinct.
  # all_distinct3 is slower than old all_distinct2 since a call to sort is
  # needed:
  # the manual of uniq says it doesn't detect non-adjacent repeated lines.
  # A few weeks ago, when testing without sort, I had no bug...
  # I may have tested all_distinct3 a a b and not all_distinct3 a b a
  # There is a sort also in all_distinct2, but uniq could have an option to
  # sort before "applying -d", or sort could have an option for printing
  # duplicates only. It would complement -u option of sort.
  # Both solutions would avoid one pipe and one more shell command call.
  # Note that old all_distinct2 was faster with pipes and read -r commands,
  # but read -r has small drawbacks. Using mapfile is a cleaner way to do
  # it, since it handles arguments with spaces from scratch.
  # Arguments with line-returns are still not handled.
  # Use C, PHP or Python for example for that.
  # Somehow Bash was made to push people to stop being lazy and do actual C
  # code, even if it needs compiling something, adding a makefile, etc. XD
  # Note that C code would be faster with all_distinct2 blueprint,
  # that's the reason I kept all 3 versions, see the (asymptotic)
  # complexity and simple algorithms ideas on this toy problem.
  if [[ $# -le 1 ]]; then
    return 1
  fi
  declare -r LFBFL_s_duplicated_arguments=$(
    printf "%s\n" "$@" \
    | sort\
    | uniq -d
  )
  [[ -n "${LFBFL_s_duplicated_arguments}" ]]
}

some_equal(){
  ! all_distinct "$@"
}

some_equal2(){
  ! all_distinct2 "$@"
}

```

```
some_equal3(){
  ! all_distinct3 "$@"
}

max(){
  # Returns the maximum string among the arguments.
  # Renvoie la chaîne la plus grande parmi celles proposées.
  # $1=sort_command with the flags you want to be applied.
  # $ max sort a b c
  # c
  if [[ $# -eq 0 ]]; then
    return 1
  fi
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  printf "%s\n" "${@:2}" \
    | eval "$1 -r"\
    | head -1
}

min(){
  # Returns the minimum string among the arguments.
  # Renvoie la chaîne la plus petite parmi celles proposées.
  # $1=sort_command with the flags you want to be applied.
  # $ min sort a b c
  # a
  if [[ $# -eq 0 ]]; then
    return 1
  fi
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  printf "%s\n" "${@:2}" \
    | eval "$1"\
    | head -1
}

is_substring(){
  # $1=string
  # $2=substring
  declare -r LFBFL_var_1=$(
    printf "%s" "$1"\
    | sed -e 's/\\/\\&/g' -e "s/'/\\\&/g"
  )
  declare -r LFBFL_var_2=$(
    printf "%s" "$2"\
    | sed -e 's/\\/\\&/g' -e "s/'/\\\&/g"
```

```
  )
  local LFBFL_var_3="\$a = '${LFBFL_var_1}';"
  LFBFL_var_3+=" \$b = preg_quote('${LFBFL_var_2}');"
  LFBFL_var_3+=" \$c = '/'.addcslashes(\$b, '/').'/';"
  LFBFL_var_3+=" die(1 - preg_match(\$c, \$a));"
  readonly LFBFL_var_3
  php -r "${LFBFL_var_3}"
  return $?
}

is_subfile(){
  # $1=file_path
  # $2=sub_file_path
  local LFBFL_var_1="\$a = file_get_contents('$1');"
  LFBFL_var_1+=" \$b = preg_quote(file_get_contents('$2'));"
  LFBFL_var_1+=" \$c = '/'.addcslashes(\$b, '/').'/';"
  LFBFL_var_1+=" die(1 - preg_match(\$c, \$a));"
  readonly LFBFL_var_1
  php -r "${LFBFL_var_1}"
  return $?
}

alias min_from_int_sort="min 'sort --numeric-sort'"
alias max_from_int_sort="max 'sort --numeric-sort'"
```

3.12 build_and_checks_dependencies/create_PDF.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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```

```
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#
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# This file was renamed from "create_PDF.sh" to "create_PDF.exec.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=generate_from_template.libr.sh
source "./${LFBFL_subdir}/generate_from_template.libr.sh"
# shellcheck source=lines_counts.libr.sh
source "./${LFBFL_subdir}/lines_counts.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=lines_maps.libr.sh
source "./${LFBFL_subdir}/lines_maps.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
# shellcheck source=strings_functions.libr.sh
source "./${LFBFL_subdir}/strings_functions.libr.sh"

create_PDF(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  declare -a LFBFL_common_options=("$@")
  keep_options LFBFL_common_options --verbose

  local LFBFL_subdir2="build_and_checks_variables"
  local LFBFL_data_file_name="${LFBFL_subdir2}/repository_data.txt"
  local LFBFL_repository_name=""
  grep_variable "${LFBFL_data_file_name}" repository_name\
    --result-variable-prefix="LFBFL_"\
```

```
66      --replace-line-returns-by=""
67
68    local LFBFL_repository_web_url=""
69    grep_variable "${LFBFL_data_file_name}" repository_web_url\
70      --result-variable-prefix="LFBFL_"\
71      --replace-line-returns-by=""
72
73    local LFBFL_repository_git_url=""
74    grep_variable "${LFBFL_data_file_name}" repository_git_url\
75      --result-variable-prefix="LFBFL_"\
76      --replace-line-returns-by=""
77
78    local LFBFL_abstract=""
79    grep_variable "${LFBFL_data_file_name}" abstract\
80      --result-variable-prefix="LFBFL_"
81
82    local LFBFL_acknowledgments=""
83    grep_variable "${LFBFL_data_file_name}" acknowledgments\
84      --result-variable-prefix="LFBFL_"
85
86    local LFBFL_author_email=""
87    grep_variable "${LFBFL_data_file_name}" author_email\
88      --result-variable-prefix="LFBFL_"\
89      --replace-line-returns-by=""
90
91    local LFBFL_author_full_name=""
92    grep_variable "${LFBFL_data_file_name}" author_full_name\
93      --result-variable-prefix="LFBFL_"\
94      --replace-line-returns-by=" "
95
96    local LFBFL_author_website=""
97    grep_variable "${LFBFL_data_file_name}" author_website\
98      --result-variable-prefix="LFBFL_"\
99      --replace-line-returns-by=""
100
101   declare -i LFBFL_i_max_line_length
102   grep_variable "${LFBFL_data_file_name}" max_line_length\
103     --result-variable-prefix="LFBFL_i_"
104
105   declare -r LFBFL_current_date=$(date -I"date")
106
107   declare -r LFBFL_current_git_SHA1=$(git rev-parse HEAD)
108
109   declare -ir LFBFL_i_number_of_commits=$(
110     git shortlog\
111     | space_starting_lines\
```

```
112       | wc -l
113     )
114
115     local LFBFL_redirect="/dev/stdout"
116     if [[ LFBFL_i_verbose -eq 0 ]]; then
117       LFBFL_redirect="/dev/null"
118     fi
119     {
120       code_lines_count_all "${LFBFL_common_options[@]}"
121       code_lines_count_not_empty "${LFBFL_common_options[@]}"
122       code_lines_count_empty "${LFBFL_common_options[@]}"
123     } &> "${LFBFL_redirect}"
124
125     local LFBFL_number_of_lines="${i_code_lines_count_all_result}"
126     LFBFL_number_of_lines+=" total lines,"
127     LFBFL_number_of_lines+=" ${i_code_lines_count_not_empty_result}"
128     LFBFL_number_of_lines+=" not empty lines,"
129     LFBFL_number_of_lines+=" ${i_code_lines_count_empty_result}"
130     LFBFL_number_of_lines+=" empty lines."
131     readonly LFBFL_number_of_lines
132
133     declare -r LFBFL_temp_path="${LFBFL_subdir2}/temp"
134
135     # see https://gitlab.com/OldManProgrammer/unix-tree/-/issues
136     # ?show=eyJpaWQiOiI0MyIsImZ1bGxfcGF0aCI6Ik9sZE1hblByb2dyYW1t
137     # ZXIvdW5peC10cmVlIiwiaWQiOjE4NTE0NTgzOH0%3D
138     tree -a --gitignore\
139       -I "node_modules/"\
140       -I "__pycache__/"\
141       -I ".mypy_cache/"\
142       -I ".ruff_cache/"\
143       -I ".git/"\
144       | replace_non_ascii_spaces\
145       > "${LFBFL_temp_path}/current_tree_light.txt"
146
147     tree -a -DFh --gitignore\
148       -I "node_modules/"\
149       -I "__pycache__/"\
150       -I ".mypy_cache/"\
151       -I ".ruff_cache/"\
152       -I ".git/"\
153       | replace_non_ascii_spaces\
154       > "${LFBFL_temp_path}/current_tree.txt.temp"
155
156     local LFBFL_s_files_paths
157     # Remove line returns that are here to keep lines short.
```

```
158   LFBFL_s_files_paths=$(
159     sed -Ez 's/(.)\\\n/\1/Mg' "${LFBFL_subdir2}/files_names_listing.txt"\
160     | grep -v '^//'
161   )
162   readonly LFBFL_s_files_paths
163   declare -a LFBFL_arr_files_paths
164   if [[ -n "${LFBFL_s_files_paths}" ]]; then
165     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
166   fi
167   readonly LFBFL_arr_files_paths
168
169   local LFBFL_temp_files_listing="${LFBFL_temp_path}/"
170   LFBFL_temp_files_listing+="files_listing.tex.sub.temp"
171   readonly LFBFL_temp_files_listing
172   : > "${LFBFL_temp_files_listing}"
173   local LFBFL_temp_files_listing2="${LFBFL_temp_path}/"
174   LFBFL_temp_files_listing2+="files_listing.html.sub.temp"
175   readonly LFBFL_temp_files_listing2
176   : > "${LFBFL_temp_files_listing2}"
177   # HTML <li> elements, hence "lis".
178   local LFBFL_temp_files_lis
179   LFBFL_temp_files_lis="${LFBFL_temp_path}/files_lis.html.sub.temp"
180   readonly LFBFL_temp_files_lis
181   : > "${LFBFL_temp_files_lis}"
182   # shellcheck disable=SC2248
183   get_split_score_simple 1 ${LFBFL_i_max_line_length} /
184   declare -r LFBFL_score_command="${get_split_score_result}"
185   declare -i LFBFL_i_score_command_properties
186   LFBFL_i_score_command_properties=${i_get_split_score_result2}
187   readonly LFBFL_i_score_command_properties
188   # shellcheck disable=SC2248
189   get_split_score_simple 1 ${LFBFL_i_max_line_length} ':'
190   declare -r LFBFL_score_command2="${get_split_score_result}"
191   declare -i LFBFL_i_score_command_properties2
192   LFBFL_i_score_command_properties2=${i_get_split_score_result2}
193   readonly LFBFL_i_score_command_properties2
194   declare -r LFBFL_suffix='%'
195   declare -i LFBFL_i=0
196   local LFBFL_cleaned_path1
197   local LFBFL_cleaned_path2
198   local LFBFL_new_lines
199   local LFBFL_new_lines2
200   local LFBFL_new_lines3
201   local LFBFL_file_path
202   local LFBFL_s_format
203   for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
```

```
    printf "Listing file for tex/HTML : %s\n" "${LFBFL_file_path}"
    # printf "Listing file for tex : %s\n" "${LFBFL_file_path}"
    # LFBFL_base_file_name=$(basename "${LFBFL_file_path}")
    LFBFL_cleaned_path1="${LFBFL_file_path//_/\\_}"
    LFBFL_cleaned_path2=$(
      printf "%s" "${LFBFL_file_path}"\
      | sed -e 's/\//:/g' -e 's/\.//g'
    )
    LFBFL_new_lines="  ${LFBFL_cleaned_path1}"
    if [[ ${#LFBFL_new_lines} -gt LFBFL_i_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines}"\
        ""\
        ${LFBFL_i_max_line_length}\
        "${LFBFL_suffix}"\
        "${LFBFL_score_command}"\
        ${LFBFL_i_score_command_properties}\
        "\\"
      LFBFL_new_lines=${repeated_split_last_line_result}
    fi
    LFBFL_new_lines2="  ${LFBFL_cleaned_path2}"
    if [[ ${#LFBFL_new_lines2} -gt LFBFL_i_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines2}"\
        ""\
        ${LFBFL_i_max_line_length}\
        "${LFBFL_suffix}"\
        "${LFBFL_score_command2}"\
        ${LFBFL_i_score_command_properties2}
      LFBFL_new_lines2=${repeated_split_last_line_result}
    fi
    LFBFL_new_lines3="${LFBFL_file_path}"
    if [[ ${#LFBFL_new_lines3} -gt LFBFL_i_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines3}"\
        ""\
        ${LFBFL_i_max_line_length}\
        "${LFBFL_suffix}"\
        "${LFBFL_score_command}"\
        ${LFBFL_i_score_command_properties}
      LFBFL_new_lines3=${repeated_split_last_line_result}
    fi
    LFBFL_s_format="\\subsection{\n%s\n}\n\\label{\n%s\n}\n\n"
    LFBFL_s_format+="\\VerbatimInput[numbers=left,xleftmargin=-5mm]"
    LFBFL_s_format+="{\n%s\n}\n\n\n"
    # shellcheck disable=SC2059
```

```
    printf "${LFBFL_s_format}"\
      "${LFBFL_new_lines}"\
      "${LFBFL_new_lines2}"\
      "${LFBFL_new_lines3}"\
      >> "${LFBFL_temp_files_listing}"

    # printf "Listing file for HTML : %s\n" "${LFBFL_file_path}"
    ((++LFBFL_i))
    LFBFL_new_lines="${LFBFL_file_path}"
    if [[ ${#LFBFL_file_path} -gt LFBFL_i_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines}"\
        "-->"\
        ${LFBFL_i_max_line_length}\
        "<!--"\
        "${LFBFL_score_command}"\
        ${LFBFL_i_score_command_properties}
      LFBFL_new_lines=${repeated_split_last_line_result}
    fi
    {
      LFBFL_s_format="<h3 id=\"subsection3.%s\">3.%s\n%s\n</h3>\n"
      LFBFL_s_format+="<pre class=\"numbered_lines\">\n"
      # shellcheck disable=SC2059
      printf "${LFBFL_s_format}"\
        "${LFBFL_i}"\
        "${LFBFL_i}"\
        "${LFBFL_new_lines}"
      sed -e 's/&/\&amp;/g' -e 's/</\&lt;/g'\
        < "${LFBFL_file_path}"
      printf "</pre>\n\n\n"
    } >> "${LFBFL_temp_files_listing2}"
    LFBFL_s_format="      <li><a href=\"#subsection3.%s\">\n%s\n"
    LFBFL_s_format+="      </a></li>\n"
    # shellcheck disable=SC2059
    printf "${LFBFL_s_format}" "${LFBFL_i}" "${LFBFL_new_lines}"\
      >> "${LFBFL_temp_files_lis}"
  done
  overwrite_if_not_equal "${LFBFL_temp_path}/files_listing.tex.sub"\
    "${LFBFL_temp_files_listing}"
  # Example of generated content of files_listing.tex.sub:
  # ----------------------------------------------------------------------
  # \subsection{
  #   build\_and\_checks\_dependencies/licenses\_templates/%
  # build\_licenses\_templates.exec.sh
  # }
  # \label{
```

```
#   build_and_checks_dependencies:licenses_templates:%
# build_licenses_templatesexecsh
# }
#
# \VerbatimInput[numbers=left,xleftmargin=-5mm]{
# build_and_checks_dependencies/licenses_templates/%
# build_licenses_templates.exec.sh
# }
# ----------------------------------------------------------------------
sed -i -e "s|\\n</pre>|</pre>|g" "${LFBFL_temp_files_listing2}"
overwrite_if_not_equal "${LFBFL_temp_path}/files_listing.html.sub"\
  "${LFBFL_temp_files_listing2}"
# Example of generated content of files_listing.html.sub:
# ----------------------------------------------------------------------
# <h3 id="subsection3.15">3.15
# build_and_checks_dependencies/licenses_templates/<!--
# -->build_licenses_templates.exec.sh
# </h3>
# ...
# <h3 id="subsection3.33">3.33
# build_and_checks_variables/.gitignore
# </h3>
# <pre class="numbered_lines">
# temp/
# upgrade_venvs_ts
# </pre>
# ----------------------------------------------------------------------
overwrite_if_not_equal "${LFBFL_temp_path}/files_lis.html.sub"\
  "${LFBFL_temp_files_lis}"
# Example of generated content of files_lis.html.sub:
# ----------------------------------------------------------------------
#       <li><a href="#subsection3.15">
# build_and_checks_dependencies/licenses_templates/<!--
# -->build_licenses_templates.exec.sh
#       </a></li>
# ----------------------------------------------------------------------

# We verify if some lines are beyond max_line_length characters
# in current_tree_light.txt and current_tree.txt.
declare -ar LFBFL_trees=(
  "current_tree_light.txt"
  "current_tree.txt.temp"
)
local LFBFL_tree_path
local LFBFL_line
local LFBFL_prefix
```

```
342   declare -i LFBFL_i_prefix_last_position
343   local LFBFL_char
344   local LFBFL_line_start
345   declare -ir LFBFL_i_overlength=$((LFBFL_i_max_line_length+1))
346   local LFBFL_file_name
347   local LFBFL_s_lines
348   declare -a LFBFL_arr_lines
349   declare -i LFBFL_i_keep_length
350   for LFBFL_tree in "${LFBFL_trees[@]}"; do
351     LFBFL_tree_path="${LFBFL_temp_path}/${LFBFL_tree}"
352     # shellcheck disable=SC2248
353     LFBFL_s_lines=$(
354       grep '.\{'${LFBFL_i_overlength}'\}' "${LFBFL_tree_path}"
355     )
356     if [[ -z "${LFBFL_s_lines}" ]]; then
357       continue
358     fi
359     mapfile -t LFBFL_arr_lines <<< "${LFBFL_s_lines}"
360     for LFBFL_line in "${LFBFL_arr_lines[@]}"; do
361       # printf "LFBFL_line: %s\n" "${LFBFL_line}"
362       LFBFL_prefix=$(
363         printf "%s" "${LFBFL_line}"\
364         | sed -E -e 's/(.*)——[^—]+$/\1/g'
365       )
366       # printf "LFBFL_prefix: %s\n" "${LFBFL_prefix}"
367       LFBFL_i_prefix_last_position=$((${#LFBFL_prefix}-1))
368       LFBFL_char="${LFBFL_prefix:${LFBFL_i_prefix_last_position}:1}"
369       # printf "LFBFL_char: %s\n" "${LFBFL_char}"
370       LFBFL_prefix=$(
371         printf "%s" "${LFBFL_prefix}"\
372         | sed -E -e 's/[^ ]+$//g'
373       )
374       if [[ "${LFBFL_char}" == "└" ]]; then
375         LFBFL_prefix+="  "
376       else
377         LFBFL_prefix+="│ "
378       fi
379       # printf "LFBFL_prefix: %s\n" "${LFBFL_prefix}"
380       LFBFL_file_name=$(
381         printf "%s" "${LFBFL_line}"\
382         | sed -E 's|.* ([^ ]+)$|\1|g'
383       )
384       # printf "LFBFL_file_name: %s\n" "${LFBFL_file_name}"
385       LFBFL_i_keep_length=$((${#LFBFL_line} - ${#LFBFL_file_name}))
386       LFBFL_line_start=${LFBFL_line:0:${LFBFL_i_keep_length}}
387       LFBFL_line_start=$(
```

```
        printf "%s" "${LFBFL_line_start}"\
        | sed -e 's/ *$//g'
      )
      # printf "LFBFL_line_start: %s\n" "${LFBFL_line_start}"
      LFBFL_new_lines="${LFBFL_prefix}${LFBFL_file_name}"
      if [[ ${#LFBFL_new_lines} -gt LFBFL_i_max_line_length ]]; then
        # shellcheck disable=SC2248
        repeated_split_last_line "${LFBFL_new_lines}"\
          "${LFBFL_prefix}"\
          ${LFBFL_i_max_line_length}\
          ""\
          ""\
          ""
        LFBFL_new_lines=${repeated_split_last_line_result}
      fi
      printf "%s\n%s\n" "${LFBFL_line_start}" "${LFBFL_new_lines}"\
        > "${LFBFL_tree_path}.some_line.temp"
      insert_file_at_token "${LFBFL_tree_path}"\
        "${LFBFL_line}"\
        "${LFBFL_tree_path}.some_line.temp"\
        ""\
        --quiet
    done
    rm -f "${LFBFL_tree_path}.some_line.temp"
  done

  overwrite_if_not_equal "${LFBFL_temp_path}/current_tree.txt"\
    "${LFBFL_temp_path}/current_tree.txt.temp" 1 1

  declare -r LFBFL_tex_path_start=\
"${LFBFL_temp_path}/${LFBFL_repository_name}.tex"
  declare -r LFBFL_html_path_start=\
"${LFBFL_temp_path}/${LFBFL_repository_name}.html"

  if [[ -f "${LFBFL_subdir2}/${LFBFL_repository_name}.tex.tpl" ]]; then
    # If there is a template, we init the process from it.
    cp "${LFBFL_subdir2}/${LFBFL_repository_name}.tex.tpl"\
      "${LFBFL_tex_path_start}.1"
  elif [[ -f "${LFBFL_subdir2}/${LFBFL_repository_name}.tex" ]]; then
    # Otherwise if there is a tex file, we init the process from it.
    cp "${LFBFL_subdir2}/${LFBFL_repository_name}.tex"\
      "${LFBFL_tex_path_start}.1"
  else
    printf "Neither .tex.tpl, nor .tex in ./%s/\n" "${LFBFL_subdir2}"
  fi

```

```
  # Same logic with repository HTML file.
  if [[ -f "${LFBFL_subdir2}/${LFBFL_repository_name}.html.tpl" ]]; then
    cp "${LFBFL_subdir2}/${LFBFL_repository_name}.html.tpl"\
      "${LFBFL_html_path_start}.1"
  elif [[ -f "${LFBFL_repository_name}.html" ]]; then
    cp "${LFBFL_repository_name}.html" "${LFBFL_html_path_start}.1"
  else
    printf "Neither .html.tpl in ./%s/, nor .html in ./\n"\
      "${LFBFL_subdir2}"
  fi

  # shellcheck disable=SC2001
  printf "%s\n" "${LFBFL_abstract}"\
    | sed -e 's/\\n/\n/g'\
    > "${LFBFL_temp_path}/abstract_temp"

  # shellcheck disable=SC2001
  printf "%s\n" "${LFBFL_acknowledgments}"\
    | sed -e 's/\\n/\n/g'\
    > "${LFBFL_temp_path}/acknowledgments_temp"

  # But the filling still occurs, in case the dev want to refill
  # part of the tex/html file that he modified with @token@
  # using some computed results.
  # Tex filling:
  if [[ -f "${LFBFL_tex_path_start}.1" ]]; then
    sed -i -e "s|@repository_name@|${LFBFL_repository_name}|g"\
           -e "s|@repository_web_url@|${LFBFL_repository_web_url}|g"\
           -e "s|@repository_git_url@|${LFBFL_repository_git_url}|g"\
           -e "s|@author_email@|${LFBFL_author_email}|g"\
           -e "s|@author_full_name@|${LFBFL_author_full_name}|g"\
           -e "s|@author_website@|${LFBFL_author_website}|g"\
           -e "s|@current_date@|${LFBFL_current_date}|g"\
           -e "s|@current_git_SHA1@|${LFBFL_current_git_SHA1}|g"\
           -e "s|@number_of_commits@|${LFBFL_i_number_of_commits}|g"\
           -e "s|@number_of_lines@|${LFBFL_number_of_lines}|g"\
      "${LFBFL_tex_path_start}.1"

    insert_file_at_token "${LFBFL_tex_path_start}.1"\
      @abstract@\
      "${LFBFL_temp_path}/abstract_temp"\
      "${LFBFL_tex_path_start}.2"

    insert_file_at_token "${LFBFL_tex_path_start}.2"\
      @acknowledgments@\
      "${LFBFL_temp_path}/acknowledgments_temp"\
```

```
      "${LFBFL_tex_path_start}.3"

    insert_file_at_token "${LFBFL_tex_path_start}.3"\
      @current_tree_light@\
      "${LFBFL_temp_path}/current_tree_light.txt"\
      "${LFBFL_tex_path_start}.4"

    insert_file_at_token "${LFBFL_tex_path_start}.4"\
      @current_tree@\
      "${LFBFL_temp_path}/current_tree.txt"\
      "${LFBFL_tex_path_start}.5"

    insert_file_at_token "${LFBFL_tex_path_start}.5"\
      @dependencies_notes@\
      "${LFBFL_temp_path}/dependencies_notes.tex.sub"\
      "${LFBFL_tex_path_start}.6"

    insert_file_at_token "${LFBFL_tex_path_start}.6"\
      @files_listing_VerbatimInput@\
      "${LFBFL_temp_path}/files_listing.tex.sub"\
      "${LFBFL_tex_path_start}.7"

    overwrite_if_not_equal\
      "${LFBFL_subdir2}/${LFBFL_repository_name}.tex"\
      "${LFBFL_tex_path_start}.7" 1
  fi

  # HTML filling:
  declare -i LFBFL_i_HTML_updated=0
  if [[ -f "${LFBFL_html_path_start}.1" ]]; then
    sed -i -e "s|@repository_name@|${LFBFL_repository_name}|g"\
           -e "s|@repository_web_url@|${LFBFL_repository_web_url}|g"\
           -e "s|@repository_git_url@|${LFBFL_repository_git_url}|g"\
           -e "s|@author_email@|${LFBFL_author_email}|g"\
           -e "s|@author_full_name@|${LFBFL_author_full_name}|g"\
           -e "s|@author_website@|${LFBFL_author_website}|g"\
           -e "s|@current_date@|${LFBFL_current_date}|g"\
           -e "s|@current_git_SHA1@|${LFBFL_current_git_SHA1}|g"\
           -e "s|@number_of_commits@|${LFBFL_i_number_of_commits}|g"\
           -e "s|@number_of_lines@|${LFBFL_number_of_lines}|g"\
      "${LFBFL_html_path_start}.1"

    insert_file_at_token "${LFBFL_html_path_start}.1"\
      @abstract@\
      "${LFBFL_temp_path}/abstract_temp"\
      "${LFBFL_html_path_start}.2"
```

```

    insert_file_at_token "${LFBFL_html_path_start}.2"\
      @acknowledgments@\
      "${LFBFL_temp_path}/acknowledgments_temp"\
      "${LFBFL_html_path_start}.3"

    insert_file_at_token "${LFBFL_html_path_start}.3"\
      @files_lis@\
      "${LFBFL_temp_path}/files_lis.html.sub"\
      "${LFBFL_html_path_start}.4"

    insert_file_at_token "${LFBFL_html_path_start}.4"\
      @current_tree_light@\
      "${LFBFL_temp_path}/current_tree_light.txt"\
      "${LFBFL_html_path_start}.5"

    insert_file_at_token "${LFBFL_html_path_start}.5"\
      @current_tree@\
      "${LFBFL_temp_path}/current_tree.txt"\
      "${LFBFL_html_path_start}.6"

    insert_file_at_token "${LFBFL_html_path_start}.6"\
      @dependencies_notes@\
      "${LFBFL_temp_path}/dependencies_notes.html.sub"\
      "${LFBFL_html_path_start}.7"

    insert_file_at_token "${LFBFL_html_path_start}.7"\
      @files_listing_HTMLPreInput@\
      "${LFBFL_temp_path}/files_listing.html.sub"\
      "${LFBFL_html_path_start}.8"

    overwrite_if_not_equal "${LFBFL_repository_name}.html"\
      "${LFBFL_html_path_start}.8" 1
    LFBFL_i_HTML_updated=$?
  fi

  # The HTML contains everything including listings with source code.
  # If it doesn't get updated,
  # then it is useless to recompute the PDF.
  if [[ LFBFL_i_HTML_updated -eq 0 ]]; then
    return
  fi

  local LFBFL_latex_command=""
  grep_variable "${LFBFL_data_file_name}" latex_command\
    --result-variable-prefix="LFBFL_"\
```

```
    --replace-line-returns-by=""

  case ${LFBFL_latex_command} in
    pdflatex) : ;;
    lualatex) : ;;
    *) printf "Unsupported latex command" ; return 1 ;;
  esac

  local LFBFL_command
  LFBFL_command="${LFBFL_latex_command}"
  LFBFL_command+=" '${LFBFL_subdir2}/${LFBFL_repository_name}.tex'"
  if [[ LFBFL_i_verbose -eq 0 ]]; then
    LFBFL_command+=" > /dev/null"
  fi
  readonly LFBFL_command

  for ((i=0; i<3; i++)); do
    eval "${LFBFL_command}"
  done

  LFBFL_files_to_temp=(\
    "${LFBFL_repository_name}.aux"\
    "${LFBFL_repository_name}.log"\
    "${LFBFL_repository_name}.out"\
    "${LFBFL_repository_name}.toc"
  )
  for LFBFL_file_name in "${LFBFL_files_to_temp[@]}"; do
    mv "${LFBFL_file_name}" "${LFBFL_temp_path}/"
  done
}

create_PDF "$@"
```

3.13 build_and_checks_dependencies/generate_from_template.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
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```
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# This file was renamed from "generate_from_template.sh" to
# "generate_from_template.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

generate_from_template_with_block_comments(){
  # $1=base_file_path
  # $2=target_file_path
  # $3=enter_block_comment
  # $4=exit_block_comment
  local LFBFL_temp
  LFBFL_temp="$2.generate_from_template_with_block_comments.temp"
  readonly LFBFL_temp
  {
    printf "%s\n" "$3"
    cat "$1"
    printf "%s\n" "$4"
  } > "${LFBFL_temp}"
  overwrite_if_not_equal "$2" "${LFBFL_temp}"
}

generate_from_template_with_line_comments(){
  # $1=base_file_path
  # $2=target_file_path
  # $3=line_comment_prefix
  # $4=optional_post_processing
  # $5=file_prefix
  local LFBFL_temp
  LFBFL_temp="$2.generate_from_template_with_line_comments.temp"
  readonly LFBFL_temp
```

```
  if [[ -n "$5" ]]; then
    printf "%s\n" "$5"\
      > "${LFBFL_temp}"
  fi
  sed -e "s/^/$3/g" "$1"\
    >> "${LFBFL_temp}"
  if [[ -n "$4" ]]; then
    eval "$4"
  fi
  overwrite_if_not_equal "$2" "${LFBFL_temp}"
}

split_file_in_two(){
  # $1=file_path
  # $2=token assume that token is the only thing on his line,
  #    except for a possible indentation.
  # $3=path_of_part1_of_file
  # $4=path_of_part2_of_file
  # We cannot add --line-regexp with --fixed-strings because of the
  # possible indentation in file.
  declare -ir LFBFL_i_line_number=$(
    grep_fixed_string_with_anchor "$1" "$2"\
      --enforce-line-ends-with-fixed-string --fixed-strings --line-number\
    | head --lines=1\
    | cut -f 1 -d ':'
  )
  declare -ir LFBFL_i_line_count=$(ll_wc -l -n "$1")
  declare -ir LFBFL_i_lines_after=$((
    LFBFL_i_line_count - LFBFL_i_line_number
  ))
  head --lines="$((LFBFL_i_line_number - 1))" "$1"\
    > "$3"
  tail --lines="${LFBFL_i_lines_after}" "$1"\
    > "$4"
}

insert_file_at_token(){
  # $1=file_path
  # $2=token assume that token is the only thing on his line,
  #    except for a possible indentation.
  # $3=path_of_file_to_insert
  # $4=path_of_result_file
  # Options:
  #   $5=--quiet to suppress diff output in overwrite_if_not_equal
  declare -r LFBFL_start_file_name="$1.insert_file_at_token1.temp"
  declare -r LFBFL_end_file_name="$1.insert_file_at_token2.temp"
```

```
  local LFBFL_result_file_name
  if [[ -n "$4" ]]; then
    LFBFL_result_file_name="$4"
  else
    LFBFL_result_file_name="$1.insert_file_at_token3.temp"
  fi
  readonly LFBFL_result_file_name
  split_file_in_two "$1" "$2" "${LFBFL_start_file_name}"\
    "${LFBFL_end_file_name}"
  cat "${LFBFL_start_file_name}" "$3" "${LFBFL_end_file_name}"\
    > "${LFBFL_result_file_name}"
  rm "${LFBFL_start_file_name}" "${LFBFL_end_file_name}"
  if [[ -z "$4" ]]; then
    overwrite_if_not_equal "$1" "${LFBFL_result_file_name}" "" "" "$5"
  fi
}
```

3.14 build_and_checks_dependencies/get_common_text_glob_patterns.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# This file was renamed from "get_common_text_glob_patterns.sh"
# to "get_common_text_glob_patterns.libr.sh".

get_COMMON_TEXT_PATHS_GLOB_PATTERNS(){
```

```
  if [[ -n "${COMMON_TEXT_PATHS_GLOB_PATTERNS}" ]]; then
    return
  fi

  declare -agr COMMON_TEXT_PATHS_GLOB_PATTERNS=(
    '**/*.c'
    '**/*.css'
    '**/*.h'
    '**/*.htm'
    '**/*.html'
    '**/*.ini'
    '**/*.js'
    '**/*.json'
    '**/*.md'
    '**/*.php'
    '**/*.py'
    '**/*.sh'
    '**/*.sql'
    '**/*.sub'
    '**/*.tex'
    '**/*.toml'
    '**/*.tpl'
    '**/*.ts'
    '**/*.txt'
    '**/*.xml'
    '**/*.yml'
    '**/COPYING'
    '**/COPYING.LESSER'
    '**/pre-commit'
    '**/post-commit'
    '**/.gitignore'
    '**/py.typed'
  )
}


get_COMMON_TEXT_FILES_GLOB_PATTERNS(){
  if [[ -n "${COMMON_TEXT_FILES_GLOB_PATTERNS}" ]]; then
    return
  fi

  declare -agr COMMON_TEXT_FILES_GLOB_PATTERNS=(
    '*.c'
    '*.css'
    '*.h'
    '*.htm'
```

```
    '*.html'
    '*.ini'
    '*.js'
    '*.json'
    '*.md'
    '*.php'
    '*.py'
    '*.sh'
    '*.sql'
    '*.sub'
    '*.tex'
    '*.toml'
    '*.tpl'
    '*.ts'
    '*.txt'
    '*.xml'
    '*.yml'
    'COPYING'
    'COPYING.LESSER'
    'pre-commit'
    'post-commit'
    '.gitignore'
    'py.typed'
  )
}
```

3.15 build_and_checks_dependencies/grammar_and_spelling_check.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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```

```
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#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# 2026-03-22T15:30+01:00: This file was renamed from
# "grammar_and_spell_check.libr.sh"
# to
# "grammar_and_spelling_check.libr.sh".
# This file was renamed from "grammar_and_spell_check.sh"
# to "grammar_and_spell_check.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

grammar_and_spelling_check(){
  # $1=configuration_path build_and_checks_variables/repository_data.txt
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  get_COMMON_TEXT_FILES_GLOB_PATTERNS
  local LFBFL_grammar_and_spelling_checker_command=""
  grep_variable "$1" grammar_and_spelling_checker_command\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  declare -r LFBFL_command="${LFBFL_grammar_and_spelling_checker_command}"
  if [[ -z "${LFBFL_command}" ]]; then
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "grammar_and_spelling_check: No command in configuration.\n"
```

```
    fi
    return
  fi
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    printf "grammar_and_spelling_check: Command found: %s.\n"\
      "${LFBFL_command}"
  fi
  local LFBFL_file_path
  local LFBFL_s_files_paths
  declare -a LFBFL_arr_files_paths
  local LFBFL_pattern
  local LFBFL_eval_string
  for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "grammar_and_spelling_check: Iterating on pattern: %s.\n"\
        "${LFBFL_pattern}"
    fi
    LFBFL_s_files_paths=$(
      find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | relevant_find
    )
    if [[ -z "${LFBFL_s_files_paths}" ]]; then
      [[ LFBFL_i_verbose -eq 1 ]]\
        && printf "grammar_and_spelling_check: No file found.\n"
      continue
    fi
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      LFBFL_eval_string="${LFBFL_command} ${LFBFL_file_path}"
      eval "${LFBFL_eval_string}"
    done
  done
}
```

3.16 build_and_checks_dependencies/licenses_templates/build_licenses_templates.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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#
```

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#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "build_licenses_templates.sh"
# to "build_licenses_templates.exec.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=check_shell_scripts_beginnings.libr.sh
source "./${LFBFL_subdir}/check_shell_scripts_beginnings.libr.sh"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=comparisons.libr.sh
source "./${LFBFL_subdir}/comparisons.libr.sh"
# shellcheck source=generate_from_template.libr.sh
source "./${LFBFL_subdir}/generate_from_template.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

build_licenses_templates(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap

  enhanced_set_shell_option pipefail --trap-unset

  local LFBFL_s_format

```

```
57    # Source
58    local LFBFL_license_source_subdir
59    LFBFL_license_source_subdir="./${LFBFL_subdir}/licenses_templates/"
60    readonly LFBFL_license_source_subdir
61    local LFBFL_source_license_prefix="${LFBFL_license_source_subdir}"
62    LFBFL_source_license_prefix+="license_file_header_"
63    readonly LFBFL_source_license_prefix
64    # Target
65    local LFBFL_license_target_subdir
66    LFBFL_license_target_subdir="${LFBFL_work_directory}/"
67    LFBFL_license_target_subdir="build_and_checks_variables/temp/"
68    readonly LFBFL_license_target_subdir
69    local LFBFL_target_license_prefix="${LFBFL_license_target_subdir}"
70    LFBFL_target_license_prefix+="license_file_header_"
71    readonly LFBFL_target_license_prefix
72
73    declare -ar LFBFL_licenses=(
74      "AGPLv3+"
75      "GPLv3+"
76      "LGPLv3+"
77    )
78
79    # ------------------------------------------------------------------
80    # Various language definitions for text file formats
81    declare -ar LFBFL_block_comment_languages=(
82      "c"
83      "py"
84    )
85    declare -ar LFBFL_block_comment_enters=(
86      '/*'
87      '"""'
88    )
89    declare -ar LFBFL_block_comment_exits=(
90      '*/'
91      '"""'
92    )
93    equal "${#LFBFL_block_comment_languages[@]}"\
94          "${#LFBFL_block_comment_enters[@]}"\
95          "${#LFBFL_block_comment_exits[@]}"
96    declare -i LFBFL_i_equal_result=$?
97    if [[ LFBFL_i_equal_result -eq 0 ]]; then
98      printf $'/!\\Problem with the definitions of languages arrays 1/!\\\n'
99      return 1
100   fi
101
102   declare -ar LFBFL_line_comment_languages=(
```

```
    "sh"
    "sql"
    "tex"
  )
  declare -ar LFBFL_line_comment_prefixes=(
    '# '
    '-- '
    '% '
  )
  equal "${#LFBFL_line_comment_languages[@]}"\
        "${#LFBFL_line_comment_prefixes[@]}"
  LFBFL_i_equal_result=$?
  if [[ LFBFL_i_equal_result -eq 0 ]]; then
    printf $'/!\\Problem with the definitions of languages arrays 2/!\\\n'
    return 1
  fi
  # ------------------------------------------------------------------

  # ------------------------------------------------------------------
  # We generate the license header templates adapted to various text
  # files formats.
  # Kamoulox do endfor, Jacques Beauregard XD
  # (C'est toujours mieux que l'inverse :) !)
  # for do endfor -> for do done (Fort Doux Donne).
  local LFBFL_some_license
  local LFBFL_license_prefix
  local LFBFL_license_prefix2
  local LFBFL_extension
  local LFBFL_license_file_name
  local LFBFL_enter_string
  local LFBFL_exit_string
  local LFBFL_temp2
  local LFBFL_prefix_string
  local LFBFL_intermediate_file_name
  local LFBFL_file_prefix
  declare -i LFBFL_i_generate_from_template_result
  declare -i LFBFL_i
  for LFBFL_some_license in "${LFBFL_licenses[@]}"; do
    LFBFL_license_prefix="${LFBFL_source_license_prefix}"
    LFBFL_license_prefix+="${LFBFL_some_license}"
    LFBFL_license_prefix2="${LFBFL_target_license_prefix}"
    LFBFL_license_prefix2+="${LFBFL_some_license}"
    for ((
      LFBFL_i = 0;
      LFBFL_i < ${#LFBFL_block_comment_languages[@]};
      ++LFBFL_i
```

```
149     )); do
150       LFBFL_extension=${LFBFL_block_comment_languages[LFBFL_i]}
151       LFBFL_license_file_name="${LFBFL_license_prefix2}"
152       LFBFL_license_file_name+=".${LFBFL_extension}"
153       LFBFL_enter_string=${LFBFL_block_comment_enters[LFBFL_i]}
154       LFBFL_exit_string=${LFBFL_block_comment_exits[LFBFL_i]}
155       generate_from_template_with_block_comments\
156         "${LFBFL_license_prefix}.tpl"\
157         "${LFBFL_license_file_name}.tpl"\
158         "${LFBFL_enter_string}" "${LFBFL_exit_string}"
159       LFBFL_i_generate_from_template_result=$?
160       if [[ LFBFL_i_verbose -eq 1 ]]; then
161         if [[ LFBFL_i_generate_from_template_result -eq 1 ]]; then
162           printf "License template %s.tpl updated.\n"\
163             "${LFBFL_license_file_name}"
164         elif [[ LFBFL_i_generate_from_template_result -eq 2 ]]; then
165           printf "License template %s.tpl created.\n"\
166             "${LFBFL_license_file_name}"
167         fi
168       fi
169     done
170     LFBFL_temp2=".generate_from_template_with_line_comments.temp"
171     for ((
172       LFBFL_i = 0;
173       LFBFL_i < ${#LFBFL_line_comment_languages[@]};
174       ++LFBFL_i
175     )); do
176       LFBFL_extension=${LFBFL_line_comment_languages[LFBFL_i]}
177       LFBFL_license_file_name="${LFBFL_license_prefix2}"
178       LFBFL_license_file_name+=".${LFBFL_extension}"
179       LFBFL_prefix_string=${LFBFL_line_comment_prefixes[LFBFL_i]}
180       LFBFL_intermediate_file_name="${LFBFL_license_file_name}.tpl"
181       LFBFL_intermediate_file_name+="${LFBFL_temp2}"
182       LFBFL_file_prefix=""
183       if [[ "${LFBFL_extension}" == "sh" ]]; then
184         LFBFL_file_prefix="${LFBFL_SHELL_SCRIPT_BEGINNING}"
185       fi
186       # The sed command is here to remove trailing spaces of previously
187       # empty lines after comment prefix is added to the line.
188       generate_from_template_with_line_comments\
189         "${LFBFL_license_prefix}.tpl"\
190         "${LFBFL_license_file_name}.tpl"\
191         "${LFBFL_prefix_string}"\
192         "sed -Ei -e 's/\s*$//g' '${LFBFL_intermediate_file_name}'"\
193         "${LFBFL_file_prefix}"
194       LFBFL_i_generate_from_template_result=$?
```

```
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        if [[ LFBFL_i_generate_from_template_result -eq 1 ]]; then
          printf "License template %s.tpl updated.\n"\
            "${LFBFL_license_file_name}"
        elif [[ LFBFL_i_generate_from_template_result -eq 2 ]]; then
          printf "License template %s.tpl created.\n"\
            "${LFBFL_license_file_name}"
        fi
      fi
    done
  done
  # ------------------------------------------------------------------

  # Now that base license templates generated languages license
  # templates in work directory, we can go there.
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  local LFBFL_data_file_name="build_and_checks_variables/"
  LFBFL_data_file_name+="repository_data.txt"
  readonly LFBFL_data_file_name
  local LFBFL_repository_name=""
  grep_variable "${LFBFL_data_file_name}" repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  local LFBFL_license=""
  grep_variable "${LFBFL_data_file_name}" license\
    --result-variable-prefix="LFBFL_"
  local LFBFL_license2=""
  grep_variable "${LFBFL_data_file_name}" license2\
    --result-variable-prefix="LFBFL_"
  local LFBFL_author_full_name=""
  grep_variable "${LFBFL_data_file_name}" author_full_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=" "

  LFBFL_license_prefix="build_and_checks_variables/temp/"
  LFBFL_license_prefix+="license_file_header_"
  readonly LFBFL_license_prefix
  LFBFL_license_prefix2="${LFBFL_license_prefix}${LFBFL_license}"
  readonly LFBFL_license_prefix2
  declare -r\
    LFBFL_license_prefix3="${LFBFL_license_prefix}${LFBFL_license2}"

  # First year according to current state of git repository.
  declare -ir LFBFL_i_first_year=$(
```

```
241     git log\
242     | grep 'Date:'\
243     | tail -1\
244     | cut -f 8 -d ' '
245   )
246   # Last year according to current state of git repository.
247   declare -ir LFBFL_i_last_year=$(
248     git log\
249     | grep 'Date:'\
250     | head -1\
251     | cut -f 8 -d ' '
252   )
253   local LFBFL_copyright_string
254   LFBFL_copyright_string="${LFBFL_i_first_year}-${LFBFL_i_last_year}"
255   LFBFL_copyright_string+=" ${LFBFL_author_full_name}"
256   readonly LFBFL_copyright_string
257
258   prepare_filled_license_file_block(){
259     # $1=license_file_name
260     sed -e "s/@repository_name@/${LFBFL_repository_name}/g"\
261         -e "s/@copyright_string@/${LFBFL_copyright_string}/g"\
262         "$1.tpl"\
263       > "$1.temp"
264     overwrite_if_not_equal "$1" "$1.temp"
265     declare -ir LFBFL_i_result=$?
266     if [[ LFBFL_i_verbose -eq 1 ]]; then
267       if [[ LFBFL_i_result -eq 1 ]]; then
268         printf "License template %s updated.\n" "$1"
269       elif [[ LFBFL_i_result -eq 2 ]]; then
270         printf "License template %s created.\n" "$1"
271       fi
272     fi
273     head --lines=-1 "$1"\
274       | tail --lines=+2\
275       > "$1.temp"
276     # Strange world: tail --lines=-1 was working
277     # to exclude first line.
278     # I tested it months ago, and saw with my eyes the start and end
279     # block comments removed with:
280     # head --lines=-1 "$1" | tail --lines=-1.
281     # Now, man page says to use --lines=+2 instead.
282     # Welcome in the world where AI change your environment and your
283     # documentation on the fly...
284   }
285
286   declare -Ar LFBFL_all_block_comment_languages=(
```

```
    ["c"]="c"
    ["php"]="c"
    ["py"]="py"
    ["js"]="c"
  )

  local LFBFL_key
  local LFBFL_dest
  local LFBFL_license_file_name2
  declare -i LFBFL_i_not_subfile
  declare -i LFBFL_i_not_subfile2
  local LFBFL_file_path
  local LFBFL_s_files_paths
  declare -a LFBFL_arr_files_paths
  for LFBFL_key in "${!LFBFL_all_block_comment_languages[@]}"; do
    LFBFL_dest=${LFBFL_all_block_comment_languages[${LFBFL_key}]}
    LFBFL_license_file_name="${LFBFL_license_prefix2}.${LFBFL_dest}"
    prepare_filled_license_file_block "${LFBFL_license_file_name}"
    if [[ -n "${LFBFL_license2}" ]]; then
      LFBFL_license_file_name2="${LFBFL_license_prefix3}"
      LFBFL_license_file_name2+=".${LFBFL_dest}"
      prepare_filled_license_file_block "${LFBFL_license_file_name2}"
    fi

    LFBFL_s_files_paths=$(
      find . -type f -iregex ".*\.\(${LFBFL_key}\|${LFBFL_key}\.tpl\)"\
        -printf '%P\n'\
      | relevant_find
    )
    if [[ -z "${LFBFL_s_files_paths}" ]]; then
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        LFBFL_s_format="build_licenses_templates: "
        LFBFL_s_format+="No file found with extension .%s or .%s.tpl.\n"
        # shellcheck disable=SC2059
        printf "${LFBFL_s_format}" "${LFBFL_key}" "${LFBFL_key}"
      fi
      continue
    fi
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      is_subfile "${LFBFL_file_path}"\
        "${LFBFL_license_file_name}.temp"
      LFBFL_i_not_subfile=$?
      LFBFL_i_not_subfile2=1
      if [[ -n "${LFBFL_license2}" ]]; then
        is_subfile "${LFBFL_file_path}"\
```

```
          "${LFBFL_license_file_name2}.temp"
        LFBFL_i_not_subfile2=$?
      fi
      if [[ LFBFL_i_not_subfile -ge 1 && LFBFL_i_not_subfile2 -ge 1 ]];
      then
        printf "File %s has no/wrong license header?\n"\
          "${LFBFL_file_path}"
      fi
    done
    rm "${LFBFL_license_file_name}.temp"
    if [[ -n "${LFBFL_license2}" ]]; then
      rm "${LFBFL_license_file_name2}.temp"
    fi
  done

  prepare_filled_license_file_line(){
    # $1=LFBFL_license_file_name
    sed -e "s/@repository_name@/${LFBFL_repository_name}/g"\
        -e "s/@copyright_string@/${LFBFL_copyright_string}/g"\
        "$1.tpl"\
      > "$1.temp"
    overwrite_if_not_equal "$1" "$1.temp"
    declare -ir LFBFL_i_result=$?
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      if [[ LFBFL_i_result -eq 1 ]]; then
        printf "License template %s updated.\n" "$1"
      elif [[ LFBFL_i_result -eq 2 ]]; then
        printf "License template %s created.\n" "$1"
      fi
    fi
  }

  declare -Ar LFBFL_all_line_comment_languages=(
    ["sh"]="sh"
    ["sql"]="sql"
    ["tex"]="tex"
  )

  for LFBFL_key in "${!LFBFL_all_line_comment_languages[@]}"; do
    LFBFL_dest=${LFBFL_all_line_comment_languages[${LFBFL_key}]}
    LFBFL_license_file_name="${LFBFL_license_prefix2}.${LFBFL_dest}"
    prepare_filled_license_file_line "${LFBFL_license_file_name}"
    if [[ -n "${LFBFL_license2}" ]]; then
      LFBFL_license_file_name2="${LFBFL_license_prefix3}"
      LFBFL_license_file_name2+=".${LFBFL_dest}"
      prepare_filled_license_file_line "${LFBFL_license_file_name2}"
```

```
    fi

    LFBFL_s_files_paths=$(
      find . -type f -iregex ".*\.\(${LFBFL_key}\|${LFBFL_key}\.tpl\)"\
        -printf '%P\n'\
      | relevant_find
    )
    if [[ -z "${LFBFL_s_files_paths}" ]]; then
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        LFBFL_s_format="build_licenses_templates: "
        LFBFL_s_format+="No file found with extension .%s or .%s.tpl.\n"
        # shellcheck disable=SC2059
        printf "${LFBFL_s_format}" "${LFBFL_key}" "${LFBFL_key}"
      fi
      continue
    fi
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      is_subfile "${LFBFL_file_path}" "${LFBFL_license_file_name}"
      LFBFL_i_not_subfile=$?
      LFBFL_i_not_subfile2=1
      if [[ -n "${LFBFL_license2}" ]]; then
        is_subfile "${LFBFL_file_path}" "${LFBFL_license_file_name2}"
        LFBFL_i_not_subfile2=$?
      fi
      if [[ LFBFL_i_not_subfile -ge 1 && LFBFL_i_not_subfile2 -ge 1 ]];
      then
        printf "File %s has no/wrong license header?\n"\
          "${LFBFL_file_path}"
      fi
    done
  done
}

build_licenses_templates "$@"
```

3.17 build_and_checks_dependencies/licenses_templates/license_file_header_AGPLv3+.tpl

```
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```

3.18 build_and_checks_dependencies/licenses_templates/license_file_header_GPLv3+.tpl

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```

3.19 build_and_checks_dependencies/licenses_templates/license_file_header_LGPLv3+.tpl

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```

```
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```

3.20 build_and_checks_dependencies/lines_counts.libr.sh

```
#!/usr/bin/env bash
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#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "lines_counts.sh" to
# "lines_counts.libr.sh".
```

```
26
27 LFBFL_subdir="build_and_checks_dependencies"
28 # shellcheck source=common_options.libr.sh
29 source "./${LFBFL_subdir}/common_options.libr.sh"
30 # shellcheck source=get_common_text_glob_patterns.libr.sh
31 source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
32 # shellcheck source=lines_filters.libr.sh
33 source "./${LFBFL_subdir}/lines_filters.libr.sh"
34
35 all_code_lines(){
36   # Options:
37   #   --verbose
38   #   --work-directory=""
39   declare -i LFBFL_i_verbose=0
40   get_verbose_option "$@"
41   local LFBFL_work_directory=""
42   get_work_directory_option "$@"
43   declare -a LFBFL_return_traps_stack
44   local LFBFL_previous_return_trap
45   init_return_trap
46   pushd_to_work_directory --trap-popd
47   can_continue_after_enhanced_pushd || return 1
48
49   enhanced_set_shell_option pipefail --trap-unset
50
51   get_COMMON_TEXT_FILES_GLOB_PATTERNS
52   local LFBFL_pattern
53   for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
54     if [[ LFBFL_i_verbose -eq 1 ]]; then
55       printf "all_code_lines: Iterating on pattern: %s.\n"\
56         "${LFBFL_pattern}"
57     fi
58     find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
59       | xargs grep --invert-match --perl-regexp --with-filename 'a(?!a)a'
60   done
61 }
62
63 all_self_code_lines(){
64   # Options:
65   #   --verbose
66   #   --work-directory=""
67   declare -i LFBFL_i_verbose=0
68   get_verbose_option "$@"
69   local LFBFL_work_directory=""
70   get_work_directory_option "$@"
71   declare -a LFBFL_return_traps_stack
```

```
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  declare -a LFBFL_common_options=("$@")
  keep_options LFBFL_common_options --verbose

  # below and another lines, see SC2145 false positive
  all_code_lines "${LFBFL_common_options[@]}" \
    | relevant_grep\
    | not_license_grep\
    | not_main_tex_grep\
    | not_main_html_grep
}

# shellcheck disable=SC2120
all_self_empty_code_lines(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  declare -a LFBFL_common_options=("$@")
  keep_options LFBFL_common_options --verbose

  all_self_code_lines "${LFBFL_common_options[@]}" \
    | empty_lines_after_file_name
}

# shellcheck disable=SC2120
all_self_not_empty_code_lines(){
  # Options:
  #   --verbose
```

```
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  declare -a LFBFL_common_options=("$@")
  keep_options LFBFL_common_options --verbose

  all_self_code_lines "${LFBFL_common_options[@]}" \
    | not_empty_lines_after_file_name
}

code_lines_count_all(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  declare -gi i_code_lines_count_all_result
  i_code_lines_count_all_result=$(
    all_self_code_lines\
    | wc --lines
  )
  # The printf is for command line use.
  # A script should use the result value instead.
  # shellcheck disable=SC2248
  printf "%s\n" ${i_code_lines_count_all_result}
}
```

```
164
165 code_lines_count_empty(){
166   # Options:
167   #   --verbose
168   #   --work-directory=""
169   declare -i LFBFL_i_verbose=0
170   get_verbose_option "$@"
171   local LFBFL_work_directory=""
172   get_work_directory_option "$@"
173   declare -a LFBFL_return_traps_stack
174   local LFBFL_previous_return_trap
175   init_return_trap
176   pushd_to_work_directory --trap-popd
177   can_continue_after_enhanced_pushd || return 1
178
179   enhanced_set_shell_option pipefail --trap-unset
180
181   declare -gi i_code_lines_count_empty_result
182   # shellcheck disable=SC2119
183   i_code_lines_count_empty_result=$(
184     all_self_empty_code_lines\
185     | wc --lines
186   )
187   # The printf is for command line use.
188   # A script should use the result value instead.
189   # shellcheck disable=SC2248
190   printf "%s\n" ${i_code_lines_count_empty_result}
191 }
192
193 code_lines_count_not_empty(){
194   # Options:
195   #   --verbose
196   #   --work-directory=""
197   declare -i LFBFL_i_verbose=0
198   get_verbose_option "$@"
199   local LFBFL_work_directory=""
200   get_work_directory_option "$@"
201   declare -a LFBFL_return_traps_stack
202   local LFBFL_previous_return_trap
203   init_return_trap
204   pushd_to_work_directory --trap-popd
205   can_continue_after_enhanced_pushd || return 1
206
207   enhanced_set_shell_option pipefail --trap-unset
208
209   declare -gi i_code_lines_count_not_empty_result
```

```
  # shellcheck disable=SC2119
  i_code_lines_count_not_empty_result=$(
    all_self_not_empty_code_lines\
    | wc --lines
  )
  # The printf is for command line use.
  # A script should use the result value instead.
  # shellcheck disable=SC2248
  printf "%s\n" ${i_code_lines_count_not_empty_result}
}
```

3.21 build_and_checks_dependencies/lines_filters.libr.sh

```
#!/usr/bin/env bash
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# This file was renamed from "lines_filters.sh" to
# "lines_filters.libr.sh".

# The goal of this file is to have some common line filters
# for bash processing and pipes.
# As is the case for other .sh files in this repository,
# it can be copied somewhere in your home directory
# and sourced in your .bashrc.

# shellcheck source=common_options.libr.sh
```

```
source "./build_and_checks_dependencies/common_options.libr.sh"

ll_wc(){
  # current options to wc:
  # -c --bytes
  # -m --chars
  # -l --lines
  # --files0-from # Incompatible with cat,
  #               # another thing intentional by ?
  #               # I don't see the meaning of "files0".
  # -L --max-line-length
  # -w --words
  # --help
  # --version
  # New option by Laurent Lyaudet (and probably other before):
  # -n --no-filenames display only number(s)
  # I think I did that the first time many years ago (Teliae ?).
  # Since then, they added it as ignored... (-n)
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  declare -a ll_wc_var_args=()
  declare -i ll_wc_var_i_number_only=0
  declare -i ll_wc_var_i=0
  for ll_wc_var_arg in "$@"; do
    # printf "%s\n" "${ll_wc_var_arg}"
    if [[ "${ll_wc_var_arg}" == "-n" ]]; then
      ll_wc_var_i_number_only=1
    elif [[ "${ll_wc_var_arg}" == "--no-filenames" ]]; then
      ll_wc_var_i_number_only=1
    elif [[ ll_wc_var_i_number_only -eq 0 ]]; then
      ll_wc_var_args[ll_wc_var_i]="${ll_wc_var_arg}"
      ll_wc_var_i=$((ll_wc_var_i + 1))
    elif [[ "${ll_wc_var_arg}" =~ -.* ]]; then
      ll_wc_var_args[ll_wc_var_i]="${ll_wc_var_arg}"
      ll_wc_var_i=$((ll_wc_var_i + 1))
    fi
  done
  # typeset -p ll_wc_var_args
  # printf "%s\n" "${ll_wc_var_i_number_only}"
  if [[ ll_wc_var_i_number_only -gt 0 ]]; then
    # Too simple code, only the base use case is handled now.
    # Convenient but incomplete.
    # With more time,
    # the simpler would be to submit a patch to GNU coreutils,
    # unless blockade from assholes.
    # Otherwise, it may be more simply handled in PHP or Python
```

```
    # (argparse, etc.).
    # You'll need to start by improving cat first,
    # with --files0-from handling.
    # wc does not display file_name when stream...
    # hence cat of last arg "${!#}"
    cat "${!#}"\
      | wc "${ll_wc_var_args[@]}"
  else
    # printf "normal\n"
    wc "${ll_wc_var_args[@]}"
  fi
}

# I submitted a patch for wc :).
# printf "https://domain/list-path/2024-05/msg00013.html" | \
#   sed --expression='s|domain|lists.gnu.org|' \
#       --expression='s|list-path|archive/html/bug-coreutils|'
# I found another way to have a long URL fit on 70 characters.
# If it is merged, the following would be better:
# If --no-filenames option available,
# we redefine ll_wc command as an alias.
wc_help_text=$(wc --help)
if [[ "${wc_help_text}" == *" --no-filenames "* ]]; then
  alias ll_wc='wc'
fi

# Trace avant le (2e ?) renommage
# de wget_sha512.sh à wget_sha512.libr.sh.
# $ ll_wc wget_sha512.sh
#  41  189 1224 wget_sha512.sh
# $ ll_wc -n wget_sha512.sh
#     41     189    1224
# $ ll_wc -l wget_sha512.sh
# 41 wget_sha512.sh
# $ ll_wc -l -n wget_sha512.sh
# 41

in_place_grep(){
  # Same arguments and options as grep.
  declare -r LFBFL_temp="${!#}.in_place_grep.temp"
  grep "$@" \
    > "${LFBFL_temp}"
  # declare -ir LFBFL_i_lines_before=$(ll_wc -l -n "${!#}")
  # declare -ir LFBFL_i_lines_after=$(ll_wc -l -n "${LFBFL_temp}")
  # printf "%s\n" "${LFBFL_i_lines_before}"
  # printf "%s\n" "${LFBFL_i_lines_after}"
```

```
  # if [[ LFBFL_i_lines_before -eq LFBFL_i_lines_after ]]; then
  # Almost as fast (on large files) and correct in all cases :).
  # Thanks Paul Eggert for the suggestion :).
  # https://debbugs.gnu.org/cgi/bugreport.cgi?bug=80547
  if cmp --silent "${LFBFL_temp}" "${!#}"; then
    rm "${LFBFL_temp}"
    return
  fi
  mv "${LFBFL_temp}" "${!#}"
}

grep_fixed_string_with_anchor(){
  # $1=file_path
  # $2=some_fixed_string
  # Options:
  #   --enforce-line-starts-with-fixed-string
  #   --enforce-line-ends-with-fixed-string
  #   Apart from $1 and $2 and the options above,
  #   all other arguments are considered grep_options like --quiet
  #   for example.
  #   But note that --fixed-strings will be added below, you don't need to
  #   supply it.
  declare -a LFBFL_grep_options=()
  declare -a LFBFL_grep_options2=()
  declare -i LFBFL_i=0
  declare -i LFBFL_i_enforce_line_starts=0
  declare -i LFBFL_i_enforce_line_ends=0
  declare -i LFBFL_i_quiet=0
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    ((++LFBFL_i))
    if [[ LFBFL_i -lt 3 ]]; then
      continue
    fi
    if [[ "${LFBFL_arg}" == "--enforce-line-starts-with-fixed-string" ]];
    then
      LFBFL_i_enforce_line_starts=1
      continue
    fi
    if [[ "${LFBFL_arg}" == "--enforce-line-ends-with-fixed-string" ]];
    then
      LFBFL_i_enforce_line_ends=1
      continue
    fi
    if [[ "${LFBFL_arg}" == "-q" || "${LFBFL_arg}" == "--quiet" ]]; then
      LFBFL_i_quiet=1
```

```
      LFBFL_grep_options+=("${LFBFL_arg}")
      continue
    fi
    LFBFL_grep_options2+=("${LFBFL_arg}")
  done

  if [[
    LFBFL_i_enforce_line_starts -eq 0
    && LFBFL_i_enforce_line_ends -eq 0
  ]]; then
    grep "${LFBFL_grep_options[@]}" --fixed-strings -- "$2" "$1"
    return
  fi
  declare -r LFBFL_s_lines=$(
    grep "${LFBFL_grep_options2[@]}" --color=never --fixed-strings --\
      "$2" "$1"
  )
  declare -i LFBFL_i_result=1
  if [[ -z "${LFBFL_s_lines}" ]]; then
    # shellcheck disable=SC2248
    return ${LFBFL_i_result}
  fi
  declare -a LFBFL_arr_lines
  mapfile -t LFBFL_arr_lines <<< "${LFBFL_s_lines}"
  readonly LFBFL_arr_lines

  declare -i LFBFL_i_fixed_string_length=${#2}
  local LFBFL_some_line
  local LFBFL_line_part
  for LFBFL_some_line in "${LFBFL_arr_lines[@]}"; do
    if [[ "${LFBFL_some_line}" == "$2" ]]; then
      LFBFL_i_result=0
      if [[ LFBFL_i_quiet -eq 1 ]]; then
        break
      else
        printf "%s\n" "${LFBFL_some_line}"
      fi
      continue
    fi
    if [[
      LFBFL_i_enforce_line_starts -eq 1
      && LFBFL_i_enforce_line_ends -eq 1
    ]]; then
      continue
    fi
    if [[ LFBFL_i_enforce_line_starts -eq 1 ]]; then
```

```
218       LFBFL_line_part=${LFBFL_some_line:0:${LFBFL_i_fixed_string_length}}
219     fi
220     if [[ LFBFL_i_enforce_line_ends -eq 1 ]]; then
221       LFBFL_line_part=${LFBFL_some_line: -${LFBFL_i_fixed_string_length}}
222     fi
223     if [[ "${LFBFL_line_part}" == "$2" ]]; then
224       LFBFL_i_result=0
225       if [[ LFBFL_i_quiet -eq 1 ]]; then
226         break
227       else
228         printf "%s\n" "${LFBFL_some_line}"
229       fi
230     fi
231   done
232   LFBFL_i_result=$((LFBFL_i_result | $?))
233   # shellcheck disable=SC2248
234   return ${LFBFL_i_result}
235 }
236
237 grep_variable(){
238   # $1=file_path
239   # $2=variable_name
240   # Options :
241   #   --replace-line-returns-by=""
242   #   --result-variable-prefix="LFBFL_" for example
243   declare -r LFBFL_regexp="(?<=^$2=).*$"
244   local LFBFL_arg
245   # printf "%s\n" "${LFBFL_regexp}"
246   local LFBFL_variable_value
247   if [[ "$*" == *--replace-line-returns-by=* ]]; then
248     local LFBFL_s_replace=""
249     for LFBFL_arg in "$@"; do
250       if [[ "${LFBFL_arg}" == --replace-line-returns-by=* ]]; then
251         LFBFL_s_replace=${LFBFL_arg#--replace-line-returns-by=}
252         break
253       fi
254     done
255     # sed -Ez -e...
256     LFBFL_variable_value=$(
257       grep --only-matching --perl-regexp "${LFBFL_regexp}" "$1"\
258       | sed --null-data --regexp-extended\
259         --expression='s/\n/'"${LFBFL_s_replace}"'/Mg'
260     )
261     # We remove the trailing replace string,
262     # since the intent is only to replace the intermediate \n.
263     if [[ -n "${LFBFL_s_replace}" ]]; then
```

```
    LFBFL_variable_value=${LFBFL_variable_value%"${LFBFL_s_replace}"}
    fi
  else
    LFBFL_variable_value=$(
      grep --only-matching --perl-regexp "${LFBFL_regexp}" "$1"
    )
  fi
  local LFBFL_prefix=""
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --result-variable-prefix=* ]]; then
      LFBFL_prefix=${LFBFL_arg#--result-variable-prefix=}
      break
    fi
  done
  declare -r LFBFL_variable_name="${LFBFL_prefix}$2"
  # printf "%s : %s\n " "${LFBFL_variable_name}" "${LFBFL_variable_value}"
  # The declare line may be dropped by a local or declare
  # in outer scope.
  # declare -g "${LFBFL_variable_name}"
  # So the assignment must be done on another line.
  printf -v "${LFBFL_variable_name}" "%s" "${LFBFL_variable_value}"
  # printf "%s\n" "${!LFBFL_variable_name}"
}

empty_lines(){
  grep '^$'
}

not_empty_lines(){
  grep --invert-match '^$'
}

space_starting_lines(){
  grep '^ '
}

not_space_starting_lines(){
  grep --invert-match '^ '
}

# Space starting or empty lines = spampty
spampty_lines(){
  grep --extended-regexp '^( |$)'
}

not_spampty_lines(){
```

```
310   grep '^[^ ]'
311   # grep --extended-regexp --invert-match '^( |$)'
312 }
313
314 all_space_lines(){
315   grep '^[ ]\+$'
316 }
317
318 not_all_space_lines(){
319   grep --invert-match '^[ ]\+$'
320 }
321
322 # All space or empty lines = aspampty
323 aspampty_lines(){
324   grep '^[ ]*$'
325 }
326
327 not_aspampty_lines(){
328   grep '[^ ]'
329   # grep --invert-match '^[ ]*$'
330 }
331
332 empty_lines_after_file_name(){
333   grep '^[^:]\+:$'
334 }
335
336 not_empty_lines_after_file_name(){
337   grep --invert-match '^[^:]\+:$'
338 }
339
340 space_starting_lines_after_file_name(){
341   grep '^[^:]\+: '
342 }
343
344 not_space_starting_lines_after_file_name(){
345   grep --invert-match '^[^:]\+: '
346 }
347
348 spampty_lines_after_file_name(){
349   grep --extended-regexp '^[^:]+:( |$)'
350 }
351
352 not_spampty_lines_after_file_name(){
353   # The two variants results differ on lines without filename before.
354   # Since the intent of the function is only on lines "blabla.sh:whatever",
355   # use the fastest one. I didn't do extensive testing and just assumed
```

```
  # that the "positive" one is faster as usual when querying database.
  # But there is no index here ;) XD.
  grep '^[^:]\+:[^ ]'
  # grep --extended-regexp --invert-match '^[^:]+:( |$)'
}

all_space_lines_after_file_name(){
  grep '^[^:]\+:[ ]\+$'
}

not_all_space_lines_after_file_name(){
  grep --invert-match '^[^:]\+:[ ]\+$'
}

aspampty_lines_after_file_name(){
  grep '^[^:]\+:[ ]*$'
}

not_aspampty_lines_after_file_name(){
  grep --invert-match '^[^:]\+:[ ]*$'
}

# If you need to test the previous 20 functions,
# look at file spampty_data.txt in miscellaneous.
# And give a try at something like:
# cat -E <(cat miscellaneous/spampty_data.txt | empty_lines)
# -E = --show-ends :). Nice command :).

not_JS_dependencies_find(){
  grep --extended-regexp --invert-match\
    --regexp="(^|/)node_modules/"\
    --regexp="(^|/)package-lock\.json$"
}

not_JS_dependencies_grep(){
  grep --extended-regexp --invert-match\
    --regexp="^([^:]+/)?node_modules/"\
    --regexp="^([^:]+/)?package-lock\.json:"
}

not_dependencies_find(){
  not_JS_dependencies_find
}

not_dependencies_grep(){
  not_JS_dependencies_grep
```

```
402 }
403
404 not_php_cache_find(){
405   grep --extended-regexp --invert-match\
406     --regexp="(^|/)\.phpdoc/cache/"
407 }
408
409 not_php_cache_grep(){
410   grep --extended-regexp --invert-match\
411     --regexp="^([^:]+/)?\.phpdoc/cache/"
412 }
413
414 not_python_cache_find(){
415   grep --extended-regexp --invert-match\
416     --regexp="(^|/)__pycache__/"\
417     --regexp="(^|/)\.mypy_cache/"\
418     --regexp="(^|/)\.ruff_cache/"
419 }
420
421 not_python_cache_grep(){
422   grep --extended-regexp --invert-match\
423     --regexp="^([^:]+/)?__pycache__/"\
424     --regexp="^([^:]+/)?\.mypy_cache/"\
425     --regexp="^([^:]+/)?\.ruff_cache/"
426 }
427
428 not_cache_find(){
429   not_php_cache_find\
430     | not_python_cache_find
431 }
432
433 not_cache_grep(){
434   not_php_cache_grep\
435     | not_python_cache_grep
436 }
437
438 not_git_find(){
439   grep --extended-regexp --invert-match "(^|/)\.git/"
440 }
441
442 not_git_grep(){
443   grep --extended-regexp --invert-match "^([^:]+/)?\.git/"
444 }
445
446 not_archive_find(){
447   grep --extended-regexp --invert-match "(\.gz|\.rar|\.tar|\.tgz|\.whl)$"
```

```
}

not_archive_grep(){
  grep --extended-regexp --invert-match\
    --regexp="^[^:]*(\.gz|\.rar|\.tar|\.tgz|\.whl):"
}

not_license_find(){
  grep --extended-regexp --invert-match\
    --regexp="(^|/)COPYING$"\
    --regexp="(^|/)COPYING\.LESSER$"
}

not_license_grep(){
  grep --extended-regexp --invert-match\
    --regexp="^([^:]+/)?COPYING:"\
    --regexp="^([^:]+/)?COPYING\.LESSER:"
}

not_main_tex_find(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep --extended-regexp --invert-match\
    --regexp="(^|/)${LFBFL_repository_name}\.tex$"
}

not_main_tex_grep(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep --extended-regexp --invert-match\
    --regexp="^([^:]+/)?${LFBFL_repository_name}\.tex:"
}

not_main_html_find(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep --extended-regexp --invert-match\
```

```
494     --regexp="(^|/)${LFBFL_repository_name}\.html$"
495 }
496
497 not_main_html_grep(){
498   local LFBFL_repository_name=""
499   grep_variable build_and_checks_variables/repository_data.txt\
500     repository_name\
501     --result-variable-prefix="LFBFL_"\
502     --replace-line-returns-by=""
503   grep --extended-regexp --invert-match\
504     --regexp="^([^:]+/)?${LFBFL_repository_name}\.html:"
505 }
506
507 not_temp_file_find(){
508   grep --extended-regexp --invert-match\
509     --regexp="(^|/)build_and_checks_variables/temp/"
510 }
511
512 not_temp_file_grep(){
513   grep --extended-regexp --invert-match\
514     --regexp="^([^:]+/)?build_and_checks_variables/temp/[^:]*:"
515 }
516
517 not_unchecked_builds_find(){
518   grep --extended-regexp --invert-match\
519     --regexp="(^|/)\.phpdoc/build/"
520 }
521
522 not_unchecked_builds_grep(){
523   grep --extended-regexp --invert-match\
524     --regexp="^([^:]+/)?\.phpdoc/build/"
525 }
526
527 relevant_find(){
528   enhanced_set_shell_option pipefail\
529     && trap 'enhanced_unset_shell_option pipefail' RETURN
530   not_dependencies_find\
531     | not_cache_find\
532     | not_git_find\
533     | not_archive_find\
534     | not_temp_file_find\
535     | not_unchecked_builds_find
536 }
537
538 relevant_grep(){
539   enhanced_set_shell_option pipefail\
```

```
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  not_dependencies_grep\
    | not_cache_grep\
    | not_git_grep\
    | not_archive_grep\
    | not_temp_file_grep\
    | not_unchecked_builds_grep
}
```

3.22 build_and_checks_dependencies/lines_maps.libr.sh

```
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# This file was renamed from "lines_maps.sh" to "lines_maps.libr.sh".

# https://en.wikibooks.org/wiki/Unicode/Character_reference/2000-2FFF
# Some invisible unicode characters that are a problem:
# 2000-200F 2028-202F 205F 2060-206F
# [
#   '2000', '2001', '2002', '2003', '2004', '2005', '2006', '2007',
#   '2008', '2009', '200A', '200B', '200C', '200D', '200E', '200F',
#   '2028', '2029', '202A', '202B', '202C', '202D', '202E', '202F',
#   '205F',
#   '2060', '2061', '2062', '2063', '2064', '2065', '2066', '2067',
#   '2068', '2069', '206A', '206B', '206C', '206D', '206E', '206F',
```

```
# ]
replace_non_ascii_spaces(){
  sed -e 's/\xc2\xa0/ /g'\
      -e 's/\xe2\x80\x80/ /g'\
      -e 's/\xe2\x80\x81/ /g'\
      -e 's/\xe2\x80\x82/ /g'\
      -e 's/\xe2\x80\x83/ /g'\
      -e 's/\xe2\x80\x84/ /g'\
      -e 's/\xe2\x80\x85/ /g'\
      -e 's/\xe2\x80\x86/ /g'\
      -e 's/\xe2\x80\x87/ /g'\
      -e 's/\xe2\x80\x88/ /g'\
      -e 's/\xe2\x80\x89/ /g'\
      -e 's/\xe2\x80\x8a/ /g'\
      -e 's/\xe2\x80\x8b/ /g'\
      -e 's/\xe2\x80\x8c/ /g'\
      -e 's/\xe2\x80\x8d/ /g'\
      -e 's/\xe2\x80\x8e/ /g'\
      -e 's/\xe2\x80\x8f/ /g'\
      -e 's/\xe2\x80\xa8/ /g'\
      -e 's/\xe2\x80\xa9/ /g'\
      -e 's/\xe2\x80\xaa/ /g'\
      -e 's/\xe2\x80\xab/ /g'\
      -e 's/\xe2\x80\xac/ /g'\
      -e 's/\xe2\x80\xad/ /g'\
      -e 's/\xe2\x80\xae/ /g'\
      -e 's/\xe2\x80\xaf/ /g'\
      -e 's/\xe2\x81\x9f/ /g'\
      -e 's/\xe2\x81\xa0/ /g'\
      -e 's/\xe2\x81\xa1/ /g'\
      -e 's/\xe2\x81\xa2/ /g'\
      -e 's/\xe2\x81\xa3/ /g'\
      -e 's/\xe2\x81\xa4/ /g'\
      -e 's/\xe2\x81\xa5/ /g'\
      -e 's/\xe2\x81\xa6/ /g'\
      -e 's/\xe2\x81\xa7/ /g'\
      -e 's/\xe2\x81\xa8/ /g'\
      -e 's/\xe2\x81\xa9/ /g'\
      -e 's/\xe2\x81\xaa/ /g'\
      -e 's/\xe2\x81\xab/ /g'\
      -e 's/\xe2\x81\xac/ /g'\
      -e 's/\xe2\x81\xad/ /g'\
      -e 's/\xe2\x81\xae/ /g'\
      -e 's/\xe2\x81\xaf/ /g'
}
```

3.23 build_and_checks_dependencies/overwrite_if_not_equal.libr.sh

```
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# This file was renamed from "overwrite_if_not_equal.sh"
# to "overwrite_if_not_equal.libr.sh".

overwrite_if_not_equal(){
  # $1=target_file_path
  # $2=temp_file_path
  # $3=keep_temp
  # $4=tree_mode : when the two files are the output of tree command,
  # we ignore if all the differences are only on directories timestamps.
  # $5=--quiet : we even don't want the message of diff saying that
  #              files are distinct
  # Return 0 target was the same
  #        1 target was different
  #        2 target did not already exist
  if [[ ! -f "$1" ]]; then
    if [[ -n "$3" ]]; then
      cp -p "$2" "$1"
    else
      mv "$2" "$1"
    fi
    return 2
```

```
45    fi
46    declare -i LFBFL_i_is_equal
47    if [[ -n "$4" ]]; then
48      # Not setting pipefail since the result would still be incorrect.
49      # shellcheck disable=SC2312
50      diff "$1" "$2"\
51        | grep --extended-regexp "^(>|<)"\
52        | grep --invert-match "/$"
53      LFBFL_i_is_equal=${PIPESTATUS[2]}
54      if [[ LFBFL_i_is_equal -eq 1 ]]; then
55        grep --only-matching "[^ ]*/$" "$1"\
56          > "overwrite_if_not_equal.file1.temp"
57        grep --only-matching "[^ ]*/$" "$2"\
58          > "overwrite_if_not_equal.file2.temp"
59        if [[ -n "$5" ]]; then
60          diff "overwrite_if_not_equal.file1.temp"\
61            "overwrite_if_not_equal.file2.temp"\
62            > /dev/null
63        else
64          diff "overwrite_if_not_equal.file1.temp"\
65            "overwrite_if_not_equal.file2.temp"
66        fi
67        LFBFL_i_is_equal=1-$?
68        rm "overwrite_if_not_equal.file1.temp"\
69          "overwrite_if_not_equal.file2.temp"
70      fi
71    else
72      if [[ -n "$5" ]]; then
73        diff --brief "$1" "$2" > /dev/null
74      else
75        diff --brief "$1" "$2"
76      fi
77      LFBFL_i_is_equal=1-$?
78    fi
79    if [[ LFBFL_i_is_equal -eq 1 ]]; then
80      if [[ -z "$3" ]]; then
81        rm "$2"
82      fi
83      return 0
84    fi
85    if [[ -n "$3" ]]; then
86      cp -p "$2" "$1"
87    else
88      mv "$2" "$1"
89    fi
90    return 1
```

```
}
```

3.24 build_and_checks_dependencies/python_black_complement.libr.sh

```
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# This file was renamed from "python_black_complement.sh"
# to "python_black_complement.libr.sh".

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

# It was rejected to enhance black with this currently.
# https://github.com/psf/black/issues/2370
# And moreover, at some point, I saw the reverse rule
# in black or pylint to add an empty line...
# This regexp may give false positives,
# but that's not the end of the world.
check_no_empty_line_after_python_function_docstring(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
```

```
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_bash_option globstar --trap-unset

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "Checking empty lines after Python function docstrings\n"
  pcre2grep -M --\
    $'def ([^"]|"(?!""))*"""([^"]|"(?!""))*"""\n\n(?!\s*def)'\
    **/*.py
}

python_black_complement(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  declare -a LFBFL_common_options=("$@")
  keep_options LFBFL_common_options --verbose

  check_no_empty_line_after_python_function_docstring\
    "${LFBFL_common_options[@]}"
}
```

3.25 build_and_checks_dependencies/python_isort_complement.libr.sh

```
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# This file was renamed from "python_isort_complement.sh"
# to "python_isort_complement.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

check_collections_abc_place(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "Checking import of _collections_abc is at the right place\n"

  enhanced_set_shell_option pipefail --trap-unset

```

```
  declare -r LFBFL_temp=".check_collections_abc_place.temp"
  local LFBFL_file_path
  declare -r LFBFL_s_files_paths=$(
    find . -type f -name "*.py" -printf '%P\n'\
    | relevant_find
  )
  if [[ -z "${LFBFL_s_files_paths}" ]]; then
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "check_collections_abc_place: No file found.\n"
    return
  fi
  declare -a LFBFL_arr_files_paths
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  readonly LFBFL_arr_files_paths
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    sed -Ez 's/\n(\nfrom _collections_abc[^\n]*)/\1\n/Mg'\
      "${LFBFL_file_path}"\
      > "${LFBFL_file_path}${LFBFL_temp}"
    overwrite_if_not_equal "${LFBFL_file_path}"\
      "${LFBFL_file_path}${LFBFL_temp}"
  done
}

python_isort_complement(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  declare -a LFBFL_common_options=("$@")
  keep_options LFBFL_common_options --verbose

  check_collections_abc_place "${LFBFL_common_options[@]}"
}
```

3.26 build_and_checks_dependencies/repository_data.libr.sh

```
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LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

# /!\ You need to activate extglob before parsing this file. /!\
# And I'm kind enough to activate it inside functions that need it below.

get_upgrade_venvs_time_interval_in_seconds(){
  # $1=data_file_path
  # Options:
  #   --verbose
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap

  enhanced_set_bash_option extglob --trap-unset
```

```

  grep_variable "$1"\
    upgrade_venvs_time_interval_in_seconds\
    --result-variable-prefix="LFBFL_"

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "You say \"%s\".\n"\
        "${LFBFL_upgrade_venvs_time_interval_in_seconds}"

  case ${LFBFL_upgrade_venvs_time_interval_in_seconds} in
    +([0-9])) : ;;
    wat) LFBFL_upgrade_venvs_time_interval_in_seconds=${RANDOM};;
    watyouwant\?) LFBFL_upgrade_venvs_time_interval_in_seconds=${SRANDOM};;
    *) printf "No wat for you?\n";; #TempsDeCerveauDisponible XD SC2249 ;)
  esac

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "I say \"%s\".\n"\
        "${LFBFL_upgrade_venvs_time_interval_in_seconds}"
}
```

3.27 build_and_checks_dependencies/shell_checks_complement.libr.sh

```
#!/usr/bin/env bash
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```

```
# This file was renamed from "python_black_complement.sh"
# to "python_black_complement.libr.sh".

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

check_no_size_of_array_first_element(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_bash_option globstar --trap-unset

  local LFBFL_s="Checking that shell scripts do not"
  LFBFL_s+=" contain wrong array size expansion."
  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "%s\n" "${LFBFL_s}"

  pcre2grep -M -- $'\$\{#.*(\b|_)arr_(?!.*\[@\])' **/*.sh
}

check_no_misplaced_then(){
  # Options:
  #   --verbose
  #   --work-directory=""
  #   --max-line-length
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  local LFBFL_arg
```

```
  declare -i LFBFL_i_max_line_length=70
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --max-line-length=* ]]; then
      LFBFL_arg=${LFBFL_arg#--max-line-length=}
      LFBFL_i_max_line_length=$((LFBFL_arg))
      break
    fi
  done
  declare -i LFBFL_i_max_previous_line_length
  LFBFL_i_max_previous_line_length=$((LFBFL_i_max_line_length - 5))

  enhanced_set_bash_option globstar --trap-unset

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf\
        "Checking that shell scripts do not contain misplaced 'then'.\n"

  local LFBFL_regexp
  LFBFL_regexp="^[^\n]{1,${LFBFL_i_max_previous_line_length}}"
  LFBFL_regexp+="(?<=\]\];)"
  LFBFL_regexp+="(?<!^\s{0,${LFBFL_i_max_previous_line_length}}\]\];)"
  LFBFL_regexp+="\n\s*then"
  # pcre2grep -M -- '^[^\n]{1,70}(?<=\]\];)(?<!^\s{0,70}\]\];)\n\s*then'
  pcre2grep -M -- "${LFBFL_regexp}" **/*.sh
}

check_no_negation_before_bash_test(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_bash_option globstar --trap-unset

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "Checking that shell scripts do not contain ! before [[.\n"

  pcre2grep -M -- $'!\s*\[\[' **/*.sh
```

```
}

shell_checks_complement(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  declare -a LFBFL_common_options=("$@")
  keep_options LFBFL_common_options --verbose --max-line-length

  check_no_size_of_array_first_element "${LFBFL_common_options[@]}"
  check_no_misplaced_then "${LFBFL_common_options[@]}"
  check_no_negation_before_bash_test "${LFBFL_common_options[@]}"
}
```

3.28 build_and_checks_dependencies/split_score.exec.php

```
#!/usr/bin/php
<?php

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```

```
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2026-01-04T13:38+01:00: This file was renamed from
"split_line_at_most.exec.php"
to
"split_score.exec.php".
This file was renamed from
"call_split_score_after_before_or_before_after.exec.php" to
"split_line_at_most.exec.php".
This file was renamed from
"call_split_score_after_before_or_before_after.php" to
"call_split_score_after_before_or_before_after.exec.php".

@category Executable
@package DevOrSysAdminScripts
@author Laurent Lyaudet <laurent.lyaudet@gmail.com>
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@license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
*/

declare(strict_types=1);
declare(encoding='UTF-8');

require_once './build_and_checks_dependencies/split_score.libr.php';

/*
php -r\
 $'require_once(\'./build_and_checks_dependencies/'\
$'split_score.libr.php\');'\
$' echo(generate_split_score'\
$'(true, 70, [\'/\'])(\'/\', 60, true));'
*/

// phpcs:disable Generic.WhiteSpace.ArbitraryParenthesesSpacing
// The first cast "(int) $func..." is to avoid false
// positive errors on possible XSS if output is unescaped.
echo(
  (int) generate_split_score(
    (bool) $argv[1],
    (int) $argv[2],
    explode(',', $argv[3]),
  )($argv[4], (int) $argv[5], (bool) $argv[6])
```

```
);
// phpcs:enable Generic.WhiteSpace.ArbitraryParenthesesSpacing
?>
```

3.29 build_and_checks_dependencies/split_score.libr.php

```
<?php

/**
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2026-01-04T13:38+01:00: This file was renamed from
"split_line_at_most.libr.php"
to
"split_score.libr.php".
This file was renamed from "split_line_at_most.php" to
"split_line_at_most.libr.php".

@category Library
@package DevOrSysAdminScripts
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*/

declare(strict_types=1);
```

```
41 declare(encoding='UTF-8');
42
43
44
45 /**
46 Given the 3 following arguments, this function returns a closure that can
47 be used to compute the split score given the current string
48 (hopefully matching a delimiter), the current cut/split position,
49 and if that split position is after or before the current string.
50
51 @param bool          $b_larger_after
52                      Score is larger when splitting after,
53                      previous name of this parameter was $b_after_before.
54 @param int           $i_max_length
55                      The maximum length allowed (goal length).
56 @param array<string> $a_delimiter_strings_domain
57                      An array of "delimiters".
58
59 @return \Closure {
60   The "split_score" closure to use repeatedly afterward.
61
62   @param string $s_delimiter_string The string tested to be a delimiter.
63   @param int    $i_cut_position     The offset of the cut in the string.
64   @param bool   $b_is_cut_after     Is the offset taken after the string?
65
66   @return int The split score.
67 }
68 */
69 function generate_split_score(
70   bool $b_larger_after,
71   int $i_max_length,
72   array $a_delimiter_strings_domain,
73 ) : Closure {
74   return function (
75     string $s_delimiter_string,
76     int $i_cut_position,
77     bool $b_is_cut_after,
78   ) use (
79     $b_larger_after,
80     $i_max_length,
81     $a_delimiter_strings_domain,
82   ) : int {
83     if(
84       !in_array(
85         $s_delimiter_string,
86         $a_delimiter_strings_domain,
```

```
        true,
      )
    ){
      // I do not handle regexps for the moment.
      // I'm too brainfucked. I just want to finish this feature fast
      // for HTML and Tex/PDF listings generation with HTML and Tex on
      // at most 70 characters.
      return 0;
    }
    if($b_is_cut_after === $b_larger_after){
      // When $b_larger_after,
      // always way better to cut after '/' than before it.
      // When !$b_larger_after (aka larger_before),
      // always way better to cut before '/' than after it.
      return 1 + $i_max_length + $i_cut_position;
    }
    return 1 + $i_cut_position;
  };
}//end generate_split_score()
?>
```

3.30 build_and_checks_dependencies/strings_functions.libr.sh

```
#!/usr/bin/env bash
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```

```
# I think I created a file strings_functions.sh or renamed to
# strings_functions.sh at some point, before committing.
# Hence, someone must have changed or reversed the name of this file
# to string_functions.sh at some point.
# This file was renamed from "string_functions.sh" to
# "strings_functions.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=comparisons.libr.sh
source "./${LFBFL_subdir}/comparisons.libr.sh"
# shellcheck source=too_long_code_lines.libr.sh
source "./${LFBFL_subdir}/too_long_code_lines.libr.sh"

split_line_at(){
  # $1=line
  # $2=position : number of characters in split_line_at_result_beginning
  declare -g split_line_at_result_beginning="${1:0:$2}"
  declare -g split_line_at_result_end="${1:$2}"
  # printf "%s\n" "$split_line_at_result_beginning"
  # printf "%s\n" "$split_line_at_result_end"
}

#---------------------------------------------------------------------------
#---------------------------------------------------------------------------
# Properties for the score for a possible split/cut in the string between
# two characters.
#---------------------------------------------------------------------------
# Some preliminary remarks:
# a) If the character before the split position is a forbidden previous
#   character, the score is -1.
#   (Usually, either we have no or one forbidden previous character that is
#   a backslash.)
# b) If the previous character is not forbidden,
#   and either the character before the position is among the delimiters,
#   or the character after the position is among the delimiters,
#   then the score is computed according to the delimiter found,
#   the position, and the fact that the position is after or before the
#   delimiter.
# c) If two consecutive delimiters are found in the string,
#   the maximum score between the one after the first delimiter
#   and before the second delimiter is taken for the score between the
#   two delimiters.
# d) Otherwise, by default,
#   the score for cutting/splitting at some position is 0.
```

```
# e) A substring is given by the goal length given for the resulting
#   strings/lines after split.
# f) When some scores are equal,
#   we split at the rightmost position with maximum score in the substring.
# g) Remark f) implies that below we do not need to consider strict
#    inequalities for the score.
# h) The score function SF(position) which will dictate our choice for
#   split/cut position is in reality
#   SF1a(
#     position, previous_character, next_character, delimiters, forbiddens
#   )
#   but at the cost of expliciting its internal and adding verbosity,
#   we can see that the score really depends on the following arguments
#   SF1b(
#     position, matched_delimiter_before, matched_delimiter_after,
#     previous_character_is_forbidden
#   )
#   If it is needed,
#   then matched_delimiter_before or matched_delimiter_after is NULL.
#   Somehow previous_character can be seen as position.previous_character,
#   and next_character can be seen as position.next_character.
#   We can see that the previous or next character without
#   knowing that it matches a delimiter is irrelevant.
#   The fact that there is 1 or more delimiters may be interesting.
#   The fact that there is 0 or more forbidden previous characters
#   may be interesting, but we haven't integrated it in the split_score
#   functions.
#   SF1a when the previous character is not forbidden
#   needs a score function SF2a with the following parameters
#   SF2a(position, current_character, delimiter, is_after):
#   SF1a(
#     position, previous_character, next_character, delimiters, forbiddens
#   ) = max(
#     For all delimiter in delimiters,
#     SF2a(position, previous_character, delimiter, TRUE),
#     SF2a(position, next_character, delimiter, FALSE),
#   )
#   =  max(
#     For all delimiter in delimiters,
#     For all couple C in {
#       (position.previous_character, TRUE),
#       (position.next_character, FALSE)
#     }
#     SF2a(position, C.1, delimiter, C.2),
#   )
#   Similarly,
```

```
#   SF1b(
#     position, matched_delimiter_before, matched_delimiter_after,
#     previous_character_is_forbidden
#   ) = max(
#     For all delimiter in delimiters,
#     SF2b(position, matched_delimiter_before, TRUE),
#     SF2b(position, matched_delimiter_after, FALSE),
#   )
#   where the score function SF2b has the following parameters
#   SF2b(cut_position, matched_delimiter, is_after).
#   We have also:
#   SF1b(
#     position, matched_delimiter_before, matched_delimiter_after,
#     previous_character_is_forbidden
#   ) = max(
#     For all delimiter in delimiters,
#     SF2c(position-1, matched_delimiter_before, TRUE),
#     SF2c(position, matched_delimiter_after, FALSE),
#   )
#   where the score function SF2c has the following parameters
#   SF2c(character_position, matched_delimiter, is_after).
#   current_character is only used to check if it is equal to the
#   delimiter, otherwise the score is zero;
#   but when expressing properties it makes sense to compare after
#   and before the same character position.
#   position, delimiter, and is_after may all three impact the score
#   once the current character matches the delimiter.
#   Position is an integer from 1 to goal length.
#   Cutting at position 0 would be stupid,
#   and moreover there would be no previous character.
#   Since we ask for a split, the current length is above goal length,
#   and there is a next character at the last considered position.
#   So, in search of properties that may have an impact on the algorithm,
#   we discard the characters to only keep matched delimiters.
#   Let us use p1 and p2 for position1 and position2,
#   mdb1 and mdb2 for matched_delimiter_before1, etc.,
#   mda1 and mda2 for matched_delimiter_after1, etc.,
#   pcif for previous_character_is_forbidden, etc.,
#   md1 and md2 for matched_delimiter1, etc.,
#   ia1 and ia2 for is_after1, etc.
#
#   What we consider as properties are of 3 forms.
#   The first form is like:
#     For all p1 in [1, goal_length], for all p2 in [1, goal_length],
#     for all mdb1 in delimiters union {NULL},
#     for all mdb2 in delimiters union {NULL},
```

```
#     for all mda1 in delimiters union {NULL},
#     for all mda2 in delimiters union {NULL},
#     RelationP(p1, p2) AND RelationM(mdb1, mdb2, mda1, mda2)
#     implies RelationS(
#       SF1b(p1, mdb1, mda1, false),
#       SF1b(p2, mdb2, mda2, false),
#     )
#   There are many choices for RelationP, RelationM and RelationS.
#   We can apply the without loss of generality argument only once when
#   breaking symmetry between equivalent properties.
#   Breaking symmetry on another relation that RelationS first
#   is a challenge that we do not tackle.
#   Indeed RelationS is the only relation where all variables appear.
#   And breaking symmetry on RelationP would not allow to break symmetry
#   on RelationM freely since some variables are correlated by the
#   suffixes 1 or 2.
#   But some choices for some relation may allow to break symmetry on
#   another relation.
#   Breaking symmetry on the result, there are only 2 choices for
#   RelationS:
#   RS= : SF1b(p1, mdb1, mda1, false) = SF1b(p2, mdb2, mda2, false),
#   RS<= : SF1b(p1, mdb1, mda1, false) <= SF1b(p2, mdb2, mda2, false).
#   But then there are 5+1 choices for RelationP:
#   RP1 : p1 = p2,
#   RP2 : p1 < p2,
#   RP3 : p1 <= p2,
#   RP4 : p1 > p2,
#   RP5 : p1 >= p2,
#   plus the choice that p1 and p2 are not necessarily related
#   (empty RP or RPe).
#   We can add RP6 : p1 = p2 + 1 and RP7 : p2 = p1 + 1, since it implies
#   some equalities between matched delimiters.
#   Note that  p1 = p2 + i or p2 = p1 + i with i >= 2 brings no additional
#   logical relations compared to p1 < p2 or p2 < p1;
#   it only yields subslices of other properties and open a slightly silly
#   door on infinity.
#   If we have RS=, by symmetry we can consider only RP1, RP2, RP3 and RPe,
#   and maybe RP6.
#   Note that RSe or empty RS would have no interest at all.
#   There are many choices for RelationM:
#   mdb1 is NULL AND mdb2 is NULL AND mda1 is NULL AND mda2 is NULL is not
#   interesting at all since both scores are 0 or -1.
#   mdb1 is NULL AND mdb2 is NULL AND mda1 is NULL AND mda2 is not NULL
#   implies that second score is always larger, that's not an additional
#   property, that is a theorem.
#   Hence, at least one of mdb1 or mda1, resp. mdb2 or mda2,
```

```
#   must (have the possibility to) be non-NULL.
#   If we start listing the remaining relations, we have:
#   1) mdb1 is NULL AND mdb2 is NULL AND mda1 is not NULL
#     AND mda2 is not NULL
#   RMA1A2
#   2) mdb1 is NULL AND mdb2 is NULL AND mda1 is not NULL
#     AND mda2 is not NULL
#     AND mda1 = mda2
#   RMA1A2=
#   3) mdb1 is NULL AND mdb2 is NULL AND mda1 is not NULL
#     AND mda2 is not NULL
#     AND mda1 != mda2 -> This case has no meaning:
#     If we only know they are different, we cannot say anything on
#     properties related to delimiters that are different unless the score
#     doesn't depend on the delimiter, and that case is already handled by
#     1). The same remark applies to some other choices.
#     The only thing that could be said would be if we had an order on
#     delimiters scores that are uniform for the position, or the score
#     before some delimiter would always be =, >=, <=, etc. the score
#     before or after another one. So it would only make sense with RPe,
#     and mda1 <= mda2 for example, but that would trivially imply RS<=.
#     It would end up with max(mda1, mdb1) <= max(mda2, mdb2) where max
#     ignores NULL value. You do not deduce or specify anything new here.
#   4) mdb1 is not NULL AND mdb2 is not NULL AND mda1 is NULL
#     AND mda2 is NULL
#   RMB1B2
#   and so on. So if we use the short notation RM followed by A1, resp. A2,
#   resp. B1, resp. B2, when mda1, resp. mda2, resp. mdb1, resp. mdb2,
#   is not null, we have the following possibilities:
#   RMA1A2, RMB1B2, RMA1B2, RMA2B1, all 4 with variant with = like RMA1A2=,
#   RMA1A2B1, RMA1A2B2, RMA1B1B2, RMA2B1B2, all 4 (RMXXYYZZ) with variants
#   with suffixes _1 (XX = YY), _2 (XX = ZZ), _3 (YY = ZZ),
#   and _4 (XX = YY = ZZ)
#   for equalities on the 1st, 2nd or 3rd couple or between the three
#   variables by transitivity,
#   RMALL where none is null, with suffixes A1A2 (A1 = A2), etc.
#   RMALL_A1A2, RMALL_A1B1, RMALL_A1B2, RMALL_A2B1, RMALL_A2B2, RMALL_B1B2,
#   for one edge;
#   RMALL_A1A2B1, RMALL_A1A2B2, RMALL_A1B1B2, RMALL_A2B1B2, one triangle;
#   RMALL_A1A2_B1B2, RMALL_A1B1_A2B2, RMALL_A1B2_A2B1, 2 disjoint edges;
#   RMALL_ALL square with diagonals.
#   So we see that we listed at least 39 possibilities only for RM.
#   We did not list yet the possibilities where the variables not present
#   in some equality can be NULL or not NULL, etc.
#   We see 9 of them : RMe (40), and the following 8:
#   RMA1A2+ = RMA1A2= OR RMA1A2B1_1 OR RMA1A2B2_1 OR RMALL_A1A2
```

```
#   RMB1B2+ = RMB1B2= OR RMA1B1B2_3 OR RMA2B1B2_3 OR RMALL_B1B2
#   RMA1B2+ = RMA1B2= OR RMA1A2B2_2 OR RMA1B1B2_2 OR RMALL_A1B2
#   RMA2B1+ = RMA2B1= OR RMA1A2B1_3 OR RMA2B1B2_1 OR RMALL_A2B1
#   RMA1A2B1+ = RMA1A2B1_4 OR RMALL_A1A2B1
#   RMA1A2B2+ = RMA1A2B2_4 OR RMALL_A1A2B2
#   RMA1B1B2+ = RMA1B1B2_4 OR RMALL_A1B1B2
#   RMA2B1B2+ = RMA2B1B2_4 OR RMALL_A2B1B2
#   We don't do complete analysis at the moment.
#   If we wanted to filter out some possibilities,
#   we would keep only the RM possibilities, among the 38,
#   that are symmetric between 1 and 2:
#   RMA1A2, RMB1B2, RMA1A2=, RMB1B2=,
#   RMALL, RMALL_A1A2, RMALL_A2B2, RMALL_A1A2_B1B2, RMALL_A1B1_A2B2,
#   RMALL_ALL. Still 10 possibilities.
#   We have 1 (RS=) times 5 times 10 + 1 (RS<=) times 8 times 10 = 130
#   combinations (life is too short ;) :) );
#   if we keep 48 RM possibilities, there is 624 combinations,
#   but we did not analyze totally which RM relations are useless.
#
#   The second form is like:
#     For all p1 in [1, goal_length], for all p2 in [1, goal_length],
#     for all md1 in delimiters union {NULL},
#     for all md2 in delimiters union {NULL},
#     for all ia1 in {TRUE,FALSE}, for all ia2 in {TRUE,FALSE},
#     RelationP(p1, p2) AND RelationMM(md1, md2) AND RelationIA(ia1, ia2)
#     implies RelationS(
#       SF2b(p1, md1, ia1)
#       SF2b(p2, md2, ia2)
#     )
#   The third form is like:
#     For all p1 in [1, goal_length], for all p2 in [1, goal_length],
#     for all md1 in delimiters union {NULL},
#     for all md2 in delimiters union {NULL},
#     for all ia1 in {TRUE,FALSE}, for all ia2 in {TRUE,FALSE},
#     RelationPP(p1, p2) AND RelationMM(md1, md2) AND RelationIA(ia1, ia2)
#     implies RelationS(
#       SF2c(p1, md1, ia1)
#       SF2c(p2, md2, ia2)
#     )
#   This time the combinatorics is much more pleasant :).
#   As above, we have RS= and RS<=, RPe, RP1 to RP7;
#   For RelationMM, assume that the possible delimiters correspond to
#   integer numbers 1 to i; hence md1, resp. md2, can be seen as an integer
#   in [0,i], either 0 for NULL or the delimiter index.
#   Clearly, if md1 = k in [1,i] and md2 = l in [1,i], then the only
#   information we can use is k = l or k != l.
```

```
#   If we allow that md1 != 0 and md2 = 0,
#   then both RS= and RS<= are false.
#   Thus RMMe is not possible.
#   If we allow that md1 = 0 and md2 != 0, then RS<= is always true,
#   whilst RS= is false.
#   Thus properties with RS<= can trivially be extended with
#   RMM-"allow-md1-0", not useful to explicit this for the moment.
#   If both md1 = 0 and md2 = 0, then both scores are 0 and we also learn
#   nothing more.
#   Thus we can focus on md1 and md2 are not NULL,
#   and are equal RMM= or not RMM!=. It seems that a property with RMM!=
#   could be extended to a property RMM=or!=/RMMNotNULL.
#   and finally RIA0 to RIA8 where TRUE = 1, FALSE = 0 and the digit is
#   3 * ia1 + ia2, and both variables can be 2 if they can be either TRUE
#   or FALSE. Wrong ! Not pleasant ! XD
#   So for RIA, we keep either: empty RIAe (RIA8),
#   ia1 = ia2 RIA= (RIA0 or RIA4),
#   ia1 = ia2 = TRUE RIA=T (RIA4), ia1 = ia2 = FALSE RIA=F (RIA0),
#   ia1 != ia2 RIA!= (RIA1 or RIA3), ia1 != ia2 = FALSE RIA!=T1 (RIA3),
#   ia1 != ia2 = TRUE RIA!=T2 (RIA1),
#   ia1 = TRUE (RIA5), ia1=FALSE (RIA2),
#   ia2 = TRUE (RIA7), ia2=FALSE (RIA6),
#   even less pleasant XD : 11 possibilities.
#   In total, we have now 1 (RS=) times 5 times 2 times 11
#   + 1 (RS<=) times 8 times 2 times 11 = 286
#   combinations (life is even shorter ;) :) ).
#   We only listed below properties of the third form.
# i) By the previous remark, there is a good chance that we missed some
#   possible property.
#   We hope we did found the properties that had a simple meaning,
#   in particular related to the algorithm where either you
#   loop on the string forward and keep last position with maximum score,
#   or you loop backward and abort as soon as possible as soon you know
#   that you found some position at the end that must have maximum score.
# j) These properties are "boolean" flags (integers that are powers of 2).
#   We first put the exact integers,
#   then wondered about switching everything to exponentiation 2**i,
#   then we took the solution that minimizes the character length between
#   [1-9][0-9]* and 2**i, then reverted to initial before committing.
#   Most efficient interpreted code is what we did first,
#   if there is no opcode cache for constants.
#   Most safe code is when the language hides the details and handles
#   families of flags,
#   but beware if some flag goes into the database.
#   Most readable code is a mix: 1 is easier to read than 2**0 ;) :P,
#   or $((2<<0))...
```

```
#   What I would love to see in many languages is a flag type, as a subtype
#   of integers, flag32, flag64 for C for example.
#   I think there is no such thing because sooner or later you play with
#   pseudo-flags that are combination of flags, hence integers.
#   But just for the declaration to let the type checker check that the
#   integer is a power of 2: flag32 MY_FLAG = 4;
#   Or better flag32 MY_FLAG = 2; as an equivalent of int32 MY_FLAG = 1<<2;
#   Another fact is that practice makes perfect, so working with the powers
#   of 2 written out in base 10 gets you accustomed to them.
#   Safe code should be favored since (pre-)compilation optimizes it,
#   but for this project, I prefer to keep the integers in decimal and
#   check that it doubles when reading my listing, because I'm familiar
#   with that and the powers of 2 doesn't go beyond what I can check fast
#   enough in my brain.
# k) Some of the properties below are union of other properties,
#   and are here to simplify not only code but understanding.
#   split_score_properties_logical_closure() function below checks that
#   all flags that are consequences of already activated flags are set.
# l) We use SSP_ prefix for SPLIT_SCORE_PROPERTY_.
# Some of these remarks are repeated below in the corresponding functions,
# and some remarks are added just after the constants.
#-----------------------------------------------------------------------
# Any score after a character that matches a delimiter
# is larger than or equal to
# any score before a character that matches a delimiter.
# RPPe RMMNotNull RIA!=T2 RS<=
declare -gir SSP_ALWAYS_LARGER_AFTER=1

# Any score before a character that matches a delimiter
# is larger than or equal to
# any score after a character that matches a delimiter.
# RPPe RMMNotNull RIA!=T1 RS<=
declare -gir SSP_ALWAYS_LARGER_BEFORE=2

# The score doesn't depend on the delimiter that is found.
# RPP1 RMMNotNull RIA= RS=
declare -gir SSP_DELIMITER_UNIFORM=4

# Whenever a character that matches a delimiter is found,
# the score after is equal to the score before.
# Or equivalently:
# For any delimiter, and for any character position,
# if the character at this position matches this delimiter,
# the score after the character
# is equal to
# the score before the character.
```

```
# RPP1 RMM= RIA!= RS=
declare -gir SSP_EQUAL_AFTER_OR_BEFORE=8
# declare -gir SSP_SAME_DELIMITER_AND_POSITION_EQUAL_AFTER_OR_BEFORE=

# Whenever a character that matches a delimiter is found,
# the score after is larger than or equal to the score before.
# Or equivalently:
# For any delimiter, and for any character position,
# if the character at this position matches this delimiter,
# the score after the character
# is larger than or equal to
# the score before the character.
# RPP1 RMM= RIA!=T2 RS<=
declare -gir SSP_LARGER_AFTER=16
# declare -gir SSP_SAME_DELIMITER_AND_POSITION_LARGER_AFTER=

# Whenever a character that matches a delimiter is found,
# the score before is larger than or equal to the score after.
# Or equivalently:
# For any delimiter, and for any character position,
# if the character at this position matches this delimiter,
# the score before the character
# is larger than or equal to
# the score after the character.
# RPP1 RMM= RIA!=T1 RS<=
declare -gir SSP_LARGER_BEFORE=32
# declare -gir SSP_SAME_DELIMITER_AND_POSITION_LARGER_BEFORE=

# The score after, resp. before, a character that matches a delimiter
# doesn't decrease when the position in the string increase.
# (RPP2 or RPP3) RMM= RIA= RS<=
declare -gir SSP_POSITION_NOT_DECREASING=64

# The score after a character that matches a delimiter doesn't decrease
# when the position in the string increases.
# (RPP2 or RPP3) RMM= RIA=T RS<=
declare -gir SSP_POSITION_NOT_DECREASING_AFTER=128

# The score before a character that matches a delimiter doesn't decrease
# when the position in the string increases.
# (RPP2 or RPP3) RMM= RIA=F RS<=
declare -gir SSP_POSITION_NOT_DECREASING_BEFORE=256

# The score after, resp. before, a character that matches a delimiter
# doesn't depend on the position in the string.
# (RPP2 or RPP3) RMM= RIA= RS=
```

```
declare -gir SSP_POSITION_UNIFORM=512

# The score after a character that matches a delimiter doesn't depend
# on the position in the string.
# (RPP2 or RPP3) RMM= RIA=T RS=
declare -gir SSP_POSITION_UNIFORM_AFTER=1024

# The score before a character that matches a delimiter doesn't depend
# on the position in the string.
# (RPP2 or RPP3) RMM= RIA=F RS=
declare -gir SSP_POSITION_UNIFORM_BEFORE=2048

# For any delimiter,
# any score after a character that matches this delimiter
# is equal to
# any score before a character that matches this delimiter.
# RPPe RMM= RIA!= RS=
# <=> RPPe RMM= RIAe RS=
declare -gir SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE=4096

# For any delimiter,
# any score after a character that matches this delimiter
# is larger than or equal to
# any score before a character that matches this delimiter.
# RPPe RMM= RIA!=T2 RS<=
declare -gir SSP_SAME_DELIMITER_LARGER_AFTER=8192

# For any delimiter,
# any score before a character that matches this delimiter
# is larger than or equal to
# any score after a character that matches this delimiter.
# RPPe RMM= RIA!=T1 RS<=
declare -gir SSP_SAME_DELIMITER_LARGER_BEFORE=16384

# For any character position,
# assuming that a character at this position matches some delimiter,
# the score after a character at this position that matches a delimiter
# is equal to
# the score before a character at this position that matches a delimiter.
# RPP1 RMMNotNull RIA!= RS=
declare -gir SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE=32768

# For any character position,
# assuming that a character at this position matches some delimiter,
# the score after a character at this position that matches a delimiter
# is larger than or equal to
```

```
# the score before a character at this position that matches a delimiter.
# RPP1 RMMNotNull RIA!=T2 RS<=
declare -gir SSP_SAME_POSITION_LARGER_AFTER=65536

# For any character position,
# assuming that a character at this position matches some delimiter,
# the score before a character at this position that matches a delimiter
# is larger than or equal to
# the score after a character at this position that matches a delimiter.
# RPP1 RMMNotNull RIA!=T1 RS<=
declare -gir SSP_SAME_POSITION_LARGER_BEFORE=131072

# For all possible split/cut positions, the score is equal as long as
# at least one of the previous and next characters is a delimiter.
# Or equivalently:
# Any score after a character that matches a delimiter
# is equal to
# any score before a character that matches a delimiter.
# is_after doesn't change anything.
# So we cut on the last position adjacent to a delimiter without
# forbidden previous character (edge case, on last position).
# RPPe RMMNotNull RIAe RS=
declare -gir SSP_UNIFORM=262144
# declare -gir SSP_ALWAYS_EQUAL_AFTER_OR_BEFORE=1

#-----------------------------------------------------------------------
# m) Note that a property telling that the score is always the same
#   even if the current character isn't among delimiters is not needed,
#   since, in that case, a null split score command will split at the goal
#   length.
#-----------------------------------------------------------------------
#-----------------------------------------------------------------------

split_score_properties_logical_closure(){
  # $1=split_score_properties
  # Since this code is not performance critical,
  # we followed lexicographic order on constants,
  # instead of an order to minimize the operations.
  # It is also safer to avoid forgetting some relation.
  declare -i LFBFL_i_result=$(($1))
  declare -i LFBFL_i_result_old=0
  declare -i LFBFL_i_all_flags=$((
    SSP_ALWAYS_LARGER_AFTER
    | SSP_ALWAYS_LARGER_BEFORE
    | SSP_DELIMITER_UNIFORM
    | SSP_EQUAL_AFTER_OR_BEFORE
```

```
530       | SSP_LARGER_AFTER
531       | SSP_LARGER_BEFORE
532       | SSP_POSITION_NOT_DECREASING
533       | SSP_POSITION_NOT_DECREASING_AFTER
534       | SSP_POSITION_NOT_DECREASING_BEFORE
535       | SSP_POSITION_UNIFORM
536       | SSP_POSITION_UNIFORM_AFTER
537       | SSP_POSITION_UNIFORM_BEFORE
538       | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
539       | SSP_SAME_DELIMITER_LARGER_AFTER
540       | SSP_SAME_DELIMITER_LARGER_BEFORE
541       | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
542       | SSP_SAME_POSITION_LARGER_AFTER
543       | SSP_SAME_POSITION_LARGER_BEFORE
544       | SSP_UNIFORM
545     ))
546     until [[ LFBFL_i_result_old -eq LFBFL_i_result ]]; do
547       LFBFL_i_result_old=$((LFBFL_i_result))
548       if ((LFBFL_i_result & SSP_ALWAYS_LARGER_AFTER)); then
549         LFBFL_i_result=$((
550           LFBFL_i_result
551           | SSP_LARGER_AFTER
552           | SSP_SAME_DELIMITER_LARGER_AFTER
553           | SSP_SAME_POSITION_LARGER_AFTER
554         ))
555       fi
556       if ((LFBFL_i_result & SSP_ALWAYS_LARGER_BEFORE)); then
557         LFBFL_i_result=$((
558           LFBFL_i_result
559           | SSP_LARGER_BEFORE
560           | SSP_SAME_DELIMITER_LARGER_BEFORE
561           | SSP_SAME_POSITION_LARGER_BEFORE
562         ))
563       fi
564       if ((LFBFL_i_result & SSP_POSITION_NOT_DECREASING)); then
565         LFBFL_i_result=$((
566           LFBFL_i_result
567           | SSP_POSITION_NOT_DECREASING_AFTER
568           | SSP_POSITION_NOT_DECREASING_BEFORE
569         ))
570       fi
571       if ((LFBFL_i_result & SSP_POSITION_UNIFORM)); then
572         LFBFL_i_result=$((
573           LFBFL_i_result
574           | SSP_POSITION_UNIFORM_AFTER
575           | SSP_POSITION_UNIFORM_BEFORE
```

```
576         | SSP_POSITION_NOT_DECREASING
577         | SSP_POSITION_NOT_DECREASING_AFTER
578         | SSP_POSITION_NOT_DECREASING_BEFORE
579       ))
580     fi
581     if ((LFBFL_i_result & SSP_POSITION_UNIFORM_AFTER)); then
582       LFBFL_i_result=$((
583         LFBFL_i_result
584         | SSP_POSITION_NOT_DECREASING_AFTER
585       ))
586     fi
587     if ((LFBFL_i_result & SSP_POSITION_UNIFORM_BEFORE)); then
588       LFBFL_i_result=$((
589         LFBFL_i_result
590         | SSP_POSITION_NOT_DECREASING_BEFORE
591       ))
592     fi
593     if ((LFBFL_i_result & SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE)); then
594       LFBFL_i_result=$((
595         LFBFL_i_result
596         | SSP_EQUAL_AFTER_OR_BEFORE
597         | SSP_LARGER_AFTER
598         | SSP_LARGER_BEFORE
599         | SSP_SAME_DELIMITER_LARGER_AFTER
600         | SSP_SAME_DELIMITER_LARGER_BEFORE
601       ))
602     fi
603     if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER)); then
604       LFBFL_i_result=$((
605         LFBFL_i_result
606         | SSP_LARGER_AFTER
607       ))
608     fi
609     if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE)); then
610       LFBFL_i_result=$((
611         LFBFL_i_result
612         | SSP_LARGER_BEFORE
613       ))
614     fi
615     if ((LFBFL_i_result & SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE)); then
616       LFBFL_i_result=$((
617         LFBFL_i_result
618         | SSP_DELIMITER_UNIFORM
619         | SSP_EQUAL_AFTER_OR_BEFORE
620         | SSP_LARGER_AFTER
621         | SSP_LARGER_BEFORE
```

```
622         | SSP_SAME_POSITION_LARGER_AFTER
623         | SSP_SAME_POSITION_LARGER_BEFORE
624       ))
625     fi
626     if ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
627       LFBFL_i_result=$((
628         LFBFL_i_result
629         | SSP_LARGER_AFTER
630       ))
631     fi
632     if ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
633       LFBFL_i_result=$((
634         LFBFL_i_result
635         | SSP_LARGER_BEFORE
636       ))
637     fi
638     if ((LFBFL_i_result & SSP_UNIFORM)); then
639       LFBFL_i_result=$((LFBFL_i_all_flags))
640       break
641     fi
642     if ((LFBFL_i_result & SSP_ALWAYS_LARGER_AFTER))\
643     && ((
644       LFBFL_i_result & (
645         SSP_ALWAYS_LARGER_BEFORE
646         | SSP_EQUAL_AFTER_OR_BEFORE
647         | SSP_LARGER_BEFORE
648         | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
649         | SSP_SAME_DELIMITER_LARGER_BEFORE
650         | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
651         | SSP_SAME_POSITION_LARGER_BEFORE
652       )
653     )); then
654       LFBFL_i_result=$((LFBFL_i_all_flags))
655       break
656     fi
657     if ((LFBFL_i_result & SSP_ALWAYS_LARGER_BEFORE))\
658     && ((
659       LFBFL_i_result & (
660         SSP_EQUAL_AFTER_OR_BEFORE
661         | SSP_LARGER_AFTER
662         | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
663         | SSP_SAME_DELIMITER_LARGER_AFTER
664         | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
665         | SSP_SAME_POSITION_LARGER_AFTER
666       )
667     )); then
```

```
      LFBFL_i_result=$((LFBFL_i_all_flags))
      break
    fi
    if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
    && ((LFBFL_i_result & SSP_EQUAL_AFTER_OR_BEFORE)); then
      LFBFL_i_result=$((
        LFBFL_i_result
        | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
      ))
    fi
    if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
    && ((LFBFL_i_result & SSP_LARGER_AFTER)); then
      LFBFL_i_result=$((
        LFBFL_i_result
        | SSP_SAME_POSITION_LARGER_AFTER
      ))
    fi
    if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
    && ((LFBFL_i_result & SSP_LARGER_BEFORE)); then
      LFBFL_i_result=$((
        LFBFL_i_result
        | SSP_SAME_POSITION_LARGER_BEFORE
      ))
    fi
    if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
    && ((LFBFL_i_result & SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE)); then
      LFBFL_i_result=$((LFBFL_i_all_flags))
      break
    fi
    if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
    && ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER)); then
      LFBFL_i_result=$((
        LFBFL_i_result
        | SSP_ALWAYS_LARGER_AFTER
      ))
    fi
    if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
    && ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE)); then
      LFBFL_i_result=$((
        LFBFL_i_result
        | SSP_ALWAYS_LARGER_BEFORE
      ))
    fi
    if ((LFBFL_i_result & SSP_LARGER_AFTER))\
    && ((LFBFL_i_result & SSP_LARGER_BEFORE)); then
      LFBFL_i_result=$((
```

```
714         LFBFL_i_result
715         | SSP_EQUAL_AFTER_OR_BEFORE
716       ))
717     fi
718     if ((LFBFL_i_result & SSP_LARGER_AFTER))\
719     && ((
720       LFBFL_i_result & (
721         SSP_POSITION_UNIFORM
722         | SSP_POSITION_UNIFORM_AFTER
723       )
724     )); then
725       LFBFL_i_result=$((
726         LFBFL_i_result
727         | SSP_SAME_DELIMITER_LARGER_AFTER
728       ))
729     fi
730     if ((LFBFL_i_result & SSP_LARGER_BEFORE))\
731     && ((
732       LFBFL_i_result & (
733         SSP_POSITION_UNIFORM
734         | SSP_POSITION_UNIFORM_BEFORE
735       )
736     )); then
737       LFBFL_i_result=$((
738         LFBFL_i_result
739         | SSP_SAME_DELIMITER_LARGER_BEFORE
740       ))
741     fi
742     if ((LFBFL_i_result & SSP_POSITION_NOT_DECREASING_AFTER))\
743     && ((LFBFL_i_result & SSP_POSITION_NOT_DECREASING_BEFORE)); then
744       LFBFL_i_result=$((
745         LFBFL_i_result
746         | SSP_POSITION_NOT_DECREASING
747       ))
748     fi
749     if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
750     && ((LFBFL_i_result & SSP_EQUAL_AFTER_OR_BEFORE)); then
751       LFBFL_i_result=$((
752         LFBFL_i_result
753         | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
754       ))
755     fi
756     if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
757     && ((LFBFL_i_result & SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE)); then
758       LFBFL_i_result=$((LFBFL_i_all_flags))
759       break
```

```
760     fi
761     if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
762     && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
763       LFBFL_i_result=$((
764         LFBFL_i_result
765         | SSP_ALWAYS_LARGER_AFTER
766       ))
767     fi
768     if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
769     && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
770       LFBFL_i_result=$((
771         LFBFL_i_result
772         | SSP_ALWAYS_LARGER_BEFORE
773       ))
774     fi
775     if ((LFBFL_i_result & SSP_POSITION_UNIFORM_AFTER))\
776     && ((LFBFL_i_result & SSP_POSITION_UNIFORM_BEFORE)); then
777       LFBFL_i_result=$((
778         LFBFL_i_result
779         | SSP_POSITION_UNIFORM
780       ))
781     fi
782     if ((LFBFL_i_result & SSP_POSITION_UNIFORM_AFTER))\
783     && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
784       LFBFL_i_result=$((
785         LFBFL_i_result
786         | SSP_ALWAYS_LARGER_AFTER
787       ))
788     fi
789     if ((LFBFL_i_result & SSP_POSITION_UNIFORM_BEFORE))\
790     && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
791       LFBFL_i_result=$((
792         LFBFL_i_result
793         | SSP_ALWAYS_LARGER_BEFORE
794       ))
795     fi
796     if ((LFBFL_i_result & SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE))\
797     && ((
798       LFBFL_i_result & (
799         SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
800         | SSP_SAME_POSITION_LARGER_AFTER
801         | SSP_SAME_POSITION_LARGER_BEFORE
802       )
803     )); then
804       LFBFL_i_result=$((LFBFL_i_all_flags))
805       break
```

```
806       fi
807       if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER))\
808       && ((
809         LFBFL_i_result & (
810           SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
811           | SSP_SAME_POSITION_LARGER_BEFORE
812         )
813       )); then
814         LFBFL_i_result=$((LFBFL_i_all_flags))
815         break
816       fi
817       if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER))\
818       && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
819         LFBFL_i_result=$((
820           LFBFL_i_result
821           | SSP_ALWAYS_LARGER_AFTER
822         ))
823       fi
824       if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE))\
825       && ((
826         LFBFL_i_result & (
827           SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
828           |  SSP_SAME_POSITION_LARGER_AFTER
829         )
830       )); then
831         LFBFL_i_result=$((LFBFL_i_all_flags))
832         break
833       fi
834       if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE))\
835       && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
836         LFBFL_i_result=$((
837           LFBFL_i_result
838           | SSP_ALWAYS_LARGER_BEFORE
839         ))
840       fi
841       if ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER))\
842       && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
843         LFBFL_i_result=$((
844           LFBFL_i_result
845           | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
846         ))
847       fi
848     done
849     declare -gi i_split_score_properties_logical_closure_result=$((
850       LFBFL_i_result
851     ))
```

```
852 }
853
854 split_score_simple(){
855   # $1=after_before
856   # $2=delimiters_strings_domain concatenated characters/delimiters
857   # $3=delimiter_string a single character
858   # $4=is_cut_after
859   declare -gi i_split_score_result
860   if [[ "$2" != *$3* ]]; then
861     i_split_score_result=0
862     return
863   fi
864   if [[ "$4" == "$1" ]]; then
865     i_split_score_result=2
866     return
867   fi
868   i_split_score_result=1
869 }
870
871 get_split_score_simple(){
872   # $1=after_before
873   # $2=max_length
874   # $3=delimiters_strings_domain
875   declare -g get_split_score_result="split_score_simple $1 '$3'"
876   declare -gi i_get_split_score_result2
877   # i_get_split_score_result2=$((11-$1*4))
878   # if [[ $1 -eq 1 ]]; then
879   #   i_get_split_score_result2=7
880   # else
881   #   i_get_split_score_result2=11
882   # fi
883   i_get_split_score_result2=$((
884     SSP_DELIMITER_UNIFORM
885     + SSP_POSITION_UNIFORM
886     + SSP_LARGER_BEFORE
887     + (
888       $1 * (
889         SSP_LARGER_AFTER
890         - SSP_LARGER_BEFORE
891       )
892     )
893   ))
894   # shellcheck disable=SC2248
895   split_score_properties_logical_closure ${i_get_split_score_result2}
896   i_get_split_score_result2=$((
897     i_split_score_properties_logical_closure_result
```

```
898   ))
899 }
900
901 get_split_score_exec(){
902   declare -g SPLIT_SCORE_EXEC="./build_and_checks_dependencies"
903   SPLIT_SCORE_EXEC+="/split_score.exec.php"
904 }
905
906 split_score(){
907   # $1=after_before
908   # $2=max_length
909   # $3=delimiters_strings_domain concatenated characters/delimiters
910   # $4=delimiter_string a single character
911   # $5=cut_position
912   # $6=is_cut_after
913   get_split_score_exec
914   declare -r LFBFL_command="${SPLIT_SCORE_EXEC} $1 $2 '$3' '$4' $5 $6"
915   declare -gi i_split_score_result
916   i_split_score_result=$(eval "${LFBFL_command}")
917 }
918
919 # This function is currently not called,
920 # hence we need this shellcheck disable.
921 # shellcheck disable=SC2034
922 get_split_score(){
923   # $1=after_before
924   # $1=max_length
925   # $2=delimiters_strings_domain
926   declare -g get_split_score_result="split_score $1 $2 '$3'"
927   declare -gi i_get_split_score_result2
928   # i_get_split_score_result2=$((9-$1*4))
929   i_get_split_score_result2=$((
930     SSP_DELIMITER_UNIFORM
931     + SSP_LARGER_BEFORE
932     + (
933       $1 * (
934         SSP_LARGER_AFTER
935         - SSP_LARGER_BEFORE
936       )
937     )
938   ))
939   # shellcheck disable=SC2248
940   split_score_properties_logical_closure ${i_get_split_score_result2}
941   i_get_split_score_result2=$((
942     i_split_score_properties_logical_closure_result
943   ))
```

```
}

split_line_at_most(){
  # $1=line
  # $2=max_length of beginning result string
  # $3=split_score_command
  #   negative score means do not consider as a possible split
  # $4=split_score_command_properties :
  #   These properties are flags.
  #   If someday some property is not a flag,
  #   then create split_score_command_properties2.
  #   Old values to get a simple idea without reading everything at the
  #   beginning of file:
  #   - null or 0 no property;
  #   - 1 split score doesn't depend on which delimiter matches;
  #   - 2 split score doesn't depend on the current position;
  #   - 4 any split score after is always larger
  #     than any split score before;
  #   - 8 any split score before is always larger
  #     than any split score after;
  # $5=forbidden_previous_character
  # $6=split_score_command_delimiter_strings_domain
  #   optimisation if small domain
  #   - this argument can be "null", then we will use $3 only instead,
  #     in that case, only mono-character strings are considered,
  #   - this argument can be an array of characters,
  #     optimize accordingly,
  #   - this argument can be an array of strings,
  #     optimize accordingly,
  #   - this argument can be an array of regexps,
  #     optimize accordingly.
  #     Not sure an array of regexps makes more sense than a single
  #     one. Maybe they could be both cases with corresponding
  #     optimizations. Same for a single character or substring.
  #
  # Note that we're always looking for the last position to split
  # among the positions with maximum score;
  # if some position gets a score from 2 delimiters,
  # its score is always the maximum of the two corresponding scores.
  declare -g split_line_at_most_result_start
  declare -g split_line_at_most_result_end
  declare -A LFBFL_positions
  # printf "%s %s %s %s\n" "$1" "$2" "$3" "$4"
  # For my use case in Bash scripts, I will need only an array of
  # characters. See get_split_score().
  if [[ -n "$6" ]]; then
```

```
990       printf $'split_line_at_most() $6 NOT IMPLEMENTED YET\n'
991     fi
992     if [[ $2 -ge ${#1} ]]; then
993       split_line_at_most_result_start=$1
994       split_line_at_most_result_end=""
995       return
996     fi
997     # We know that ${#1} > $2, hence ${#1} - 1 >= $2
998     # Edge cases:
999     # There is always at least one character that is overlength,
1000    # but we need to look only at the score before that character that is
1001    # overlength, the score after that character is invalid,
1002    # whatever ${#1} - 1 = $2, or ${#1} - 1 > $2.
1003    # That first character that is overlength has index $2.
1004    # The cut position after that character has index $2 + 1.
1005    # Looping from character position going from 0 to $2, we need to drop
1006    # score before character 0 and drop score after character $2.
1007    # We do this with forbidden score -1.
1008    declare -ir LFBFL_i_max=$(($2))
1009    declare -i LFBFL_i
1010    declare -i LFBFL_j
1011    local LFBFL_current_char
1012
1013    # At the beginning, each position has score 0.
1014    LFBFL_positions["0"]="-1"
1015    for ((LFBFL_i = 1; LFBFL_i <= LFBFL_i_max; ++LFBFL_i)) do
1016      LFBFL_positions["${LFBFL_i}"]="0"
1017    done
1018    LFBFL_positions["${LFBFL_i}"]="-1"
1019
1020    # If we have a forbidden previous character,
1021    # we mark the forbidden positions with -1.
1022    if [[ -n "$5" ]]; then
1023      for ((LFBFL_i = 0; LFBFL_i < LFBFL_i_max;)) do
1024        LFBFL_current_char="${1:${LFBFL_i}:1}"
1025        ((++LFBFL_i))
1026        if [[ "${LFBFL_current_char}" == "$5" ]]; then
1027          LFBFL_positions["${LFBFL_i}"]="-1"
1028        fi
1029      done
1030    fi
1031
1032    declare -r LFBFL_sort_command="sort --numeric-sort"
1033
1034    if (($4 & SSP_DELIMITER_UNIFORM))\
1035    && (($4 & SSP_LARGER_AFTER))\
```

```
1036    && (($4 & SSP_POSITION_NOT_DECREASING_AFTER)); then
1037      for ((LFBFL_i = LFBFL_i_max; LFBFL_i >= 0; --LFBFL_i)) do
1038        LFBFL_current_char="${1:${LFBFL_i}:1}"
1039        eval "$3 '${LFBFL_current_char}' 1"
1040        # needs j=i+1 printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_j}"
1041        if [[ i_split_score_result -ge 1 ]]; then
1042          LFBFL_j=$((LFBFL_i + 1))
1043          # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_j}"
1044          if [[ ${LFBFL_positions["${LFBFL_j}"]} != "-1" ]]; then
1045            LFBFL_positions["${LFBFL_j}"]=$(
1046              max "${LFBFL_sort_command}"\
1047                "${LFBFL_positions["${LFBFL_j}"]}"\
1048                "${i_split_score_result}"
1049            )
1050            break
1051          fi
1052        fi
1053        eval "$3 '${LFBFL_current_char}' 0"
1054        # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1055        if [[ i_split_score_result -ge 1 ]]; then
1056          # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1057          if [[ ${LFBFL_positions["${LFBFL_i}"]} != "-1" ]]; then
1058            LFBFL_positions["${LFBFL_i}"]=$(
1059              max "${LFBFL_sort_command}"\
1060                "${LFBFL_positions["${LFBFL_i}"]}"\
1061                "${i_split_score_result}"
1062            )
1063          fi
1064        fi
1065      done
1066    elif (($4 & SSP_DELIMITER_UNIFORM))\
1067    && (($4 & SSP_LARGER_BEFORE))\
1068    && (($4 & SSP_POSITION_NOT_DECREASING_BEFORE)); then
1069      for ((LFBFL_i = LFBFL_i_max; LFBFL_i >= 0; --LFBFL_i)) do
1070        LFBFL_current_char="${1:${LFBFL_i}:1}"
1071        # We still need to break in second if, after computing score after,
1072        # because SSP_LARGER_BEFORE implies large inequality instead of
1073        # strict inequality.
1074        # And since the algorithm is to take the largest position with
1075        # maximum score, there is a possibility that we'll break on a found
1076        # score before but for splitting on a score after.
1077        eval "$3 '${LFBFL_current_char}' 1"
1078        # needs j=i+1 printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_j}"
1079        if [[ i_split_score_result -ge 1 ]]; then
1080          LFBFL_j=$((LFBFL_i + 1))
1081          # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_j}"
```

```
1082          if [[ ${LFBFL_positions["${LFBFL_j}"]} != "-1" ]]; then
1083            LFBFL_positions["${LFBFL_j}"]=$(
1084              max "${LFBFL_sort_command}"\
1085                "${LFBFL_positions["${LFBFL_j}"]}"\
1086                "${i_split_score_result}"
1087            )
1088          fi
1089        fi
1090        eval "$3 '${LFBFL_current_char}' 0"
1091        # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1092        if [[ i_split_score_result -ge 1 ]]; then
1093          # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1094          if [[ ${LFBFL_positions["${LFBFL_i}"]} != "-1" ]]; then
1095            LFBFL_positions["${LFBFL_i}"]=$(
1096              max "${LFBFL_sort_command}"\
1097                "${LFBFL_positions["${LFBFL_i}"]}"\
1098                "${i_split_score_result}"
1099            )
1100            break
1101          fi
1102        fi
1103      done
1104    else
1105      for ((LFBFL_i = 0; LFBFL_i <= LFBFL_i_max;)) do
1106        LFBFL_current_char="${1:${LFBFL_i}:1}"
1107        eval "$3 '${LFBFL_current_char}' ${LFBFL_i} 0"
1108        # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1109        if [[ i_split_score_result -ge 1 ]]; then
1110          # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1111          if [[ ${LFBFL_positions["${LFBFL_i}"]} != "-1" ]]; then
1112            LFBFL_positions["${LFBFL_i}"]=$(
1113              max "${LFBFL_sort_command}"\
1114                "${LFBFL_positions["${LFBFL_i}"]}"\
1115                "${i_split_score_result}"
1116            )
1117          fi
1118        fi
1119        ((++LFBFL_i))
1120        eval "$3 '${LFBFL_current_char}' ${LFBFL_i} 1"
1121        # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1122        if [[ i_split_score_result -ge 1 ]]; then
1123          # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1124          if [[ ${LFBFL_positions["${LFBFL_i}"]} != "-1" ]]; then
1125            LFBFL_positions["${LFBFL_i}"]=$(
1126              max "${LFBFL_sort_command}"\
1127                "${LFBFL_positions["${LFBFL_i}"]}"\
```

```
1128                "${i_split_score_result}"
1129            )
1130          fi
1131        fi
1132      done
1133    fi
1134    declare -ir LFBFL_i_max_score=$(
1135      max 'sort --numeric-sort' "${LFBFL_positions[@]}"
1136    )
1137    local LFBFL_value
1138    declare -i LFBFL_i_best_position
1139    for ((LFBFL_i = LFBFL_i_max; LFBFL_i >= 0; --LFBFL_i)) do
1140      if [[ LFBFL_i -eq 0 ]]; then
1141        printf $'/!\\split_line_at_most bug: no split position found./!\\\n'
1142      fi
1143      local LFBFL_value=${LFBFL_positions[${LFBFL_i}]}
1144      # printf "%s %s\n" "${LFBFL_i} ${LFBFL_value}"
1145      if [[ "${LFBFL_value}" == "${LFBFL_i_max_score}" ]]; then
1146        LFBFL_i_best_position=$((LFBFL_i))
1147        break
1148      fi
1149    done
1150    # shellcheck disable=SC2248
1151    split_line_at "$1" ${LFBFL_i_best_position}
1152    split_line_at_most_result_start="${split_line_at_result_beginning}"
1153    split_line_at_most_result_end="${split_line_at_result_end}"
1154    # printf "%s\n" "$split_line_at_most_result_start"
1155    # printf "%s\n" "$split_line_at_most_result_end"
1156  }
1157
1158  split_last_line(){
1159    # $1=new_lines
1160    # $2=prefix
1161    # $3=max_length
1162    # $4=suffix
1163    # $5=split_score_command
1164    # $6=split_score_command_properties
1165    # $7=forbidden_previous_character ('\' usually)
1166    # Options:
1167    #   --verbose
1168    # In general, a prefix/suffix can be a required line prefix/suffix
1169    # for all final lines like a comment prefix,
1170    # or it can be a continuation line prefix/suffix applied only
1171    # when splitting a line in 2 lines,
1172    # or it can be a prefix/suffix for the string made of all the lines.
1173    # Clearly, there is no gain to couple handling a prefix/suffix for
```

```
  # the string made of all the lines in this function.
  # Since we add a prefix/suffix when splitting,
  # it is easy to see that we do not need to distinguish between
  # required line prefix/suffix and continuation line prefix/suffix.
  # The $2 and $4 arguments are given as concatenation of the
  # required and continuation line prefixes/suffixes as needed by the
  # user.
  # Thus lines arguments should be given already prefixed
  # and suffixed.
  # This is logic, since it would be cumbersome to handle prefixing
  # or not already splitted lines in this function
  # (see repeated_split_last_line).
  # The given max_length is the goal.
  # This function adapts the effective max position for the split by
  # taking into account the length of the suffix.
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap

  enhanced_set_shell_option pipefail --trap-unset

  declare -g split_last_line_result="$1"
  declare -ir LFBFL_i_overlength=$(($3 + 1))
  declare -ir LFBFL_i_length2=$(($3 - ${#4}))
  # shellcheck disable=SC2248
  get_overlength_regexp ${LFBFL_i_overlength}

  # Testing if some line is too long.
  printf "%s" "$1"\
    | grep -q "${get_overlength_regexp_result}";
  if [[ ${PIPESTATUS[1]} -eq 1 ]]; then
    return
  fi

  declare -r LFBFL_start=$(
    printf "%s" "$1"\
    | head --lines=-1
  )
  # printf "start: %s\n" "${LFBFL_start}"
  declare -r LFBFL_last_line=$(
    printf "%s" "$1"\
    | tail --lines=1
  )
  # printf "last_line: %s\n" "${LFBFL_last_line}"
```

```
  split_last_line_result=""
  if [[ -n "${LFBFL_start}" ]]; then
    split_last_line_result="${LFBFL_start}"
    # Last line-return of LFBFL_start seems dropped by $() and not ="${}".
    # printf "%s %s\n" "${#split_last_line_result}"
    #   "${split_last_line_result}"
    split_last_line_result+=$'\n'
    # something="${split_last_line_result}"
    # printf "%s %s\n" "${#something}" "${something}"
  fi
  if [[ -n "$5" ]]; then
    # shellcheck disable=SC2248
    split_line_at_most "${LFBFL_last_line}" ${LFBFL_i_length2}\
      "$5" "$6" "$7"
    split_last_line_result+="${split_line_at_most_result_start}"
    split_last_line_result+="$4"
    split_last_line_result+=$'\n'
    split_last_line_result+="$2${split_line_at_most_result_end}"
  else
    split_last_line_result+="${LFBFL_last_line:0:${LFBFL_i_length2}}"
    split_last_line_result+="$4"
    split_last_line_result+=$'\n'
    split_last_line_result+="$2${LFBFL_last_line:${LFBFL_i_length2}}"
  fi
  # printf "result: %s\n" "${split_last_line_result}"
}

repeated_split_last_line(){
  # See split_last_line for the first arguments.
  # $8=final_suffix to handle when the end of the original string
  #    is the suffix.
  declare -g repeated_split_last_line_result="$1"
  local LFBFL_result="$1x"
  until
    [[ "${LFBFL_result}" == "${repeated_split_last_line_result}" ]];
  do
    split_last_line "${repeated_split_last_line_result}" "$2" "$3"\
      "$4" "$5" "$6" "$7"
    LFBFL_result="${repeated_split_last_line_result}"
    repeated_split_last_line_result="${split_last_line_result}"
    # printf "%s\n" "${repeated_split_last_line_result}"
  done
  if [[ -n "$4" && -n "$8" ]]; then
    declare -ir LFBFL_i_offset=$((-${#4}))
    local LFBFL_lines_end
    LFBFL_lines_end="${repeated_split_last_line_result: ${LFBFL_i_offset}}"
```

```
    if [[ "${LFBFL_lines_end}" == "$4" ]]; then
      repeated_split_last_line_result+="$8"
    fi
  fi
}
```

3.31 build_and_checks_dependencies/too_long_code_lines.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# This file was renamed from "too_long_code_lines.sh"
# to "too_long_code_lines.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

get_overlength_regexp(){
  declare -g get_overlength_regexp_result=".\{$1\}"
  # declare -g get_overlength_regexp_result="^.\{$1\}"
  # declare -g get_overlength_regexp_result=".\{$1\}\$"
```

```
39    # First one seems to be faster by a few percents when testing like this:
40    # time for ((i = 0; i < 1000; ++i)); do grep '.{70}' COPYING; done
41    # time for ((i = 0; i < 1000; ++i)); do grep '^.{70}' COPYING; done
42    # time for ((i = 0; i < 1000; ++i)); do grep '.{70}$' COPYING; done
43  }
44
45  too_long_code_lines(){
46    # Options:
47    #   --verbose
48    #   --work-directory=""
49    #   --max-line-length
50    declare -i LFBFL_i_verbose=0
51    get_verbose_option "$@"
52    local LFBFL_work_directory=""
53    get_work_directory_option "$@"
54    declare -a LFBFL_return_traps_stack
55    local LFBFL_previous_return_trap
56    init_return_trap
57    pushd_to_work_directory --trap-popd
58    can_continue_after_enhanced_pushd || return 1
59
60    local LFBFL_arg
61    declare -i LFBFL_i_max_line_length=70
62    for LFBFL_arg in "$@"; do
63      if [[ "${LFBFL_arg}" == --max-line-length=* ]]; then
64        LFBFL_arg=${LFBFL_arg#--max-line-length=}
65        LFBFL_i_max_line_length=$((LFBFL_arg))
66        break
67      fi
68    done
69
70    declare -ir LFBFL_i_overlength=$((LFBFL_i_max_line_length+1))
71    # shellcheck disable=SC2248
72    get_overlength_regexp ${LFBFL_i_overlength}
73
74    enhanced_set_shell_option pipefail --trap-unset
75
76    get_COMMON_TEXT_FILES_GLOB_PATTERNS
77    local LFBFL_pattern
78    local LFBFL_long_line
79    local LFBFL_file_path
80    local LFBFL_line
81    local LFBFL_extension
82    local LFBFL_base_name
83    local LFBFL_s_long_lines
84    declare -a LFBFL_arr_long_lines
```

```
for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    printf "too_long_code_lines: Iterating on pattern: %s.\n"\
      "${LFBFL_pattern}"
  fi
  # shellcheck disable=SC2248
  LFBFL_s_long_lines=$(
    find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
    | relevant_find\
    | not_license_find\
    | xargs grep --line-number --with-filename\
      "${get_overlength_regexp_result}"
  )
  if [[ -z "${LFBFL_s_long_lines}" ]]; then
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "too_long_code_lines: No file found with long lines.\n"
    continue
  fi
  mapfile -t LFBFL_arr_long_lines <<< "${LFBFL_s_long_lines}"
  for LFBFL_long_line in "${LFBFL_arr_long_lines[@]}"; do
    LFBFL_file_path=${LFBFL_long_line%%:*} # Drop long ':*' suffix
    # Remove file_path and line_number
    LFBFL_line=${LFBFL_long_line#*:} # Drop short '*:' prefix
    LFBFL_line=${LFBFL_line#*:} # Drop short '*:' prefix
    # LFBFL_line="${LFBFL_line/%\\/}" # Drop/Replace '\' suffix by ''
    LFBFL_extension=${LFBFL_file_path##*.} # Drop long '*.' prefix
    LFBFL_base_name=${LFBFL_file_path%.*} # Drop short '.*' suffix
    if [[ "${LFBFL_extension}" == "html" ]]; then
      if [[ -f "${LFBFL_base_name}.md" ]]; then
        continue
      fi
      if grep --quiet --fixed-strings -- "${LFBFL_line}"\
        "./build_and_checks_variables/temp/files_listing.html.sub"
      then
        continue
      fi
      printf "%s\n" "${LFBFL_long_line}"
    elif [[ "${LFBFL_extension}" == "md" ]]; then
      if [[ ! -f "${LFBFL_base_name}.md.tpl" ]]; then
        printf "%s\n" "${LFBFL_long_line}"
      fi
    else
      printf "%s\n" "${LFBFL_long_line}"
    fi
  done
done
```

```
}
```

3.32 build_and_checks_dependencies/update_common_build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# of the GNU General Public License
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#
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# This file was renamed from "update_common_build_and_checks.sh"
# to "update_common_build_and_checks.exec.sh".

source "overwrite_if_not_equal.libr.sh"

update_common_build_and_checks(){
  declare -i LFBFL_i_verbose=0
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_i_verbose=1
  fi
  readonly LFBFL_i_verbose

  if [[ ! -o pipefail ]]; then
    [[ LFBFL_i_verbose -eq 1 ]] && printf "pipefail option activated\n"
    set -o pipefail
    trap 'set +o pipefail' RETURN
  fi

```

```
43    declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
44    cp "./${LFBFL_common_file_name}.tpl"\
45      "./${LFBFL_common_file_name}.temp"
46
47    declare -ar LFBFL_files_names=(
48      "build_dependencies_notes.exec.php"
49      "build_dependencies_notes.libr.php"
50      "build_md_from_printable_md.exec.sh"
51      "check_shell_scripts_beginnings.libr.sh"
52      "check_shell_scripts_indentation.libr.sh"
53      "check_URLs.libr.sh"
54      "common_options.libr.sh"
55      "comparisons.libr.sh"
56      "create_PDF.exec.sh"
57      "generate_from_template.libr.sh"
58      "get_common_text_glob_patterns.libr.sh"
59      "grammar_and_spelling_check.libr.sh"
60      "licenses_templates/build_licenses_templates.exec.sh"
61      "licenses_templates/license_file_header_AGPLv3+.tpl"
62      "licenses_templates/license_file_header_GPLv3+.tpl"
63      "licenses_templates/license_file_header_LGPLv3+.tpl"
64      "lines_counts.libr.sh"
65      "lines_filters.libr.sh"
66      "lines_maps.libr.sh"
67      "overwrite_if_not_equal.libr.sh"
68      "python_black_complement.libr.sh"
69      "python_isort_complement.libr.sh"
70      "repository_data.libr.sh"
71      "shell_checks_complement.libr.sh"
72      "split_score.exec.php"
73      "split_score.libr.php"
74      "strings_functions.libr.sh"
75      "too_long_code_lines.libr.sh"
76      "update_or_check_files_names_listing.exec.sh"
77    )
78
79    local LFBFL_file_name
80    local LFBFL_file_sha512
81    local LFBFL_file_sha512_all
82    local LFBFL_base_file_name
83    for LFBFL_file_name in "${LFBFL_files_names[@]}"; do
84      LFBFL_file_sha512=$(
85        sha512sum "./${LFBFL_file_name}"\
86        | cut -f1 -d' '
87      )
88      LFBFL_file_sha512_all="LFBFL_correct_sha512="
```

```
    LFBFL_file_sha512_all+="'${LFBFL_file_sha512:0:45}'\n"
    LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
    LFBFL_file_sha512_all+="'${LFBFL_file_sha512:45:44}'\n"
    LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
    LFBFL_file_sha512_all+="'${LFBFL_file_sha512:89}'"
    LFBFL_base_file_name=$(basename "${LFBFL_file_name}")
    sed -i\
      "s|@sha512_${LFBFL_base_file_name}@|${LFBFL_file_sha512_all}|g"\
      "./${LFBFL_common_file_name}.temp"
  done

  overwrite_if_not_equal "./${LFBFL_common_file_name}"\
    "./${LFBFL_common_file_name}.temp"

  declare -r LFBFL_main_file_name="build_and_checks.exec.sh"
  cp "./${LFBFL_main_file_name}.tpl" "../${LFBFL_main_file_name}.temp"

  LFBFL_file_sha512=$(
    sha512sum "./${LFBFL_common_file_name}"\
    | cut -f1 -d' '
  )
  LFBFL_file_sha512_all="local LFBFL_correct_sha512\n"
  LFBFL_file_sha512_all+="  LFBFL_correct_sha512="
  LFBFL_file_sha512_all+="'${LFBFL_file_sha512:0:45}'\n"
  LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
  LFBFL_file_sha512_all+="'${LFBFL_file_sha512:45:44}'\n"
  LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
  LFBFL_file_sha512_all+="'${LFBFL_file_sha512:89}'"
  sed -i\
    "s|@sha512_${LFBFL_common_file_name}@|${LFBFL_file_sha512_all}|g"\
    "../${LFBFL_main_file_name}.temp"

  overwrite_if_not_equal "../${LFBFL_main_file_name}"\
    "../${LFBFL_main_file_name}.temp"
}

update_common_build_and_checks "$@"
```

3.33 build_and_checks_dependencies/update_or_check_files_names_listing.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
```

```
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# either version 3 of the License,
# or (at your option) any later version.
#
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# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet

# 2025-12-30T21:33+01:00: This file was moved from
# "./build_and_checks_dependencies/listings"
# to
# "./build_and_checks_dependencies".
# This file was renamed from "update_or_check_files_names_listing.sh"
# to "update_or_check_files_names_listing.exec.sh".
# This file was renamed from "update_or_check_file_names_listing.sh"
# to "update_or_check_files_names_listing.sh"?

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=strings_functions.libr.sh
source "./${LFBFL_subdir}/strings_functions.libr.sh"

update_or_check_files_names_listing(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
```

```
53    pushd_to_work_directory --trap-popd
54    can_continue_after_enhanced_pushd || return 1
55
56    declare -i LFBFL_i_write=0
57    if [[ "$*" == *--write* ]]; then
58      LFBFL_i_write=1
59    fi
60    readonly LFBFL_i_write
61
62    declare -i LFBFL_i_append=0
63    if [[ "$*" == *--append* ]]; then
64      LFBFL_i_append=1
65    fi
66    readonly LFBFL_i_append
67
68    enhanced_set_shell_option pipefail --trap-unset
69
70    LFBFL_subdir2="build_and_checks_variables"
71    local LFBFL_listing="./${LFBFL_subdir2}/files_names_listing.txt"
72    readonly LFBFL_listing
73    local LFBFL_data_file_name="./${LFBFL_subdir2}/repository_data.txt"
74    readonly LFBFL_data_file_name
75
76    declare -i LFBFL_i_max_line_length
77    grep_variable "${LFBFL_data_file_name}" max_line_length\
78      --result-variable-prefix="LFBFL_i_"
79
80    if [[ LFBFL_i_write -eq 1 ]]; then
81      : > "${LFBFL_listing}"
82    fi
83    # Remove line returns that are here to keep lines short.
84    sed -Ez 's/(.)\\\n/\1/Mg' "${LFBFL_listing}"\
85      > "${LFBFL_listing}.temp"
86    # shopt -s dotglob was needed at some point but I don't see why now.
87    # shellcheck disable=SC2248
88    get_split_score_simple 1 ${LFBFL_i_max_line_length} /
89    declare -r LFBFL_split_score_command="${get_split_score_result}"
90    declare -i LFBFL_i_split_score_command_properties
91    LFBFL_i_split_score_command_properties=${i_get_split_score_result2}
92    readonly LFBFL_i_split_score_command_properties
93    local LFBFL_suffix=\\ # instead of '\' to avoid shellcheck SC1003
94    readonly LFBFL_suffix
95    # There is always a $'\n' final suffix from >>.
96    # But when origin string ends with the suffix we must add something.
97    declare -r LFBFL_final_suffix=$'\n\\'
98
```

```
  local LFBFL_s_files_paths
  LFBFL_s_files_paths=$(
    find . -type f -printf '%P\n'\
    | relevant_find\
    | sort
  )
  # printf "%s\n" "${LFBFL_s_files_paths}"
  declare -a LFBFL_arr_files_paths
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf\
        "update_or_check_files_names_listing: Some files to process.\n"
    fi
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  fi
  # printf "%s\n" "${LFBFL_arr_files_paths[@]}"

  local LFBFL_file_path
  local LFBFL_base_file_name
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    # if [[ LFBFL_i_verbose -eq 1 ]]; then
    #   printf\
    #     "update_or_check_files_names_listing: Maybe ignored file: %s\n"\
    #     "${LFBFL_file_path}"
    # fi
    git check-ignore -q "${LFBFL_file_path}" && continue
    LFBFL_base_file_name=$(basename "${LFBFL_file_path}")
    [[ "${LFBFL_base_file_name}" != files_names_listing.txt* ]]\
      || continue
    if [[ "${LFBFL_base_file_name}" == *.md ]]; then
      if [[ -f "${LFBFL_file_path}.tpl" ]]; then
        continue
      fi
    fi
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "update_or_check_files_names_listing: Non-ignored file: %s\n"\
        "${LFBFL_file_path}"
    fi
    # shellcheck disable=SC2248
    repeated_split_last_line "${LFBFL_file_path}"\
      ""\
      ${LFBFL_i_max_line_length}\
      "${LFBFL_suffix}"\
      "${LFBFL_split_score_command}"\
      ${LFBFL_i_split_score_command_properties}\
      ""\
```

```
145         "${LFBFL_final_suffix}"
146       if [[ LFBFL_i_write -eq 1 ]]; then
147         # shellcheck disable=SC2001
148         printf "%s\n" "${repeated_split_last_line_result}"\
149           >> "${LFBFL_listing}"
150         continue
151       fi
152       if grep_fixed_string_with_anchor "${LFBFL_listing}.temp"\
153           "${LFBFL_file_path}" --quiet --enforce-line-ends-with-fixed-string;
154       then
155         :
156       else
157         printf "The file %s is not listed in %s.\n"\
158           "${LFBFL_file_path}"\
159           "${LFBFL_listing}"
160         if [[ LFBFL_i_append -eq 1 ]]; then
161           # shellcheck disable=SC2001
162           printf "%s\n" "${repeated_split_last_line_result}"\
163             >> "${LFBFL_listing}"
164         fi
165       fi
166     done
167
168     LFBFL_s_files_paths=$(cat "${LFBFL_listing}.temp")
169     if [[ -n "${LFBFL_s_files_paths}" ]]; then
170       mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
171     fi
172
173     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
174       [[ "${LFBFL_file_path}" != '//'* ]] || continue
175       if [[ ! -f "${LFBFL_file_path}" ]]; then
176         printf "The non-file %s is listed in %s.\n"\
177           "${LFBFL_file_path}"\
178           "${LFBFL_listing}"
179       fi
180     done
181
182     rm "${LFBFL_listing}.temp"
183
184     if [[ LFBFL_i_verbose -eq 1 ]]; then
185       cat "${LFBFL_listing}"
186     fi
187   }
188
189   update_or_check_files_names_listing "$@"
```

3.34 build_and_checks_variables/.gitignore

```
temp/
upgrade_venvs_ts
```

3.35 build_and_checks_variables/bandit.ini

```
# FILE: .bandit
[bandit]
```

3.36 build_and_checks_variables/bandit_baseline.json

```
{"results":[]}
```

3.37 build_and_checks_variables/black.toml

```
[tool.black]
line-length = 75
```

3.38 build_and_checks_variables/dependencies_data.json

```
[
  { "dependency_name": "Bash", "URLs": [
      "https://www.gnu.org/software/bash/",
      "https://tiswww.case.edu/php/chet/bash/bashtop.html",
      "https://savannah.gnu.org/projects/bash/"
  ] },
  { "dependency_name": "GNU coreutils", "URLs": [
      "https://www.gnu.org/software/coreutils/",
      "https://savannah.gnu.org/projects/coreutils/"
  ] },
  { "dependency_name": "uutils coreutils", "URLs": [
      "https://uutils.github.io/coreutils/",
      "https://github.com/uutils/coreutils"
  ] },
  { "dependency_name": "diff (GNU diffutils)", "URLs": [
      "https://www.gnu.org/software/diffutils/",
      "https://savannah.gnu.org/projects/diffutils/"
  ] },
  { "dependency_name": "find, xargs (GNU findutils)", "URLs": [
      "https://www.gnu.org/software/findutils/",
      "https://savannah.gnu.org/projects/findutils/"
  ] },
  { "dependency_name": "grep (GNU grep)", "URLs": [
      "https://www.gnu.org/software/grep/",
      "https://savannah.gnu.org/projects/grep/"
```

```
26      ] },
27      { "dependency_name": "sed (GNU sed)", "URLs": [
28          "https://www.gnu.org/software/sed/",
29          "https://savannah.gnu.org/projects/sed/"
30      ] },
31      { "dependency_name": "ShellCheck", "URLs": [
32          "https://www.shellcheck.net/",
33          "https://github.com/koalaman/shellcheck"
34      ] },
35      { "dependency_name": "GNU Wget", "URLs": [
36          "https://www.gnu.org/software/wget/",
37          "https://savannah.gnu.org/projects/wget/"
38      ] },
39      { "dependency_name": "more of util-linux", "URLs": [
40          "https://www.kernel.org/pub/linux/utils/util-linux/",
41          "https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/",
42          {
43            "URL": "https://github.com/util-linux/util-linux",
44            "comment": "official?"
45          }
46      ] },
47      { "dependency_name": "pandoc", "URLs": [
48          "https://pandoc.org/",
49          "https://github.com/jgm/pandoc"
50      ] },
51      { "dependency_name": "pdftex/pdflatex", "URLs": [
52          "https://www.tug.org/applications/pdftex/",
53          "https://tug.org/texlive/svn/"
54      ] },
55      { "dependency_name": "tree", "URLs": [
56          "https://oldmanprogrammer.net/source.php?dir=projects/tree",
57          "https://gitlab.com/OldManProgrammer/unix-tree",
58          "https://github.com/Old-Man-Programmer/tree"
59      ] },
60      { "dependency_name": "git", "URLs": [
61          "https://git-scm.com/about",
62          "https://github.com/git/git"
63      ] },
64      { "dependency_name": "pylint", "URLs": [
65          "https://www.pylint.org/",
66          "https://pylint.readthedocs.io/en/stable/",
67          "https://github.com/pylint-dev/pylint",
68          "https://pypi.org/project/pylint/"
69      ] },
70      { "dependency_name": "notify-send", "URLs": [
71          "https://gitlab.gnome.org/GNOME/libnotify/",
```

```
72          "https://gnome.pages.gitlab.gnome.org/libnotify/"
73      ] },
74      { "dependency_name": "black", "URLs": [
75          "https://github.com/psf/black",
76          "https://black.readthedocs.io/en/stable/",
77          "https://pypi.org/project/black/"
78      ] },
79      { "dependency_name": "isort", "URLs": [
80          "https://pycqa.github.io/isort/",
81          "https://github.com/pycqa/isort/",
82          "https://pypi.org/project/isort/"
83      ] },
84      { "dependency_name": "mypy", "URLs": [
85          "https://mypy-lang.org/",
86          "https://mypy.readthedocs.io/en/stable/",
87          "https://github.com/python/mypy",
88          "https://pypi.org/project/mypy/"
89      ] },
90      { "dependency_name": "phpdoc (phpDocumentor)", "URLs": [
91          "https://phpdoc.org/",
92          "https://docs.phpdoc.org/",
93          "https://github.com/phpdocumentor/phpdocumentor"
94      ] },
95      { "dependency_name": "pip", "URLs": [
96          "https://pip.pypa.io/en/stable/",
97          "https://github.com/pypa/pip",
98          {
99            "URL": "https://pypi.org/project/pip/",
100           "comment": "(egg and chicken ;) )"
101         }
102     ] },
103     { "dependency_name": "ruff", "URLs": [
104         "https://docs.astral.sh/ruff/",
105         "https://github.com/astral-sh/ruff",
106         "https://pypi.org/project/ruff/"
107     ] },
108     { "dependency_name": "bandit", "URLs": [
109         "https://bandit.readthedocs.io/en/latest/",
110         "https://github.com/PyCQA/bandit",
111         "https://pypi.org/project/bandit/"
112     ] },
113     { "dependency_name": "phpcs (PHP_CodeSniffer)", "URLs": [
114         "https://github.com/PHPCSStandards/PHP_CodeSniffer",
115         "https://packagist.org/packages/squizlabs/php_codesniffer"
116     ] },
117     { "dependency_name": "phpmd", "URLs": [
```

```
        "https://phpmd.org/",
        "https://github.com/phpmd/phpmd",
        "https://packagist.org/packages/phpmd/phpmd"
    ] },
    { "dependency_name": "php", "URLs": [
        "https://www.php.net/",
        "https://github.com/php/php-src"
    ] },
    { "dependency_name": "tex", "URLs": [
        "https://ctan.org/tex-archive/systems/knuth/dist/tex"
    ] },
    { "dependency_name": "pcregrep, pcre2grep", "URLs": [
        "https://www.pcre.org/",
        "https://pcre2project.github.io/pcre2/",
        "https://github.com/PCRE2Project/pcre1",
        "https://github.com/PCRE2Project/pcre2",
        "https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c"
    ] },
    { "dependency_name": "python3", "URLs": [
        "https://www.python.org/",
        "https://docs.python.org/3/"
    ] }
]
```

3.39 build_and_checks_variables/dependencies_notes.txt

```
Bash can be found at the following URLs:
https://www.gnu.org/software/bash/
https://tiswww.case.edu/php/chet/bash/bashtop.html
https://savannah.gnu.org/projects/bash/

GNU coreutils can be found at the following URLs:
https://www.gnu.org/software/coreutils/
https://savannah.gnu.org/projects/coreutils/

uutils coreutils can be found at the following URLs:
https://uutils.github.io/coreutils/
https://github.com/uutils/coreutils

diff (GNU diffutils) can be found at the following URLs:
https://www.gnu.org/software/diffutils/
https://savannah.gnu.org/projects/diffutils/

find, xargs (GNU findutils) can be found at the following URLs:
https://www.gnu.org/software/findutils/
https://savannah.gnu.org/projects/findutils/
```

```

grep (GNU grep) can be found at the following URLs:
https://www.gnu.org/software/grep/
https://savannah.gnu.org/projects/grep/

sed (GNU sed) can be found at the following URLs:
https://www.gnu.org/software/sed/
https://savannah.gnu.org/projects/sed/

ShellCheck can be found at the following URLs:
https://www.shellcheck.net/
https://github.com/koalaman/shellcheck

GNU Wget can be found at the following URLs:
https://www.gnu.org/software/wget/
https://savannah.gnu.org/projects/wget/

more of util-linux can be found at the following URLs:
https://www.kernel.org/pub/linux/utils/util-linux/
https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/
https://github.com/util-linux/util-linux official?

pandoc can be found at the following URLs:
https://pandoc.org/
https://github.com/jgm/pandoc

pdftex/pdflatex can be found at the following URLs:
https://www.tug.org/applications/pdftex/
https://tug.org/texlive/svn/

tree can be found at the following URLs:
https://oldmanprogrammer.net/source.php?dir=projects/tree
https://gitlab.com/OldManProgrammer/unix-tree
https://github.com/Old-Man-Programmer/tree

git can be found at the following URLs:
https://git-scm.com/about
https://github.com/git/git

pylint can be found at the following URLs:
https://www.pylint.org/
https://pylint.readthedocs.io/en/stable/
https://github.com/pylint-dev/pylint
https://pypi.org/project/pylint/

notify-send can be found at the following URLs:
```

```
https://gitlab.gnome.org/GNOME/libnotify/
https://gnome.pages.gitlab.gnome.org/libnotify/

black can be found at the following URLs:
https://github.com/psf/black
https://black.readthedocs.io/en/stable/
https://pypi.org/project/black/

isort can be found at the following URLs:
https://pycqa.github.io/isort/
https://github.com/pycqa/isort/
https://pypi.org/project/isort/

mypy can be found at the following URLs:
https://mypy-lang.org/
https://mypy.readthedocs.io/en/stable/
https://github.com/python/mypy
https://pypi.org/project/mypy/

phpdoc (phpDocumentor) can be found at the following URLs:
https://phpdoc.org/
https://docs.phpdoc.org/
https://github.com/phpdocumentor/phpdocumentor

pip can be found at the following URLs:
https://pip.pypa.io/en/stable/
https://github.com/pypa/pip
https://pypi.org/project/pip/ (egg and chicken ;) )

ruff can be found at the following URLs:
https://docs.astral.sh/ruff/
https://github.com/astral-sh/ruff
https://pypi.org/project/ruff/

bandit can be found at the following URLs:
https://bandit.readthedocs.io/en/latest/
https://github.com/PyCQA/bandit
https://pypi.org/project/bandit/

phpcs (PHP_CodeSniffer) can be found at the following URLs:
https://github.com/PHPCSStandards/PHP_CodeSniffer
https://packagist.org/packages/squizlabs/php_codesniffer

phpmd can be found at the following URLs:
https://phpmd.org/
https://github.com/phpmd/phpmd
```

```
https://packagist.org/packages/phpmd

php can be found at the following URLs:
https://www.php.net/
https://github.com/php/php-src

tex can be found at the following URL:
https://ctan.org/tex-archive/systems/knuth/dist/tex

pcregrep, pcre2grep can be found at the following URLs:
https://www.pcre.org/
https://pcre2project.github.io/pcre2/
https://github.com/PCRE2Project/pcre1
https://github.com/PCRE2Project/pcre2
https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c

python3 can be found at the following URLs:
https://www.python.org/
https://docs.python.org/3/

If you see something that I missed regarding my dependencies,
please tell me/email me :).
```

3.40 build_and_checks_variables/DevOrSysAdminScripts.html.tpl

```
<!DOCTYPE html>
<html lang="en">
<!--
This file is part of DevOrSysAdminScripts library.

DevOrSysAdminScripts is free software:
you can redistribute it and/or modify it under the terms
of the GNU Lesser General Public License
as published by the Free Software Foundation,
either version 3 of the License,
or (at your option) any later version.

DevOrSysAdminScripts is distributed in the hope
that it will be useful,
but WITHOUT ANY WARRANTY;
without even the implied warranty of
MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
See the GNU Lesser General Public License for more details.

You should have received a copy of
the GNU Lesser General Public License
```

```
along with DevOrSysAdminScripts.
If not, see <https://www.gnu.org/licenses/>.

©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
-->
<head>
<meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
<meta name="generator" content="DevOrSysAdminScripts">
<style>
.title{
  margin: 2ex auto;
  text-align: center;
}
.titlemain{
  margin: 1ex 2ex 2ex 1ex;
}
.titlerest{
  font-size: 1.17em;
  margin: 0ex 2ex;
}
table{
  border-collapse: collapse;
}
td{
  padding: 0;
}
pre{
  text-align: left;
  margin-left: 0ex;
  margin-right: auto;
}
pre.numbered_lines:before{
  background: snow;
  border-right: 1px black solid;
  color: #010101;
  content: attr(line-numbers);
  float: left;
  padding: 0 1ch;
  text-align: right;
}
blockquote{
  margin-left: 4ex;
  margin-right: 4ex;
  text-align: left;
}
ol.level1, ol.level2{
```

```
  list-style-type: none;
  /*
  list-style-type: decimal;
  Previous rule is shorter and more efficient if you need only level1.
  But you don't have an accessible counter if you use decimal.
  */
}
ol.level1{
  counter-reset: level1;
}
ol.level2{
  counter-reset: level2;
}
ol.level1 li:before{
  content: counter(level1) ". ";
  counter-increment: level1;
}
ol.level2 li:before{
  content: counter(level1) "." counter(level2) " ";
  counter-increment: level2;
}
</style>
<title>@repository_name@</title>
</head>
<body >
<table class="title">
  <tr>
    <td style="padding:1ex">
      <h1 class="titlemain">@repository_name@</h1>
      <h2 class="titlerest">
        @author_full_name@
        <br>
        <a href="@author_website@">
          <span style="font-family:monospace">
          @author_website@
          </span>
        </a>
        <br>
        <a href="mailto:@author_email@">
          <span style="font-family:monospace">
          @author_email@
          </span>
        </a>
      </h2>
    </td>
  </tr>
```

```
</table>

<blockquote class="abstract">
  <span style="font-weight:bold">Abstract:</span>
  @abstract@
</blockquote>
<p>Current version: @current_date@</p>
<p>Current number of commits: @number_of_commits@</p>
<p>Current git SHA1: @current_git_SHA1@</p>
<p>
Code lines: @number_of_lines@
</p>


<h2>Contents</h2>

<ol class="level1">
  <li><a href="#section1">Files Tree</a></li>
  <li><a href="#section2">Notes on licenses and dependencies</a></li>
  <li><a href="#section3">Listing of files</a><br>
    <ol class="level2">
      @files_lis@
    </ol>
  </li>
</ol>


<h2 id="section1">1 Files tree</h2>
<pre class="verbatim">
@current_tree_light@
</pre>
<pre class="verbatim">
@current_tree@
</pre>


<h2 id="section2">2 Notes on licenses and dependencies</h2>

<p>
This repository is dual licensed.
All the Bash scripts source code is covered by GPLv3+.
All the PHP scripts source code is covered by LGPLv3+.
All the Python scripts source code is covered by LGPLv3+.
</p>

<p>
```

```
The text of the licenses is available at:
<a href="https://www.gnu.org/licenses/"><span
  style="font-family:monospace"
>https://www.gnu.org/licenses/</span></a>.
The git repository of this source code is also available at:
<a href="@repository_web_url@"><span
 style="font-family:monospace"
>@repository_web_url@</span></a>.
Clone it with this URL:
<span style="font-family:monospace"
>@repository_git_url@</span>.
</p>

<div>
<p>
Here is a listing of the dependencies of the code in this repository:
</p>
<ul>
<li> The following dependencies are GPLv3+ licensed:<br>
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
<li> The following dependencies are GPLv2+ licensed:<br>
more, pandoc, pdftex, pdflatex, tree;
<li> The following dependencies are GPLv2 licensed:<br>
git, pylint;
<li> The following dependencies are LGPLv2.1+ licensed:<br>
notify-send;
<li> The following dependencies are MIT licensed:<br>
black, isort, mypy, phpdoc, pip, ruff;
<li> The following dependencies are Apache-2.0 licensed:<br>
bandit;
<li> The following dependencies are BSD-3-Clause licensed:<br>
phpcs, phpmd;
<li> php is licensed under PHP License v3.01;
<li> tex is licensed under Knuth license;
<li> pcregrep, pcre2grep has a custom license from Cambridge
University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception.
<li> python3 is licensed under PSF License Version 2 and the Zero-Clause
BSD license.
</ul>
</div>

<p>
Remark: Some commands listed under GPLv3+ are POSIX
and can have different implementations under another license;
```

```
for example, uutils coreutils is MIT licensed,
whilst GNU coreutils is GPLv3+ licensed.
</p>

@dependencies_notes@


<h2 id="section3">3 Listing of files</h2>

@files_listing_HTMLPreInput@

<p>
@acknowledgments@
</p>

<script>
"use strict";
const pres = Array.from(
  document.getElementsByClassName('numbered_lines')
);
pres.forEach(
  (pre) => {
    const nb = Array.from(pre.innerText.matchAll("\n")).length + 1;
    let numbers = "";
    for(let i = 1; i <= nb; ++i){
      numbers += i.toString();
      if(i < nb){ numbers += "\n "; }
    }
    pre.setAttribute('line-numbers', numbers);
  }
);
</script>
</html>
```

3.41 build_and_checks_variables/DevOrSysAdminScripts.tex.tpl

```
% This file is part of DevOrSysAdminScripts library.
%
% DevOrSysAdminScripts is free software:
% you can redistribute it and/or modify it under the terms
% of the GNU Lesser General Public License
% as published by the Free Software Foundation,
% either version 3 of the License,
% or (at your option) any later version.
%
% DevOrSysAdminScripts is distributed in the hope
```

```
% that it will be useful,
% but WITHOUT ANY WARRANTY;
% without even the implied warranty of
% MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
% See the GNU Lesser General Public License for more details.
%
% You should have received a copy of
% the GNU Lesser General Public License
% along with DevOrSysAdminScripts.
% If not, see <https://www.gnu.org/licenses/>.
%
% ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet

% 2025-12-29T17:08+01:00: This file was moved from
% "./latex"
% to
% "./build_and_checks_variables".

\documentclass{article}

\usepackage[utf8]{inputenc}
\usepackage[T1]{fontenc}
\usepackage{subfigure}
\usepackage{amsmath}
\usepackage{amssymb}
\usepackage{amsthm}
\usepackage[pdftex]{hyperref}
\usepackage{tikz}
\usepackage{caption}
\usepackage[round]{natbib}
\usepackage{fancyhdr}
\usepackage{amsfonts}
\usepackage{times}
\usepackage{ifpdf}
\usepackage{latexsym}
\usepackage{graphicx}
\usepackage{enumerate}
\usepackage{pmboxdraw}
\usepackage{fancyvrb}
\usepackage[french,english]{babel}

% *** les environnements ***
%\theoremstyle{break}
\newtheorem{definition}{Definition}[section]
\newtheorem{proposition}[definition]{Proposition}
\newtheorem{theorem}[definition]{Theorem}
```

```
\newtheorem{lemma}[definition]{Lemma}
\newtheorem{corollary}[definition]{Corollary}
\newtheorem{remark}[definition]{Remark}
\newtheorem{openproblem}[definition]{Open problem}

\begin{document}

\author{
  @author_full_name@\\
  \url{@author_website@}\\
  \href{mailto:@author_email@}{@author_email@}
}
\title{@repository_name@}

\maketitle
\begin{abstract}
@abstract@
\end{abstract}

Current version: @current_date@

Current number of commits: @number_of_commits@

Current git SHA1: @current_git_SHA1@

Code lines: @number_of_lines@

\tableofcontents
\newpage

\section{Files tree}
\label{section:tree}

\begin{verbatim}
@current_tree_light@
\end{verbatim}

\begin{verbatim}
@current_tree@
\end{verbatim}

\section{Notes on licenses and dependencies}
\label{section:notes_on}

This repository is dual licensed.
All the Bash scripts source code is covered by GPLv3+.
```

```
All the PHP scripts source code is covered by LGPLv3+.
All the Python scripts source code is covered by LGPLv3+.

The text of the licenses is available at:
\url{https://www.gnu.org/licenses/}.
The git repository of this source code is also available at:
\url{@repository_web_url@}.
Clone it with this URL:
\url{@repository_git_url@}.

Here is a listing of the dependencies of the code in this repository:
\begin{itemize}
\item The following dependencies are GPLv3+ licensed:\newline
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
\item The following dependencies are GPLv2+ licensed:\newline
more, pandoc, pdftex, pdflatex, tree;
\item The following dependencies are GPLv2 licensed:\newline
git, pylint;
\item The following dependencies are LGPLv2.1+ licensed:\newline
notify-send;
\item The following dependencies are MIT licensed:\newline
black, isort, mypy, phpdoc, pip, ruff;
\item The following dependencies are Apache-2.0 licensed:\newline
bandit;
\item The following dependencies are BSD-3-Clause licensed:\newline
phpcs, phpmd;
\item php is licensed under PHP License v3.01;
\item tex is licensed under Knuth license;
\item pcregrep, pcre2grep has a custom license from Cambridge
University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception,
\item python3 is licensed under PSF License Version 2 and the Zero-Clause
BSD license.
\end{itemize}

Remark: Some commands listed under GPLv3+ are POSIX
and can have different implementations under another license;
for example, uutils coreutils is MIT licensed,
whilst GNU coreutils is GPLv3+ licensed.

@dependencies_notes@

\section{Listing of files}
\label{section:listing}

```

```
@files_listing_VerbatimInput@

@acknowledgments@

\end{document}
```

3.42 build_and_checks_variables/files_names_listing.txt

```
// This file is part of DevOrSysAdminScripts library.
//
// DevOrSysAdminScripts is free software:
// you can redistribute it and/or modify it under the terms
// of the GNU Lesser General Public License
// as published by the Free Software Foundation,
// either version 3 of the License,
// or (at your option) any later version.
//
// DevOrSysAdminScripts is distributed in the hope
// that it will be useful,
// but WITHOUT ANY WARRANTY;
// without even the implied warranty of
// MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
// See the GNU Lesser General Public License for more details.
//
// You should have received a copy of
// the GNU Lesser General Public License
// along with DevOrSysAdminScripts.
// If not, see <https://www.gnu.org/licenses/>.
//
// ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
// 2024-10-31T21:37+01:00: This file was moved from
// "./build_and_checks_dependencies/listings"
// to
// "./build_and_checks_variables".
// Blasé20240517Résultat.txt
// Blasé20240517.sh
// Blasé20240518_2.txt
// Blasé20240518_3.txt
build_and_checks_dependencies/build_and_checks.exec.sh.tpl
build_and_checks_dependencies/build_dependencies_notes.exec.php
build_and_checks_dependencies/build_dependencies_notes.libr.php
build_and_checks_dependencies/build_md_from_printable_md.exec.sh
build_and_checks_dependencies/check_shell_scripts_beginnings.libr.sh
build_and_checks_dependencies/check_shell_scripts_indentation.libr.sh
build_and_checks_dependencies/check_URLs.libr.sh
build_and_checks_dependencies/common_build_and_checks.exec.sh
```

```
build_and_checks_dependencies/common_build_and_checks.exec.sh.tpl
build_and_checks_dependencies/common_options.libr.sh
build_and_checks_dependencies/comparisons.libr.sh
build_and_checks_dependencies/create_PDF.exec.sh
build_and_checks_dependencies/generate_from_template.libr.sh
build_and_checks_dependencies/get_common_text_glob_patterns.libr.sh
build_and_checks_dependencies/grammar_and_spelling_check.libr.sh
build_and_checks_dependencies/licenses_templates/\
build_licenses_templates.exec.sh
build_and_checks_dependencies/licenses_templates/\
license_file_header_AGPLv3+.tpl
build_and_checks_dependencies/licenses_templates/\
license_file_header_GPLv3+.tpl
build_and_checks_dependencies/licenses_templates/\
license_file_header_LGPLv3+.tpl
build_and_checks_dependencies/lines_counts.libr.sh
build_and_checks_dependencies/lines_filters.libr.sh
build_and_checks_dependencies/lines_maps.libr.sh
// build_and_checks_dependencies/listings/files_listing.tex.tpl
build_and_checks_dependencies/overwrite_if_not_equal.libr.sh
build_and_checks_dependencies/python_black_complement.libr.sh
build_and_checks_dependencies/python_isort_complement.libr.sh
build_and_checks_dependencies/repository_data.libr.sh
build_and_checks_dependencies/shell_checks_complement.libr.sh
build_and_checks_dependencies/split_score.exec.php
build_and_checks_dependencies/split_score.libr.php
build_and_checks_dependencies/strings_functions.libr.sh
build_and_checks_dependencies/too_long_code_lines.libr.sh
build_and_checks_dependencies/update_common_build_and_checks.exec.sh
build_and_checks_dependencies/\
update_or_check_files_names_listing.exec.sh
build_and_checks_variables/.gitignore
build_and_checks_variables/bandit.ini
build_and_checks_variables/bandit_baseline.json
build_and_checks_variables/black.toml
build_and_checks_variables/dependencies_data.json
build_and_checks_variables/dependencies_notes.txt
build_and_checks_variables/DevOrSysAdminScripts.html.tpl
// build_and_checks_variables/DevOrSysAdminScripts.tex
build_and_checks_variables/DevOrSysAdminScripts.tex.tpl
build_and_checks_variables/files_names_listing.txt
build_and_checks_variables/licenses_notes.txt
build_and_checks_variables/phpcs_ruleset.xml
build_and_checks_variables/phpmd_baseline.xml
build_and_checks_variables/pylintrc.toml
build_and_checks_variables/repository_data.txt
```

```
build_and_checks_variables/repository_data.txt.tpl
build_and_checks_variables/ruff.toml
build_and_checks_variables/shellcheck.ini
build_and_checks_variables/shellcheck_notes.txt
// build_and_checks_variables/upgrade_venvs_ts
build_and_checks.exec.sh
COPYING
COPYING.LESSER
// DevOrSysAdminScripts.html
// DevOrSysAdminScripts.pdf
// EncoreBlasé20240517-20240518.txt
git_helpers/git_checks.libr.sh
git_helpers/post-commit
git_helpers/post_commit_renaming_comment.libr.sh
git_helpers/pre-commit
IMAP_transfert.py
LiensUtiles.txt
// Uncomment some files in miscellaneous to do some testing.
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123456789012345
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123456789012345\
56
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123456789012345\
56\
\
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123456789012345\
5\
\
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123456789012345\
\
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123456789012345\
\
// miscellaneous/\
a_long_long_long_long_long_long_long_long_long_filename_so_that_you_can_se\
e_that_it_works.txt
miscellaneous/funny.sh
miscellaneous/spampty_data.txt
miscellaneous/string_escaping.html
miscellaneous/string_escaping.xhtml
// miscellaneous/\
yet_another_long_long_long_long_long_long_long_long_long_filename_so_that_\
```

```
you_can_see_that_it_works.txt
// README.html
// README.md
README.md.tpl
string_escaping.libr.php
unicode.libr.php
upgrade_build_and_checks.exec.sh
wget_sha512.libr.sh
```

3.43 build_and_checks_variables/licenses_notes.txt

```
This repository is dual licensed.
All the Bash scripts source code is covered by GPLv3+.
All the PHP scripts source code is covered by LGPLv3+.
All the Python scripts source code is covered by LGPLv3+.

Here is a listing of the dependencies of the code in this repository:
- The following dependencies are GPLv3+ licensed:
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
- The following dependencies are GPLv2+ licensed:
more, pandoc, pdftex, pdflatex, tree;
- The following dependencies are GPLv2 licensed:
git, pylint;
- The following dependencies are LGPLv2.1+ licensed:
notify-send;
- The following dependencies are MIT licensed:
black, isort, mypy, phpdoc, pip, ruff;
- The following dependencies are Apache-2.0 licensed:
bandit;
- The following dependencies are BSD-3-Clause licensed:
phpcs, phpmd;
- php is licensed under PHP License v3.01;
- tex is licensed under Knuth license;
- pcregrep, pcre2grep has a custom license from Cambridge University,
but PCRE2 has BSD-3-Clause WITH PCRE2-exception,
- python3 is licensed under PSF License Version 2 and the Zero-Clause BSD
license.

Remark: Some commands listed under GPLv3+ are POSIX
and can have different implementations under another license;
for example, uutils coreutils is MIT licensed,
whilst GNU coreutils is GPLv3+ licensed.
```

3.44 build_and_checks_variables/phpcs_ruleset.xml

```
<?xml version="1.0"?>
<ruleset name="LFBFLCustomStandard" namespace="LFBFL\CS\CustomStandard">
  <description>A custom coding standard</description>
  <config name="tab_width" value="2"/>

  <file>../</file>

  <arg name="extensions" value="php"/>
  <arg name="report" value="summary"/>
  <arg name="colors"/>
  <arg value="sp"/>

  <!--
  I added all the standards, then excluded all the rules that gave me
  false positives or that seemed unappropriate when looking at the
  documentation.
  Some are bugs, some are conflicting with my own coding style.
  You may delete an exclude and see what pops up to maybe understand
  my choices.
  -->

  <rule ref="Generic">
    <!-- You have to allow long or short or both syntaxes. -->
    <exclude name="Generic.Arrays.DisallowShortArraySyntax"/>
    <exclude name="Generic.Commenting.DocComment.ParamGroup"/>
    <exclude name="Generic.Commenting.DocComment.TagsNotGrouped"/>
    <exclude name="Generic.Commenting.DocComment.TagValueIndent"/>
    <!-- Back to my habits at Teliae with always a closing tag. -->
    <!-- <exclude name="Generic.PHP.ClosingPHPTag.NotFound"/> -->
    <!--
    Not standard in Unix/Linux world to keep files not ending with \n.
    This is always an option of the code editors.
    Since PHP doesn't output the final \n after ?>, it is ok.
    -->
    <exclude name="Generic.Files.EndFileNoNewline.Found"/>
    <exclude name="Generic.Files.ExecutableFile.Executable"/>
    <exclude name="Generic.Formatting.MultipleStatementAlignment"/>
    <exclude
     name="Generic.Functions.OpeningFunctionBraceBsdAllman.BraceOnSameLine"
    />
    <exclude
      name="Generic.NamingConventions.CamelCapsFunctionName.NotCamelCaps"
    />
    <!-- Applies only to true, false and null. -->
```

```
    <exclude name="Generic.PHP.LowerCaseConstant.Found"/>
    <exclude name="Generic.PHP.UpperCaseConstant.Found"/>
    <!-- const_bidule_HTML_truc would be better. Everything is prefix. -->
    <exclude name="Generic.WhiteSpace.DisallowSpaceIndent.SpacesUsed"/>
  </rule>
  <rule ref="PEAR">
    <exclude name="PEAR.Commenting.FileComment.MissingVersion"/>
    <exclude name="PEAR.Commenting.FileComment.MissingLinkTag"/>
    <!--
    Look at the comment before generate_split_score.
    PEAR and Squiz require to align the parameter names parts in the same
    comment, and also the parameter comments parts in the same comment,
    but with a long parameter name like $a_delimiter_strings_domain
    and a line length limit at 75 characters, parameter comments parts
    have only 24 characters left per line, which is ridiculous.
    Our choice to put the parameter comments parts starting on another line
    with same "indent" as parameter names parts is way better in those
    cases. Alignment has always been a trap with short lines.
    Since phpDocumentor handles the parameter comments parts correctly,
    it is enough for me.
    Worse is the handling of "nested doc block comment" for the Closure
    returned by generate_split_score().
    Our solution makes at least the source code readable.
    Maybe one day the dev tools will handle it correctly.
    -->
    <exclude name="PEAR.Commenting.FunctionComment.DuplicateReturn"/>
    <exclude name="PEAR.Commenting.FunctionComment.ExtraParamComment"/>
    <exclude name="PEAR.Commenting.FunctionComment.MissingParamComment"/>
    <exclude name="PEAR.Commenting.FunctionComment.SpacingAfterParamName"/>
    <exclude name="PEAR.Commenting.FunctionComment.SpacingAfterParamType"/>
    <exclude name="PEAR.ControlStructures.ControlSignature.Found"/>
    <exclude name="PEAR.ControlStructures.MultiLineCondition.Alignment"/>
    <exclude
      name="PEAR.ControlStructures.MultiLineCondition.SpaceBeforeOpenBrace"
    />
    <exclude
     name="PEAR.ControlStructures.MultiLineCondition.SpacingAfterOpenBrace"
    />
    <exclude
      name="PEAR.ControlStructures.MultiLineCondition.StartWithBoolean"
    />
    <!--
    Don’t laugh when looking at the false positives of the two exclude
    below ;) XD.
    They don’t check for multiline function declarations apparently.
    -->
```

```
91      <exclude
92        name="PEAR.Functions.FunctionDeclaration.NewlineBeforeOpenBrace"
93      />
94      <exclude
95        name="PEAR.Functions.FunctionDeclaration.BraceOnSameLine"
96      />
97      <exclude
98        name="PEAR.NamingConventions.ValidFunctionName.FunctionNoCapital"
99      />
100     <exclude
101       name="PEAR.NamingConventions.ValidFunctionName.FunctionNameInvalid"
102     />
103   </rule>
104   <rule ref="PSR1">
105   </rule>
106   <rule ref="PSR2">
107     <!-- Back to my habits at Teliae with always a closing tag. -->
108     <exclude name="PSR2.Files.ClosingTag.NotAllowed"/>
109     <exclude
110 name="PSR2.ControlStructures.ControlStructureSpacing.SpacingAfterOpenBrace"
111     />
112   </rule>
113   <rule ref="PSR12">
114     <exclude
115  name="PSR12.ControlStructures.ControlStructureSpacing.FirstExpressionLine"
116     />
117     <exclude name="PSR12.Files.FileHeader.SpacingAfterDeclareBlock"/>
118     <!--
119     3 false positives, see:
120     https://github.com/PHPCSStandards/PHP_CodeSniffer/issues/1402
121     -->
122     <exclude
123       name="PSR12.Files.DeclareStatement.SpaceFoundAfterDirectiveValue"
124     />
125     <exclude
126       name="PSR12.Files.DeclareStatement.SpaceFoundBeforeDirectiveValue"
127     />
128     <exclude
129       name="PSR12.Files.DeclareStatement.SpaceFoundBeforeSemicolon"
130     />
131     <!-- and even a fourth and fifth false positives -->
132     <exclude
133       name="PSR12.Files.DeclareStatement.CloseBracketNotAligned"
134     />
135     <exclude
136       name="PSR12.Files.DeclareStatement.ExtraSpaceFoundAfterBracket"
```

```
    />
    <exclude name="PSR12.Classes.ClosingBrace.StatementAfter"/>
    <exclude
      name="PSR12.Functions.ReturnTypeDeclaration.SpaceBeforeColon"
    />
    <exclude name="PSR2.Namespaces.NamespaceDeclaration.BlankLineAfter"/>
    <exclude name="PSR2.Namespaces.UseDeclaration.SpaceAfterLastUse"/>
    <!-- I agree except for string concatenation. -->
    <exclude name="PSR12.Operators.OperatorSpacing.NoSpaceBefore"/>
    <exclude name="PSR12.Operators.OperatorSpacing.NoSpaceAfter"/>
  </rule>
  <rule ref="Squiz">
    <exclude name="Squiz.Arrays.ArrayDeclaration.SingleLineNotAllowed"/>
    <!--
    It doesn't know the //*/ trick...
    https://github.com/PHPCSStandards/PHP_CodeSniffer/issues/1403
    -->
    <exclude name="Squiz.Commenting.BlockComment.CloserSameLine"/>
    <exclude name="Squiz.Commenting.BlockComment.FirstLineIndent"/>
    <exclude name="Squiz.Commenting.BlockComment.LineIndent"/>
    <exclude name="Squiz.Commenting.BlockComment.NoCapital"/>
    <exclude name="Squiz.Commenting.BlockComment.NoEmptyLineAfter"/>
    <exclude name="Squiz.Commenting.BlockComment.NoEmptyLineBefore"/>
    <!-- Problem for doc block comment before namespace -->
    <exclude name="Squiz.Commenting.BlockComment.WrongStart"/>
    <exclude name="Squiz.Commenting.DocCommentAlignment.SpaceBeforeStar"/>
    <exclude name="Squiz.Commenting.FileComment.CopyrightTagOrder"/>
    <exclude name="Squiz.Commenting.FileComment.IncorrectAuthor"/>
    <exclude name="Squiz.Commenting.FileComment.IncorrectCopyright"/>
    <exclude name="Squiz.Commenting.FileComment.MissingSubpackageTag"/>
    <exclude name="Squiz.Commenting.FileComment.PackageTagOrder"/>
    <exclude name="Squiz.Commenting.FileComment.SpacingAfterOpen"/>
    <exclude name="Squiz.Commenting.FileComment.SubpackageTagOrder"/>
    <exclude name="Squiz.Commenting.FunctionComment.DuplicateReturn"/>
    <exclude name="Squiz.Commenting.FunctionComment.ExtraParamComment"/>
    <!-- asks for boolean instead of bool which is incorrect -->
    <exclude
      name="Squiz.Commenting.FunctionComment.IncorrectParamVarName"
    />
    <!-- asks for boolean instead of bool which is incorrect -->
    <exclude name="Squiz.Commenting.FunctionComment.InvalidReturn"/>
    <exclude name="Squiz.Commenting.FunctionComment.MissingParamComment"/>
    <exclude name="Squiz.Commenting.FunctionComment.ParamCommentFullStop"/>
    <exclude
      name="Squiz.Commenting.FunctionComment.SpacingAfterParamName"
    />
```

```
183     <exclude
184       name="Squiz.Commenting.FunctionComment.SpacingAfterParamType"
185     />
186     <!-- Problem for doc block comment before namespace -->
187     <exclude name="Squiz.Commenting.InlineComment.DocBlock"/>
188     <exclude name="Squiz.Commenting.InlineComment.NotCapital"/>
189     <!-- It doesn't know the //*/ trick... -->
190     <exclude name="Squiz.Commenting.InlineComment.NoSpaceBefore"/>
191     <exclude name="Squiz.Commenting.InlineComment.InvalidEndChar"/>
192     <exclude name="Squiz.Commenting.LongConditionClosingComment.Invalid"/>
193     <exclude name="Squiz.Commenting.PostStatementComment.Found"/>
194     <exclude
195       name="Squiz.ControlStructures.ControlSignature.SpaceAfterCloseBrace"
196     />
197     <exclude
198   name="Squiz.ControlStructures.ControlSignature.SpaceAfterCloseParenthesis"
199     />
200     <exclude
201       name="Squiz.ControlStructures.ControlSignature.SpaceAfterKeyword"
202     />
203     <exclude name="Squiz.Formatting.OperatorBracket.MissingBrackets"/>
204     <exclude name="Squiz.Functions.GlobalFunction.Found"/>
205     <exclude
206       name="Squiz.Functions.MultiLineFunctionDeclaration.BraceOnSameLine"
207     />
208     <exclude
209   name="Squiz.Functions.MultiLineFunctionDeclaration.NewlineBeforeOpenBrace"
210     />
211     <exclude
212       name="Squiz.NamingConventions.ValidFunctionName.NotCamelCaps"
213     />
214     <exclude
215       name="Squiz.NamingConventions.ValidVariableName.NotCamelCaps"
216     />
217     <exclude name="Squiz.Operators.ComparisonOperatorUsage.NotAllowed"/>
218     <exclude name="Squiz.Operators.ComparisonOperatorUsage.ImplicitTrue"/>
219     <exclude name="Squiz.PHP.CommentedOutCode"/>
220     <exclude name="Squiz.Strings.EchoedStrings.HasBracket"/>
221     <exclude name="Squiz.WhiteSpace.OperatorSpacing.NoSpaceBefore"/>
222     <exclude name="Squiz.WhiteSpace.OperatorSpacing.NoSpaceAfter"/>
223     <exclude name="Squiz.WhiteSpace.OperatorSpacing.SpacingBefore"/>
224     <exclude
225       name="Squiz.WhiteSpace.FunctionClosingBraceSpace.SpacingBeforeClose"
226     />
227   </rule>
228   <rule ref="Zend">
```

```
229       <!-- Back to my habits at Teliae with always a closing tag. -->
230       <exclude name="Zend.Files.ClosingTag.NotAllowed"/>
231       <exclude
232         name="Zend.NamingConventions.ValidVariableName.NotCamelCaps"
233       />
234     </rule>
235
236     <rule ref="Generic.Arrays.ArrayIndent">
237       <properties>
238         <property name="indent" value="2" />
239       </properties>
240     </rule>
241     <rule ref="Generic.Files.LineLength">
242       <properties>
243         <property name="lineLimit" value="75"/>
244         <property name="absoluteLineLimit" value="75"/>
245       </properties>
246     </rule>
247     <rule ref="Generic.Formatting.SpaceAfterNot">
248       <properties>
249         <property name="spacing" value="0" />
250       </properties>
251     </rule>
252     <rule ref="Generic.Functions.OpeningFunctionBraceKernighanRitchie">
253       <properties>
254         <property name="checkFunctions" value="true"/>
255         <property name="checkClosures" value="true"/>
256       </properties>
257     </rule>
258     <rule ref="Generic.Metrics.CyclomaticComplexity">
259       <properties>
260         <property name="complexity" value="20" />
261         <property name="absoluteComplexity" value="40" />
262       </properties>
263     </rule>
264     <rule ref="Generic.Strings.UnnecessaryStringConcat">
265       <properties>
266         <property name="allowMultiline" value="true" />
267       </properties>
268     </rule>
269     <rule ref="Generic.WhiteSpace.ArbitraryParenthesesSpacing">
270       <properties>
271         <property name="ignoreNewlines" value="true" />
272       </properties>
273     </rule>
274     <rule ref="Generic.WhiteSpace.ScopeIndent">
```

```
275       <properties>
276         <property name="ignoreIndentationTokens" type="array">
277           <element value="T_COMMENT"/>
278           <element value="T_DOC_COMMENT_OPEN_TAG"/>
279         </property>
280         <property name="indent" value="2"/>
281       </properties>
282     </rule>
283
284     <rule ref="PEAR.Functions.FunctionCallSignature">
285       <properties>
286         <property name="indent" value="2"/>
287       </properties>
288     </rule>
289     <rule ref="PEAR.Functions.FunctionDeclaration">
290       <properties>
291         <property name="indent" value="2"/>
292       </properties>
293     </rule>
294     <rule ref="PEAR.WhiteSpace.ScopeIndent">
295       <properties>
296         <property name="ignoreIndentationTokens" type="array">
297           <element value="T_COMMENT"/>
298           <element value="T_DOC_COMMENT_OPEN_TAG"/>
299         </property>
300         <property name="indent" value="2"/>
301       </properties>
302     </rule>
303
304     <rule ref="PSR2.Methods.FunctionCallSignature">
305       <properties>
306         <property name="indent" value="2"/>
307       </properties>
308     </rule>
309
310     <rule ref="PSR12.ControlStructures.ControlStructureSpacing">
311       <properties>
312         <property name="indent" value="2"/>
313       </properties>
314     </rule>
315
316     <rule ref="Squiz.Functions.MultiLineFunctionDeclaration">
317       <properties>
318         <property name="indent" value="2"/>
319       </properties>
320     </rule>
```

```
  <rule ref="Squiz.Strings.ConcatenationSpacing">
    <properties>
      <property name="ignoreNewlines" value="true" />
    </properties>
  </rule>
  <rule ref="Squiz.WhiteSpace.FunctionSpacing">
    <properties>
      <property name="spacing" value="3" />
      <property name="spacingBeforeFirst" value="0" />
      <property name="spacingAfterLast" value="0" />
    </properties>
  </rule>
</ruleset>
```

3.45 build_and_checks_variables/phpmd_baseline.xml

```
<?xml version="1.0"?>
<phpmd-baseline>
  <violation rule="PHPMD\Rule\CyclomaticComplexity"
    file="build_and_checks_dependencies/build_dependencies_notes.libr.php"
  />
  <violation rule="PHPMD\Rule\Design\NpathComplexity"
    file="build_and_checks_dependencies/build_dependencies_notes.libr.php"
  />
  <violation rule="PHPMD\Rule\Design\LongMethod"
    file="build_and_checks_dependencies/build_dependencies_notes.libr.php"
  />
  <violation rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
    file="build_and_checks_dependencies/build_dependencies_notes.libr.php"
  />
  <violation rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
    file="build_and_checks_dependencies/build_dependencies_notes.libr.php"
  />
  <violation rule="PHPMD\Rule\CleanCode\ElseExpression"
    file="build_and_checks_dependencies/build_dependencies_notes.libr.php"
  />
  <violation rule="PHPMD\Rule\Naming\LongVariable"
    file="build_and_checks_dependencies/build_dependencies_notes.libr.php"
  />
  <violation rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
    file="build_and_checks_dependencies/split_score.libr.php"
  />
  <violation rule="PHPMD\Rule\Naming\LongVariable"
    file="build_and_checks_dependencies/split_score.libr.php"
  />
  <violation rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
```

```
    file="build_and_checks_dependencies/split_score.libr.php"
  />
  <violation rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
    file="string_escaping.libr.php"
  />
  <violation rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
    file="string_escaping.libr.php"
  />
  <violation rule="PHPMD\Rule\Design\LongMethod"
    file="unicode.libr.php"
  />
  <violation rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
    file="unicode.libr.php"
  />
  <violation rule="PHPMD\Rule\Naming\LongVariable"
    file="unicode.libr.php"
  />
  <violation rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
    file="unicode.libr.php"
  />
  <violation rule="PHPMD\Rule\CleanCode\MissingImport"
    file="unicode.libr.php"
  />
  <violation rule="PHPMD\Rule\CyclomaticComplexity"
    file="unicode.libr.php"
  />
  <violation rule="PHPMD\Rule\Design\NpathComplexity"
    file="unicode.libr.php"
  />
</phpmd-baseline>
```

3.46 build_and_checks_variables/pylintrc.toml

```
[tool.pylint.main]
exit-zero = true
ignore = ["node_modules"]
jobs = 0
[tool.pylint.basic]
# We use uppercase acronyms like HTML, IMAP, etc.
module-rgx = "^[_a-zA-Z0-9+]+$"
function-rgx = "^[_a-zA-Z0-9+]+$"
[tool.pylint."messages control"]
# Code like R0912 is better to order disables by category.
disable = [
  "C1804",  # use-implicit-booleaness-not-comparison-to-string
  "I0011",  # locally-disabled
```

```
  "I0020",  # suppressed-message
  "R0912",  # too-many-branches
  "R0913",  # too-many-arguments
  "R0914",  # too-many-locals
  "R0915",  # too-many-statements
  "R0917",  # too-many-positional-arguments
]
enable = "all"
[tool.pylint.reports]
output-format = "colorized"
```

3.47 build_and_checks_variables/repository_data.txt

```
Look at grep_variable to understand the structure of this file.
Basically, it has almost no structure apart from some lines
starting with a "variable_name=".
Anything else is comments.

repository_name=DevOrSysAdminScripts
repository_web_url=https://github.com/LLyaudet/DevOrSysAdminScripts/
repository_git_url=git@github.com:LLyaudet/DevOrSysAdminScripts.git

abstract=Some “small” useful scripts
abstract=for enhancing development or system-administrators tasks

license=GPLv3+
license2=LGPLv3+

author_email=Laurent.Lyaudet@gmail.com
author_full_name=Laurent Frédéric Bernard François Lyaudet
author_last_name=Lyaudet
author_name=Laurent Lyaudet
author_website=https://lyaudet.eu/laurent/

acknowledgments=Merci Dieu !
acknowledgments=Merci Père ! Merci Seigneur ! Merci Saint Esprit !

Great day! Now I allow myself lines of 75 characters instead of 70 :).
I checked how many characters fit in the PDF file generated by tex with
default paper size,
when using verbatim environment with line numbers for source code listing.
With a screen capture in GIMP,
I mesured 197px between two-digits line numbers and the left border of
the page.
(It is less than 197px for three-digits or more line numbers.)
The same distance from the right border yields 75 characters allowed.
```

```
Since when printing the HTML file, the number of allowed characters in
<pre> tags is way larger, 75 is the right limit with the tools I use.

max_line_length=75

latex_command=pdflatex

grammar_and_spelling_checker_command=


Why one venv for each Python dependency ?
Because no language of my knowledge did it right to handle all
possible dependency conflicts.
So in case of such a conflict, one specialized venv can get you out
of any problem.
The best solution would have been to allow install of different
versions of the same package, and have a mechanism:
- first intent, use the latest available version of a package among
those that satisfy the declared dependency constraints of the package
that has this dependency;
- second intent, check a dependency overload configuration file
with tree structure: if calling python package is A version 1.1.1
with dependency B itself with dependency C, use B 2.2.2 and C 3.3.3;
you could have along that another rule saying that when called
directly B 2.2.2 should use C 4.4.4;
and even more complicated:
if calling python package is A version 1.1.1 with dependency B and D
both themselves with dependency C, use B 2.2.2, D 0.1.0,
and C 3.3.3 for B, and C 4.4.4 for D;
once you've seen that it can never be more complicated than that once
the same version number guarantees that the code is the same,
you create a config language "à la CSS" for cascading dependencies ;)
XD fit into JSON the dumbest possible way and you stop bothering
about dependency conflicts for the rest of history of informatics.
{
  "A":{
    "1.1.1": {
      "B": {
        "2.2.2": {
          "C": {
            // Dumb, you did not found it by yourself? Failed exam.
            "3.3.3" : {}
          }
        }
      },
      "D": {
```

```
          "0.1.0": {
            "C": {
              // Dumb, you did not found it by yourself? Failed exam.
              "4.4.4" : {}
            }
          }
        }
      }
    },
    "B": {
      "2.2.2": {
        "C": {
          // Dumb, you did not found it by yourself? Failed exam.
          "4.4.4" : {}
        }
      }
    }
  }
  Wonderful someone rediscovered nested structures XD
  Seriously so many engineers or computer scientists are dumb to cry.

  bandit_venv=../venv_bandit
  black_venv=../venv_black
  isort_venv=../venv_isort
  mypy_venv=../venv_mypy
  pylint_venv=../venv_pylint
  ruff_venv=../venv_ruff


  upgrade venvs can be auto or ask, any other value is interpreted
  as ask.

  upgrade_venvs=ask


  time interval can be 0 to always upgrade or ask to upgrade,
  or a number at most bash max int
  https://superuser.com/questions/1030122/
  what-is-the-maximum-value-of-a-numeric-bash-shell-variable
  it should be enough to simulate never "upgrade or ask for upgrade"
  unless you claim to live many billions years ;)
  If it is wat, this is a random int XD (see RANDOM).
  Note that the probability that it upgrades your dependencies is not
  1/2, it increases over time, and is 1 after slightly
  less than one day.
  If it is watyouwant?, see SRANDOM XD.
```

```

upgrade_venvs_time_interval_in_seconds=wat


For the Nu HTML, CSS, and SVG checker.

vnu_jar_path=../jar_venv/vnu.jar

Not fully implemented yet, it will only warn that vnu jar was not
updated since some time.
"wat" works, but "watyouwant?" joke was not repeated ;).

upgrade_vnu_jar_time_interval_in_seconds=604800

use_vnu_for_xhtml=false
```

3.48 build_and_checks_variables/repository_data.txt.tpl

```
Look at grep_variable to understand the structure of this file.
Basically, it has almost no structure apart from some lines
starting with a "variable_name=".
Anything else is comments.
The goal of this template is to list all possible variables.

repository_name=
repository_web_url=
repository_git_url=

abstract=
abstract=

license=
license2=

author_email=
author_full_name=
author_last_name=
author_name=
author_website=

acknowledgments=
acknowledgments=

max_line_length=

latex_command=
```

```

grammar_and_spelling_checker_command=

bandit_venv=
black_venv=
isort_venv=
mypy_venv=
pylint_venv=
ruff_venv=

upgrade_venvs=

upgrade_venvs_time_interval_in_seconds=

vnu_jar_path=

upgrade_vnu_jar_time_interval_in_seconds=

use_vnu_for_xhtml=

npm_lint_directories=
npm_lint_directories=
```

3.49 build_and_checks_variables/ruff.toml

```
[lint]
select = [
  "ALL",
  "D211",  # blank-line-before-class no...before-class-docstring!
]
# ignore = ["D203", "D212", "D404"]
ignore = [
  "C901",  # complex-structure AKA give-me-toy-code
  "D203",  # (want-)?incorrect-blank-line-before-class
  "D205",  # missing-blank-line-after-summary So so cute|true ;) XD.
  "D212",  # multi-line-summary-first-line
  "D404",  # docstring-starts-with-this Shame! XD

  # raw-string-in-exception may be useful for a real project
  # with "dictionnaries" of errors made of constants.
  "EM101",

  # commented-out-code Of course git log is always faster...
  # Note that if you're tired or someone else in your team is,
  # it may catch some commit that lacked some review,
  # or automated tests: "Last manual test succeeded, ah but when I
```

```
  # debugged at some point I commented a part that was needed in
  # another case.",...
  "ERA001",

  "N802",  # invalid-function-name IMAP, HTML, etc.
  "PLR0912",  # too-many-branches AKA ...
  "PLR0913",  # too-many-arguments AKA ...
  "PLR0915",  # too-many-statements AKA ...
  # Not enough reading when young makes brain slow to read and
  # understand lengthy functions.
  # If the project is finished, you may play with AKA rules.
]
[lint.per-file-ignores]
"build_and_checks_dependencies/licenses_templates/*.py" = [
  "INP001"  # implicit-namespace-package
]
"IMAP_transfert.py" = [
  "T201",  # print Use log instead in full applications.
  "TRY003",  # raise-vanilla-args
]
```

3.50 build_and_checks_variables/shellcheck.ini

```
enable=all
check-sourced=true
external-sources=true
source-path="."
source-path="SCRIPTDIR"
source-path="./build_and_checks_dependencies"
source-path="./git_helpers"
```

3.51 build_and_checks_variables/shellcheck_notes.txt

```
A listing of the current suppressed warnings/errors from shellcheck,
and also some of the warnings that were suppressed in some previous
state of the git repository (in that case, we use X suffix, see SC2162X):
- SC1090 : Can't follow non-constant source.
           Use a directive to specify location
           # shellcheck source=bla/bli.sh
           Only remaining cases are in common_build_and_checks
           when sourcing some Python venv
           source venv/bin/active
- SC1091 : Not following (file not found most of the time).
           Only remaining cases are in common_build_and_checks
           when sourcing some Python venv
           source venv/bin/active
```

```
- SC1111 : Use of slanted double quotes (valid inside single quotes)
           False positive, but may catch some problem.
- SC2001 : Replace some sed with bash builtin ${a//b/c}
           False positive when c contains \n.
- SC2034 : Unused variable
           False positive for globals that are declared in functions
           that are not currently called.
           Somehow it can be used to detect dead code.
- SC2059 : Don't use variables in the printf format string.
           Use printf '..%s..' "$foo".
           False positive when using a variable to build the format string.
- SC2086 : Double quote to prevent globbing and word splitting.
           False positive when returning integer variable
           via an indirection like return ${!foo}.
           See SC2248 below and the corresponding issue on GitHub.
- SC2119 : Use foo "$@" if function's $1 should mean script's $1.
           False positive when passing arguments would yield
           a wrong result.
- SC2120 : foo references arguments, but none are ever passed.
           False positives when some arguments are meant for debug in CLI.
- SC2125 : Brace expansions and globs are literal.
           False positive when assigning a variable declared with -i.
           https://github.com/koalaman/shellcheck/issues/914
- SC2145 : Argument mixes string and array.
           False positive when using "$@" line return and pipes.
           https://github.com/koalaman/shellcheck/issues/3192
           Example:
             # shellcheck disable=SC2145
             all_self_code_lines "$@"\
               | empty_lines_after_file_name
- SC2154 : var is referenced but not assigned.
           Very useful but false positive when I play with
           variable variable names.
- SC2162X: read without -r will mangle backslashes.
           False positive when read is used without variable:
           https://github.com/koalaman/shellcheck/issues/623
- SC2248 : Prefer double quoting
           False positive on returning integer values.
           https://github.com/koalaman/shellcheck/issues/3405
- SC2309 : -eq treats this as a variable.
             Use = to compare as string (or expand explicitly with $var).
           Very useful but false positive when I play with
           variable variable names.
- SC2312 : Some command is not invoked separately and thus its return
           value is masked. Frequent with pipes.
- SCNNNNX: Use printf instead of echo.
```

```
            https://github.com/koalaman/shellcheck/issues/3419

The command to check if there is anything else:
grep -r "shellcheck disable" . | grep -v -e '.git' -e "=SC1090$"\
  -e "=SC1091$" -e "=SC1090,SC1091$" -e "=SC1111$" -e "=SC2001$"\
  -e "=SC2034$" -e "=SC2059$" -e "=SC2086$" -e "=SC2119$" -e "=SC2120$"\
  -e "=SC2125$" -e "=SC2145$" -e "=SC2154$" -e "=SC2248$" -e "=SC2309$"\
  -e "=SC2312$" -e "=SC2154,SC2309$"
```

3.52 build_and_checks.exec.sh

```
#!/usr/bin/env bash
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# The file "build_and_checks.exec.sh.tpl" was renamed from
# "build_and_checks.sh.tpl" to "build_and_checks.exec.sh.tpl".
# The file "build_and_checks.exec.sh" generated from the file
# "build_and_checks.sh.tpl" or "build_and_checks.exec.sh.tpl"
# was renamed from
# "build_and_checks.sh" to "build_and_checks.exec.sh".

build_and_checks(){
  # $1=work_directory
  # Options:
  #   --check-download (=keep to keep the downloaded files)
  #   --verbose
```

```
36    local LFBFL_work_directory="."
37    if [[ -n "$1" ]]; then
38      LFBFL_work_directory="$1"
39    fi
40    readonly LFBFL_work_directory
41
42    local LFBFL_verbose=""
43    if [[ "$*" == *--verbose* ]]; then
44      printf "%s %s\n" "$0" "$*"
45      LFBFL_verbose="--verbose"
46    fi
47    readonly LFBFL_verbose
48
49    local LFBFL_check_download=""
50    if [[ "$*" == *--check-download=keep* ]]; then
51      LFBFL_check_download="--check-download=keep"
52    elif [[ "$*" == *--check-download* ]]; then
53      LFBFL_check_download="--check-download"
54    fi
55    readonly LFBFL_check_download
56
57    source ./wget_sha512.libr.sh
58
59    LFBFL_subdir="build_and_checks_dependencies"
60    mkdir --parents "${LFBFL_subdir}/licenses_templates"
61    local LFBFL_variables_directory
62    LFBFL_variables_directory="${LFBFL_work_directory}/"
63    LFBFL_variables_directory+="build_and_checks_variables"
64    readonly LFBFL_variables_directory
65    mkdir --parents "${LFBFL_variables_directory}/temp"
66    if [[ ! -f "${LFBFL_variables_directory}/.gitignore" ]]; then
67      printf "temp/\nupgrade_venvs_ts\n"\
68        > "${LFBFL_variables_directory}/.gitignore"
69    fi
70
71    local LFBFL_dependencies_URL
72    LFBFL_dependencies_URL="https://raw.githubusercontent.com/LLyaudet/"
73    LFBFL_dependencies_URL+="DevOrSysAdminScripts/main/${LFBFL_subdir}"
74    readonly LFBFL_dependencies_URL
75
76    declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
77    local LFBFL_script_download_URL
78    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/"
79    LFBFL_script_download_URL+="${LFBFL_common_file_name}"
80    readonly LFBFL_script_download_URL
81    declare -r LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_common_file_name}"
```

```
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='090b7303680a32178fb541969cbb106ed908193a9ff2e'
  LFBFL_correct_sha512+='b0288700096e38d3c1d661d352e35c8916cdc4203e55'
  LFBFL_correct_sha512+='ee9eadfa421a8c40a73dbda491ae82b4a82ce6f'
  wget_sha512 "${LFBFL_file_path}"\
    "${LFBFL_script_download_URL}"\
    "${LFBFL_correct_sha512}"\
    "${LFBFL_verbose}"\
    "${LFBFL_check_download}"
  chmod +x "${LFBFL_file_path}"

  "${LFBFL_file_path}" "${LFBFL_work_directory}"\
    "${LFBFL_dependencies_URL}"\
    "${LFBFL_verbose}"\
    "${LFBFL_check_download}"
}

if [[ "$*" == *--fixed-point-build* ]]; then
  printf -- "--fixed-point-build\n"
  source "./build_and_checks_dependencies/lines_filters.libr.sh"
  LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  LFBFL_data_file_name="${LFBFL_variables_directory}/"
  LFBFL_data_file_name+="repository_data.txt"
  LFBFL_repository_name=""
  grep_variable "${LFBFL_data_file_name}" repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  # Touching the 3 following files first let us iterate
  # upgrade_build_and_checks 2 times instead of 3 to reach
  # a fixed point.
  touch "${LFBFL_variables_directory}/${LFBFL_repository_name}.tex"
  touch "${LFBFL_work_directory}/${LFBFL_repository_name}.pdf"
  touch "${LFBFL_work_directory}/${LFBFL_repository_name}.html"
  # The two iterations of build_and_checks needs to complete
  # during the same minute to have a fixed point.
  build_and_checks "$@"
  # build_and_checks "$@" second iteration below
  # Then calling this script without --fixed-point-build should not
  # yield any new superficial modifications.
fi

```

```
build_and_checks "$@"
```

3.53 COPYING

```
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```

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```

```

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```
    ensure that, in the event an Application does not supply the
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```

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```

3.55 git_helpers/git_checks.libr.sh

```
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# 2024-05-31T01:57+02:00: This file was renamed from
# "git_checks.libr.sh"
```

```
# to
# "git_checks.libr.sh".
# 2024-05-31T01:57+02:00: This file was moved from
# "."
# to
# "./git_helpers".
# 2024-05-31T01:49+02:00: This file was renamed from
# "git_checks.libr.sh"
# to
# "git_checks.libr.sh".
# 2024-05-31T01:49+02:00: This file was moved from
# "."
# to
# "./git_helpers".
# This file was renamed from "git_checks.sh" to "git_checks.libr.sh".

check_files(){
  # Checks used on Django projects.
  # Use `git commit --no-verify` after check.
  declare -r LFBFL_send_summary_1="ATTENTION : models modifiés"
  declare -r LFBFL_send_body_1="Vérifiez si besoin de migrations"
  local LFBFL_send_body_2="Pensez aussi aux contraintes d'unicités"
  LFBFL_send_body_2+=" dès la création du modèle"
  readonly LFBFL_send_body_2
  declare -r LFBFL_send_summary_2="ATTENTION : serializers modifiés"
  declare -r LFBFL_send_body_3=\
"Les nouveaux preloadings sont interdits -> #Prefetch()"
  declare -i LFBFL_i_error=0
  git diff --cached --name-only\
    > /tmp/DOSAS_django_git_check1.temp
  if grep models /tmp/DOSAS_django_git_check1.temp;
  then
    notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_1}"
    notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_2}"
    LFBFL_i_error=1
  fi
  if grep serializer /tmp/DOSAS_django_git_check1.temp;
  then
    notify-send "${LFBFL_send_summary_2}" "${LFBFL_send_body_3}"
    LFBFL_i_error=1
  fi
  rm /tmp/DOSAS_django_git_check1.temp
  # shellcheck disable=SC2248
  return ${LFBFL_i_error}
}

```

```
72
73 check_no_abusive_trailing_comma(){
74   # Useful for Python code, although it regularly yields
75   # false positives in Python code
76   # (but it is easy to distinguish by hand).
77   # Use `git commit --no-verify` after check.
78   declare -r LFBFL_send_summary_1="ATTENTION"
79   local LFBFL_send_body_1="Il semblerait que vous affectiez un tuple"
80   LFBFL_send_body_1+=" au lieu d'une autre valeur dans une variable."
81   readonly LFBFL_send_body_1
82   declare -i LFBFL_i_error=0
83   git diff --cached -r\
84     > /tmp/DOSAS_django_git_check2.temp
85   if grep ' = .*,\s*$' /tmp/DOSAS_django_git_check2.temp;
86   then
87     notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_1}"
88     LFBFL_i_error=1
89   fi
90   rm /tmp/DOSAS_django_git_check2.temp
91   # shellcheck disable=SC2248
92   return ${LFBFL_i_error}
93 }
94
95
96 check_black_code_formatting(){
97   # attention ça ne marche que sur les fichiers "stagés" avec git add
98   declare -r LFBFL_s_files_paths=$(
99     git diff --cached --name-only\
100    | grep '\.py'
101  )
102  printf "%s\n" "${LFBFL_s_files_paths}"
103  declare -a LFBFL_arr_files_paths
104  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
105  readonly LFBFL_arr_files_paths
106  declare -i LFBFL_i_error=0
107  # Not very useful since LFBFL_file should be local to the loop by
108  # default.
109  local LFBFL_file_path
110  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
111    printf "Black will check formatting on file %s\n" "${LFBFL_file_path}"
112    if ! black --check --diff "${LFBFL_file_path}";
113    then
114      LFBFL_i_error=1
115    fi
116  done
117  # shellcheck disable=SC2248
```

```
  return ${LFBFL_i_error}
}
```

3.56 git_helpers/post-commit

```
#!/usr/bin/env bash
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# 2024-05-31T01:49+02:00: This file was renamed from
# "post-commit"
# to
# "post-commit".
# 2024-05-31T01:49+02:00: This file was moved from
# "."
# to
# "./git_helpers".

LFBFL_some_directory="../DevOrSysAdminScripts"
declare -i LFBFL_i_pushd_result3
pushd "${LFBFL_some_directory}" || {
  LFBFL_i_pushd_result3=$?
  printf "post-commit no such directory\n"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_pushd_result3}
}
# shellcheck source=post_commit_renaming_comment.libr.sh
```

```
source "./git_helpers/post_commit_renaming_comment.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./build_and_checks_dependencies/lines_filters.libr.sh"
declare -i LFBFL_i_popd_result3
popd || {
  LFBFL_i_popd_result3=$?
  printf "post-commit no popd\n"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_popd_result3}
}

LFBFL_data_file_name="build_and_checks_variables/repository_data.txt"
LFBFL_max_comment_line_length=""
if [[ -f "${LFBFL_data_file_name}" ]]; then
  LFBFL_max_line_length=""
  grep_variable "${LFBFL_data_file_name}" "max_line_length"\
    --result-variable-prefix="LFBFL_"
  LFBFL_max_comment_line_length="--max-comment-line-length="
  LFBFL_max_comment_line_length+="${LFBFL_max_line_length}"
fi

commit_a_file_renamed_comment --verbose --log-directory-change\
  "${LFBFL_max_comment_line_length}"
exit 0
```

3.57 git_helpers/post_commit_renaming_comment.libr.sh

```
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```
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#
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# 2024-05-31T01:49+02:00: This file was renamed from
# "post_commit_renaming_comment.libr.sh"
# to
# "post_commit_renaming_comment.libr.sh".
# 2024-05-31T01:49+02:00: This file was moved from
# "."
# to
# "./git_helpers".
# This file was renamed from "post_commit_renaming_comment"
# to "post_commit_renaming_comment.libr.sh".
# (Before first commit it seems. But I had the old name in listing.)

LFBFL_this_file_path=$(realpath "${BASH_SOURCE[0]}")
LFBFL_this_file_directory=$(dirname "${LFBFL_this_file_path}")
LFBFL_some_directory="${LFBFL_this_file_directory}/../"
declare -i LFBFL_i_pushd_result2
pushd "${LFBFL_some_directory}" || {
  LFBFL_i_pushd_result2=$?
  printf "post_commit_renaming_comment.libr.sh no such directory\n"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_pushd_result2}
}
# shellcheck source=common_options.libr.sh
source "build_and_checks_dependencies/common_options.libr.sh"
# shellcheck source=strings_functions.libr.sh
source "build_and_checks_dependencies/strings_functions.libr.sh"
declare -i LFBFL_i_popd_result2
popd || {
  LFBFL_i_popd_result2=$?
  printf "post_commit_renaming_comment.libr.sh no popd\n"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_popd_result2}
}

# Never forget that the worse security is the illusion of security
# (Kevin Mitnick). So, do not expect to have an accurate trackability
# of code history just because you use this post-commit hook or
# something else more evolved. This is useful but not a guarantee.
# For this reason, this function will only handle when all files
# impacted in the preceding commit are only renamed.
# It assumes that `diff` is the current default difference utility
# parametered with `git`. This point could be enhanced.
```

```
commit_a_file_renamed_comment(){
  # Options:
  #   --verbose
  #   --log-directory-change
  #   --max-comment-line-length
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_arg
  local LFBFL_s_format

  declare -i LFBFL_i_log_directory_change=0
  if [[ "$*" == *--log-directory-change* ]]; then
    LFBFL_i_log_directory_change=1
  fi
  readonly LFBFL_i_log_directory_change

  declare -i LFBFL_i_max_comment_line_length=70
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --max-comment-line-length=* ]]; then
      LFBFL_arg=${LFBFL_arg#--max-comment-line-length=}
      LFBFL_i_max_comment_line_length=$((LFBFL_arg))
      break
    fi
  done
  readonly LFBFL_i_max_comment_line_length

  if [[ LFBFL_i_verbose -eq 1 ]]; then
    LFBFL_s_format="LFBFL_i_log_directory_change: %s\n"
    LFBFL_s_format+="LFBFL_i_max_comment_line_length: %s\n"
    # shellcheck disable=SC2059,SC2248
    printf "${LFBFL_s_format}"\
      "${LFBFL_i_log_directory_change}"\
      ${LFBFL_i_max_comment_line_length}
  fi
  #-------------------------------------------------------------------

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  declare -r LFBFL_renaming_lines=$(
    git log -p -1\
    | grep '^diff --git' -A 1
  )
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    printf "LFBFL_renaming_lines: %s\n" "${LFBFL_renaming_lines}"
  fi
```

```
112
113     declare -a LFBFL_renaming_lines_array
114     mapfile -t LFBFL_renaming_lines_array <<< "${LFBFL_renaming_lines}"
115     readonly LFBFL_renaming_lines_array
116
117     declare -r LFBFL_timestamp=$(date --iso-8601="minute")
118     if [[ LFBFL_i_verbose -eq 1 ]]; then
119       printf "LFBFL_timestamp: %s\n" "${LFBFL_timestamp}"
120     fi
121     local LFBFL_split_score_command1
122     declare -i LFBFL_i_split_score_command_properties1
123     # shellcheck disable=SC2248
124     get_split_score_simple 1 ${LFBFL_i_max_comment_line_length} _
125     LFBFL_split_score_command1="${get_split_score_result}"
126     LFBFL_i_split_score_command_properties1=${i_get_split_score_result2}
127     readonly LFBFL_split_score_command1
128     readonly LFBFL_i_split_score_command_properties1
129     local LFBFL_split_score_command2
130     declare -i LFBFL_i_split_score_command_properties2
131     # shellcheck disable=SC2248
132     get_split_score_simple 1 ${LFBFL_i_max_comment_line_length} /
133     LFBFL_split_score_command2="${get_split_score_result}"
134     LFBFL_i_split_score_command_properties2=${i_get_split_score_result2}
135     readonly LFBFL_split_score_command2
136     readonly LFBFL_i_split_score_command_properties2
137     declare -i LFBFL_i_renaming_happened=0
138     declare -i LFBFL_i
139     declare -i LFBFL_j
140     # declare -r LFBFL_special_file_name="files_names_listing.txt"
141     declare -r LFBFL_sfn="files_names_listing.txt"
142     local LFBFL_diff_line
143     local LFBFL_similarity_line
144     local LFBFL_old_file_path
145     local LFBFL_old_file_name
146     local LFBFL_old_file_directory
147     local LFBFL_new_file_path
148     local LFBFL_new_file_name
149     local LFBFL_new_file_directory
150     local LFBFL_useful_file_name
151     local LFBFL_extension
152     local LFBFL_comment_prefix
153     local LFBFL_comment_prefix2
154     local LFBFL_old_file_name2
155     local LFBFL_old_file_directory2
156     local LFBFL_new_file_name2
157     local LFBFL_new_file_directory2
```

```
158   local LFBFL_new_comment
159   declare -i LFBFL_i_copyright_line_number
160   declare -r LFBFL_send_summary_2="ATTENTION: No line with copyright"
161   declare -i LFBFL_i_line_count
162   declare -i LFBFL_i_lines_after
163   local LFBFL_temp_file_path
164   for ((LFBFL_i=0; LFBFL_i<${#LFBFL_renaming_lines_array[@]};)); do
165     LFBFL_diff_line="${LFBFL_renaming_lines_array[${LFBFL_i}]}"
166     LFBFL_i=$((LFBFL_i + 1))
167     LFBFL_similarity_line="${LFBFL_renaming_lines_array[${LFBFL_i}]}"
168     LFBFL_i=$((LFBFL_i + 1))
169     # We need to increment 3 times since grep adds "--" lines between.
170     LFBFL_i=$((LFBFL_i + 1))
171     if [[ LFBFL_i_verbose -eq 1 ]]; then
172       printf "LFBFL_diff_line: %s\nLFBFL_similarity_line: %s\n"\
173         "${LFBFL_diff_line}"\
174         "${LFBFL_similarity_line}"
175     fi
176     if [[ "${LFBFL_similarity_line:17:4}" != "100%" ]]; then
177       continue
178     fi
179
180     # Extract old file name from diff line. --------------------------
181     LFBFL_old_file_path=$(
182       printf "%s" "${LFBFL_diff_line}"\
183       | cut -d ' ' -f 3
184     )
185     # It starts by "a/".
186     LFBFL_old_file_path=".${LFBFL_old_file_path:1}"
187     LFBFL_old_file_name=$(basename "${LFBFL_old_file_path}")
188     LFBFL_old_file_directory=$(dirname "${LFBFL_old_file_path}")
189     # Extract new file name from diff line. --------------------------
190     LFBFL_new_file_path=$(
191       printf "%s" "${LFBFL_diff_line}"\
192       | cut -d ' ' -f 4
193     )
194     # It starts by "b/".
195     LFBFL_new_file_path=".${LFBFL_new_file_path:1}"
196     LFBFL_new_file_name=$(basename "${LFBFL_new_file_path}")
197     LFBFL_new_file_directory=$(dirname "${LFBFL_new_file_path}")
198     if [[ LFBFL_i_verbose -eq 1 ]]; then
199       LFBFL_s_format="LFBFL_old_file_path: %s\n"
200       LFBFL_s_format+="LFBFL_old_file_name: %s\n"
201       LFBFL_s_format+="LFBFL_old_file_directory: %s\n"
202       LFBFL_s_format+="LFBFL_new_file_path: %s\n"
203       LFBFL_s_format+="LFBFL_new_file_name: %s\n"
```

```
      LFBFL_s_format+="LFBFL_new_file_directory: %s\n"
      # shellcheck disable=SC2059
      printf "${LFBFL_s_format}"\
        "${LFBFL_old_file_path}"\
        "${LFBFL_old_file_name}"\
        "${LFBFL_old_file_directory}"\
        "${LFBFL_new_file_path}"\
        "${LFBFL_new_file_name}"\
        "${LFBFL_new_file_directory}"
    fi
    # Extract "LFBFL_extension" with smart guessing for .tpl ---------
    # and set prefix for line with ©Copyright. -----------------------
    LFBFL_useful_file_name="${LFBFL_new_file_name}"
    LFBFL_extension="${LFBFL_useful_file_name##*.}"
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_useful_file_name: %s\nLFBFL_extension: %s\n"\
        "${LFBFL_useful_file_name}"\
        "${LFBFL_extension}"
    fi
    for ((LFBFL_j=0; LFBFL_j<3; ++LFBFL_j)); do
      if [[ "${LFBFL_extension}" != "tpl" ]]; then
        break
      fi
      LFBFL_useful_file_name="${LFBFL_useful_file_name:0:-4}"
      LFBFL_extension="${LFBFL_useful_file_name##*.}"
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        printf "LFBFL_useful_file_name: %s\nLFBFL_extension: %s\n"\
          "${LFBFL_useful_file_name}"\
          "${LFBFL_extension}"
      fi
    done
    LFBFL_comment_prefix=""
    if [[ "${LFBFL_extension}" == "sh" ]]; then
      LFBFL_comment_prefix="# "
    fi
    if [[ "${LFBFL_extension}" == "sql" ]]; then
      LFBFL_comment_prefix="-- "
    fi
    if [[ "${LFBFL_extension}" == "tex" ]]; then
      LFBFL_comment_prefix="% "
    fi
    if [[ "${LFBFL_useful_file_name}" == "${LFBFL_sfn}" ]]; then
      LFBFL_comment_prefix="// "
    fi
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_comment_prefix: %s\n" "${LFBFL_comment_prefix}"
```

```
    fi
    #-------------------------------------------------------------------
    # Split if necessary old_file_name, new_file_name, etc. to have
    # comment lines below max_comment_line_length.
    LFBFL_comment_prefix2="${LFBFL_comment_prefix}"'"'
    # -----
    LFBFL_old_file_name2="${LFBFL_comment_prefix2}"
    LFBFL_old_file_name2+="${LFBFL_old_file_name}"
    # shellcheck disable=SC2248
    repeated_split_last_line "${LFBFL_old_file_name2}"\
      "${LFBFL_comment_prefix2}"\
      ${LFBFL_i_max_comment_line_length}\
      '"'\
      "${LFBFL_split_score_command1}"\
      ${LFBFL_i_split_score_command_properties1}
    LFBFL_old_file_name2="${repeated_split_last_line_result}"
    # -----
    LFBFL_old_file_directory2="${LFBFL_comment_prefix2}"
    LFBFL_old_file_directory2+="${LFBFL_old_file_directory}"
    # shellcheck disable=SC2248
    repeated_split_last_line "${LFBFL_old_file_directory2}"\
      "${LFBFL_comment_prefix2}"\
      ${LFBFL_i_max_comment_line_length}\
      '"'\
      "${LFBFL_split_score_command2}"\
      ${LFBFL_i_split_score_command_properties2}
    LFBFL_old_file_directory2="${repeated_split_last_line_result}"
    # -----
    LFBFL_new_file_name2="${LFBFL_comment_prefix2}"
    LFBFL_new_file_name2+="${LFBFL_new_file_name}"
    # shellcheck disable=SC2248
    repeated_split_last_line "${LFBFL_new_file_name2}"\
      "${LFBFL_comment_prefix2}"\
      ${LFBFL_i_max_comment_line_length}\
      '"'\
      "${LFBFL_split_score_command1}"\
      ${LFBFL_i_split_score_command_properties1}
    LFBFL_new_file_name2="${repeated_split_last_line_result}"
    # -----
    LFBFL_new_file_directory2="${LFBFL_comment_prefix2}"
    LFBFL_new_file_directory2+="${LFBFL_new_file_directory}"
    # shellcheck disable=SC2248
    repeated_split_last_line "${LFBFL_new_file_directory2}"\
      "${LFBFL_comment_prefix2}"\
      ${LFBFL_i_max_comment_line_length}\
      '"'\
```

```
      "${LFBFL_split_score_command2}"\
      ${LFBFL_i_split_score_command_properties2}
    LFBFL_new_file_directory2="${repeated_split_last_line_result}"
    # Create new comments strings. ----------------------------------
    LFBFL_new_comment=""
    if [[ "${LFBFL_new_file_name}" != "${LFBFL_old_file_name}" ]]; then
      LFBFL_new_comment="${LFBFL_comment_prefix}"
      LFBFL_new_comment+="${LFBFL_timestamp}"
      LFBFL_new_comment+=": This file was renamed from"
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_old_file_name2}"'"'
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_comment_prefix}to"
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_new_file_name2}"'".'
    fi
    if [[ LFBFL_i_log_directory_change -eq 1
      && "${LFBFL_new_file_directory}" != "${LFBFL_old_file_directory}"
    ]];
    then
      if [[ "${LFBFL_new_file_name}" != "${LFBFL_old_file_name}" ]]; then
        LFBFL_new_comment+=$'\n'
      fi
      LFBFL_new_comment+="${LFBFL_comment_prefix}"
      LFBFL_new_comment+="${LFBFL_timestamp}"
      LFBFL_new_comment+=": This file was moved from"
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_old_file_directory2}"'"'
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_comment_prefix}to"
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_new_file_directory2}"'".'
    fi
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_new_comment: %s\n" "${LFBFL_new_comment}"
    fi
    # Find line with ©Copyright. -------------------------------------
    LFBFL_i_copyright_line_number=$(
      grep -n '©Copyright' "${LFBFL_new_file_path}"\
      | head --lines=1\
      | cut -d ':' -f 1
    )
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_i_copyright_line_number: %s\n"\
        "${LFBFL_i_copyright_line_number}"
    fi
```

```
    if [[ LFBFL_i_copyright_line_number -eq 0 ]]; then
      notify-send "${LFBFL_send_summary_2}"\
        "${LFBFL_send_summary_2} for: ${LFBFL_new_comment}"
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        printf "No copyright line. Skipping file.\n"
      fi
      continue
    fi
    # Get total number of lines. -------------------------------------
    LFBFL_i_line_count=$(
      wc -l\
      < "${LFBFL_new_file_path}"
    )
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_i_line_count: %s\n" "${LFBFL_i_line_count}"
    fi
    # Compute number of lines after. ---------------------------------
    LFBFL_i_lines_after=$((
      LFBFL_i_line_count - LFBFL_i_copyright_line_number
    ))
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_i_lines_after: %s\n" "${LFBFL_i_lines_after}"
    fi
    # Update file. ---------------------------------------------------
    LFBFL_temp_file_path="${LFBFL_new_file_path}.LFBFL.temp"
    {
      head --lines="${LFBFL_i_copyright_line_number}"\
        "${LFBFL_new_file_path}"
      printf "%s\n" "${LFBFL_new_comment}"
      tail --lines="${LFBFL_i_lines_after}" "${LFBFL_new_file_path}"
    } >> "${LFBFL_temp_file_path}"
    mv "${LFBFL_temp_file_path}" "${LFBFL_new_file_path}"
    # Add file to stage. ---------------------------------------------
    git add "${LFBFL_new_file_path}"
    LFBFL_i_renaming_happened=1
  done
  readonly LFBFL_i_renaming_happened
  if [[ LFBFL_i_renaming_happened -eq 1 ]]; then
    # Commit ---------------------------------------------------------
    local LFBFL_message="commit_a_file_renamed_comment()"
    LFBFL_message+=" post-commit hook, ${LFBFL_timestamp}."
    readonly LFBFL_message
    printf "Commit message:\n%s\n" "${LFBFL_message}"
    git commit --no-verify --message="${LFBFL_message}"
  fi
}
```

3.58 git_helpers/pre-commit

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# of the GNU General Public License
# as published by the Free Software Foundation,
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# or (at your option) any later version.
#
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# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
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# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2024 Laurent Frédéric Bernard François Lyaudet
# 2024-05-31T01:57+02:00: This file was renamed from
# "pre-commit"
# to
# "pre-commit".
# 2024-05-31T01:57+02:00: This file was moved from
# "."
# to
# "./git_helpers".

LFBFL_some_directory="../DevOrSysAdminScripts"
declare -i LFBFL_i_pushd_result3
pushd "${LFBFL_some_directory}" || {
  LFBFL_i_pushd_result3=$?
  printf "pre-commit no such directory\n"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_pushd_result3}
}
# shellcheck source=git_checks.libr.sh
source "./git_helpers/git_checks.libr.sh"
declare -i LFBFL_i_popd_result3
popd || {
```

```
  LFBFL_i_popd_result3=$?
  printf "pre-commit no popd\n"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_popd_result3}
}

declare -gi LFBFL_i_git_hook_error=0
if ! check_files; then
  LFBFL_i_git_hook_error=1
fi
if ! check_no_abusive_trailing_comma; then
  LFBFL_i_git_hook_error=1
fi
if ! check_black_code_formatting; then
  LFBFL_i_git_hook_error=1
fi
readonly LFBFL_i_git_hook_error
# shellcheck disable=SC2248
exit ${LFBFL_i_git_hook_error}
```

3.59 IMAP_transfert.py

```
"""
This file is part of DevOrSysAdminScripts library.

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```

```
Pour transférer d'un compte IMAP à un autre
Penser à autoriser les apps moins sûres sur Gmail si nécessaire
"""

import argparse
import email
import imaplib


class IMAPTransfertBadArgumentError(Exception):
    """
    A class to avoid a linter warning,
    that needs an Error suffix to avoid another linter warning,
    that needs a docstring to avoid another linter warning.
    If you don't see its intent, try badminton.
    """


class SurpriseError(Exception):
    """
    A class to avoid a linter warning,
    that needs an Error suffix to avoid another linter warning,
    that needs a docstring to avoid another linter warning.
    If you don't see its intent after seeing its use, try volleyball.
    But "Cheer up!", now you don't need "pass" XD.
    """


def IMAP_transfert(
    from_login: str,
    from_password: str,
    from_host: str,
    from_port: int,
    to_login: str,
    to_password: str,
    to_host: str,
    to_port: int,
    *,
    to_new_folder: str = "",
    verbose: bool = False,
) -> None:
    """
    Transfert all emails from some IMAP account
    to another IMAP account.
    """
    empty = ""
```

```
    if from_login == "":
        raise IMAPTransfertBadArgumentError("From login is empty.")
    # if from_password == "":
    if from_password == empty:
        # https://github.com/PyCQA/bandit/issues/714
        # Dumb
        raise IMAPTransfertBadArgumentError("From password is empty.")
    if from_host == "":
        raise IMAPTransfertBadArgumentError("From host is empty.")
    if from_port <= 0:
        raise IMAPTransfertBadArgumentError(
            "From port is not positive.",
        )
    if to_login == "":
        raise IMAPTransfertBadArgumentError("To login is empty.")
    if to_password == empty:
        # And Dumber The infinite collection XD
        raise IMAPTransfertBadArgumentError("To password is empty.")
    if to_host == "":
        raise IMAPTransfertBadArgumentError("To host is empty.")
    if to_port <= 0:
        raise IMAPTransfertBadArgumentError(
            "To port is not positive.",
        )
    # pylint: disable=pointless-string-statement
    """
    Maybe one day Python will ship something like that in its standard
    library:
    https://www.php.net/manual/fr/function.filter-var.php
    But now urlparse doesn't do any validation.
    You may want to check that the hosts or other fields are not
    "anything".
    """

    # pylint: disable=unreachable
    raise SurpriseError(
        "I haven't tested this version of my script... Oooooh!\n"
        "But I linted it without this exception ;). Aaaaaaaah!\n"
        "XD XD XD\n"
        "Untested Scream Horror Show (Soon on BrainfuckedFlix)",
    )
    # """
    # I haven't tested any version with an empty to_new_folder...

    if verbose:
        print("Connecting to from account.")
```

```
    from_account = imaplib.IMAP4_SSL(host=from_host, port=from_port)
    from_account.login(from_login, from_password)

    if verbose:
        print('Connecting to "to account".')
    to_account = imaplib.IMAP4_SSL(host=to_host, port=to_port)
    to_account.login(to_login, to_password)

    if to_new_folder != "":
        if verbose:
            print("Creating new folder in to account.")
        to_account.create(to_new_folder)

    if verbose:
        print("Fetching all emails ids in from account.")
    from_account.select()
    result, ids = from_account.uid("search", "all")
    if verbose:
        print(f'from_account.uid("search") result:{result} ids:{ids}')
    ids = ids[0].decode().split()
    if verbose:
        print("Fetching and transfering each email in from account.")
    for my_id in ids:
        result, raw_message = from_account.uid(
            "fetch",
            my_id,
            "(RFC822)",
        )
        if verbose:
            print(f'from_account.uid("fetch", {my_id}) result:{result}')
        message = raw_message[0][1]
        my_email = email.message_from_string(message.decode("utf-8"))
        flags = imaplib.ParseFlags(raw_message[0][0])
        # flags = " ".join(flags) mypy is unhappy but IMAP flags are
        # ASCII strings, we give mypy what it wants.
        flags_as_strs = [flag.decode("ascii") for flag in flags]
        flags_str = " ".join(flags_as_strs)
        # date = imaplib.Internaldate2tuple(raw_message[0][0])
        date = '"' + my_email["date"] + '"'
        if verbose:
            print(
                f'from_account.uid("fetch", {my_id})'
                f" flags:{flags} date:{date}",
            )
        to_account.append(to_new_folder, flags_str, date, message)

```

```
def main() -> None:
    """
    Parse command-line arguments and
    transfert all emails from some IMAP account
    to another IMAP account.
    """
    parser = argparse.ArgumentParser(
        description=(
            "Transfert all emails from some IMAP account"
            " to another IMAP account"
        ),
    )
    parser.add_argument(
        "from_login",
        type=str,
        help="Email account login of user one.",
    )
    parser.add_argument(
        "from_password",
        type=str,
        help="Email account password of user one.",
    )
    parser.add_argument(
        "to_login",
        type=str,
        help="Email account login of user two.",
    )
    parser.add_argument(
        "to_password",
        type=str,
        help="Email account password of user two.",
    )
    parser.add_argument(
        "--to_new_folder",
        type=str,
        default="",
        help="Transfert emails into a new folder of this name.",
    )
    parser.add_argument(
        "--from_host",
        type=str,
        default="imap.gmail.com",
        help="Host of first email account",
    )
    parser.add_argument(
```

```
        "--from_port",
        type=int,
        default=993,
        help="Port on host of first email account",
    )
    parser.add_argument(
        "--to_host",
        type=str,
        default="imap.gmail.com",
        help="Host of second email account",
    )
    parser.add_argument(
        "--to_port",
        type=int,
        default=993,
        help="Port on host of second email account",
    )
    parser.add_argument(
        "-v",
        "--verbose",
        action="store_true",
        help="Print details of operations",
    )

    args = parser.parse_args()

    if args.verbose:
        print(
            "-----------------------------------------------------\n"
            "IMAP transfert:\n"
            "-----------------------------------------------------\n"
            "From account:\n"
            f" - Login: {args.from_login}\n"
            f" - Password: {args.from_password}\n"
            f" - Host: {args.from_host}\n"
            f" - Port: {args.from_port}\n"
            "To account:\n"
            f" - Login: {args.to_login}\n"
            f" - Password: {args.to_password}\n"
            f" - Host: {args.to_host}\n"
            f" - Port: {args.to_port}\n"
            "-----------------------------------------------------\n",
        )
        if args.to_new_folder is not None:
            print(
                f"Emails will go in folder {args.to_new_folder}"
```

```
                " after it is created.",
            )
    try:
        IMAP_transfert(
            args.from_login,
            args.from_password,
            args.from_host,
            args.from_port,
            args.to_login,
            args.to_password,
            args.to_host,
            args.to_port,
            to_new_folder=args.to_new_folder,
            verbose=args.verbose,
        )
    except (IMAPTransfertBadArgumentError, SurpriseError) as e:
        print(e)


if __name__ == "__main__":
    main()
```

3.60 LiensUtiles.txt

```
https://support.google.com/a/answer/9476255?hl=fr#zippy=%2C%C3%A9tape\
-configurez-le-service-de-migration-de-donn%C3%A9es
----------------------------------------------------------------------
https://stackoverflow.com/a/78413334/5796086
https://stackoverflow.com/questions/21612980/\
why-is-usr-bin-env-bash-superior-to-bin-bash/78413334
https://web.archive.org/web/20240501105801/https://stackoverflow.com\
/questions/21612980/why-is-usr-bin-env-bash-superior-to-bin-bash/\
78413334#78413334
----------------------------------------------------------------------
https://news.ycombinator.com/item?id=22214690
https://github.com/koalaman/shellcheck/issues/2005
https://opensource.stackexchange.com/questions/6271/\
calling-gpld-programs-from-a-shell-script
----------------------------------------------------------------------
```

3.61 miscellaneous/funny.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
```

```
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unprotected_funny(){
  # Works in a script but not in shell if job control is active.
  shopt -s lastpipe
  declare -i LFBFL_i_result=0
  local LFBFL_line
  printf "hahaha\n" | while read -r LFBFL_line; do
    if [[ "${LFBFL_line}" == "hahaha" ]]; then
      LFBFL_i_result=1
    fi
  done
  [[ LFBFL_i_result -eq 1 ]] && printf "I had a good laugh.\n"
}

protected_funny(){
  # Always works regarding the echoes.
  printf "hahaha\n" | while read -r LFBFL_line; do
    unprotected_funny
  done
}
```

3.62 miscellaneous/spampty_data.txt

```


a
```

```
 a
empty.txt:
one_space.txt:
all_space.txt:
normal.txt:a
normal_again.txt: a
```

3.63 miscellaneous/string_escaping.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <title>Truc</title>
</head>
<body>
<header id="title-block-header">
<h1 class="title">Truc</h1>
</header>
&lt;abc> &amp; &lt;de>
<p title="<&>'
 abcd`\ machin">
This is some text
</p>

<pre>
"&lt;&amp;>'&quot;
abcd`\ machin"
This is some text
</pre>

<script>
let a = 'abcd & < > \\ \n  \"\'';
alert(a);
let b = "abcd & < > \\ \n  \"\'";
alert(b);
</script>

</body>
</html>
```

3.64 miscellaneous/string_escaping.xhtml

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
          "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml" xml:lang="fr" lang="fr">
```

```
<head>
<meta content="text/html; charset=UTF-8" http-equiv="Content-Type" />
  <title>Truc</title>
</head>
<body>
<h1 class="title">Truc</h1>
&lt;abc> &amp; &lt;de>

<p title="&lt;&amp;>'&#10;
 abcd`\ machin">
This is some text
</p>

<pre>
"&lt;&amp;>'&quot;
abcd`\ machin"
This is some text
</pre>

<script type="text/javascript">
let a = 'abcd &amp; &lt; > \\ \n  \"\'';
alert(a);
let b = "abcd &amp; &lt; > \\ \n  \"\'";
alert(b);
</script>

</body>
</html>
```

3.65 README.md.tpl

```
# DevOrSysAdminScripts

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[![Linting:pylint:badge]][Pylint-GitHub.com]
[![Linting:ruff:badge]][Ruff-GitHub.com]

|                **Some "small" useful scripts**               |
|:-------------------------------------------------------------:|
```

```
| **for enhancing development or system-administrators tasks** |

Currently, the biggest part of this repository is in
a suite of shell scripts about build and checks that
I use in some of my other repositories.

I tried to follow some conventions for my bash code:

- I tried to encapsulate most of the code in functions.
- Unless used as return values, variables in functions are defined
  as local variables (`local` keyword or `declare`)
  with prefix "LFBFL_".
- When used as return values, variables in functions are defined
  as global variables (`declare -g`) with prefix
  "@function_name@_result_".
- Whenever a variable is no more modified after some point,
  add the keyword `readonly` or use `declare -r`.
- Whenever a variable will only contain integer values (or boolean
  values as 0 or 1), use `declare -i`.
- I named bash files with suffixes ".exec.sh" when the script can
  be runned
  (executable, abbreviation stops before a vowel).
- I named bash files with suffixes ".libr.sh" when the script contains
  only functions definitions
  (library, abbreviation stops before a vowel).
- But the files "pre-commit" and "post-commit" were not renamed,
  since it is not possible to give them other names to use them as
  pre-commit/post-commit hooks in `git`.
- When some ".exec.sh" code is encapsulated into functions,
  either these functions can be reused and go in some ".libr.sh" file,
  either they are truly specific to this script and they are kept
  in ".exec.sh" file.
- When using `source` builtin, although any path/filename with a slash
  will only be searched for from the current directory (tested),
  we add the prefix "./", to make it explicit.
- We use mapfile instead of read -r most of the time, but with the
  following exceptions: real use of read -r to read the command line of the
  shell, comments (of course), and a file funny.sh where it is kind of
  required to use read -r.
- We replaced all echo calls by printf calls to follow
  "The Open Group Base Specifications" since at least 2 decades:
  "New applications are encouraged to use printf instead of echo.",
  <https://pubs.opengroup.org/onlinepubs/000095399/>.
- We used `_?i_` or `_?j_`, etc. in integer variables names in Bash
  scripts.

```

```
[CodeFactor-badge]: https://www.codefactor.io/repository/github/\
llyaudet/DevOrSysAdminScripts/badge

[CodeFactor-package-page]: https://www.codefactor.io/repository/\
github/llyaudet/DevOrSysAdminScripts

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3dd6ef85e8db4460991d1ebc404b68dd

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DevOrSysAdminScripts/dashboard?utm_source=gh&utm_medium=referral\
&utm_content=&utm_campaign=Badge_grade

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languages/top/llyaudet/DevOrSysAdminScripts

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languages/code-size/llyaudet/DevOrSysAdminScripts

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security-bandit-yellow.svg

[Bandit-GitHub.com]: https://github.com/PyCQA/bandit

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code%20style-black-000000.svg

[Black-GitHub.com]: https://github.com/psf/black

[Imports:isort:badge]: https://img.shields.io/badge/\
%20imports-isort-%231674b1?style=flat&labelColor=ef8336

[Isort-GitHub.io]: https://pycqa.github.io/isort/

[Typecheck:mypy:badge]: https://www.mypy-lang.org/static/\
mypy_badge.svg

[Typecheck-mypy-lang.org]: https://mypy-lang.org/

[Linting:pylint:badge]: https://img.shields.io/badge/\
linting-pylint-yellowgreen

[Pylint-GitHub.com]: https://github.com/pylint-dev/pylint

[Linting:ruff:badge]: https://img.shields.io/endpoint?url=\
https://raw.githubusercontent.com/astral-sh/ruff/main/assets/\
```

```
badge/v2.json

[Ruff-GitHub.com]: https://github.com/astral-sh/ruff
```

3.66 string_escaping.libr.php

```
<?php

/**
This file is part of DevOrSysAdminScripts library.

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as published by the Free Software Foundation,
either version 3 of the License,
or (at your option) any later version.

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that it will be useful,
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@category Library
@package DevOrSysAdminScripts
@author Laurent Lyaudet <laurent.lyaudet@gmail.com>
@copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
@license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
*/

declare(strict_types=1);
declare(encoding='UTF-8');



/**
The namespace string_escaping is splitted into subnamespaces for each
```

```
language.
This subnamespace is for HTML (surprised?)...

@package DevOrSysAdminScripts
@subpackage string_escaping\HTML
*/
namespace string_escaping\HTML {
  /**
  When you want that the string appears as is in HTML text block.

  @param string $s_string The string you want to escape.

  @return string The escaped string.
  */
  function escape_text(string $s_string) : string {
    return str_replace(
      [
        '&',
        '<',
      ],
      [
        '&amp;',
        '&lt;',
      ],
      $s_string,
    );
  }//end escape_text()



  /**
  When you want that the string appears as is in an HTML tag attribute
  that starts and ends with double quotes: my_attribute="something".
  Example: a title attribute.
  /!\ in double quotes /!\ my_attribute='something' is not supported.

  @param string $s_string The string you want to escape.

  @return string The escaped string.
  */
  function escape_attribute(string $s_string) : string {
    return str_replace(
      ['"'],
      ['&quot;'],
      $s_string,
    );
```

```
  }//end escape_attribute()



  /**
  When you want that the string appears verbatim inside HTML <pre> tags.

  @param string $s_string The string you want to escape.

  @return string The escaped string.
  */
  function escape_pre(string $s_string) : string {
    return str_replace(
      [
        '&',
        '<',
      ],
      [
        '&amp;',
        '&lt;',
      ],
      $s_string,
    );
  }//end escape_pre()
}//end namespace string_escaping\HTML




/**
The namespace string_escaping is splitted into subnamespaces for each
language.
This subnamespace is for XHTML (still surprised?)...

@package DevOrSysAdminScripts
@subpackage string_escaping\XHTML
*/
namespace string_escaping\XHTML {
  /**
  When you want that the string appears as is in XHTML text block.

  @param string $s_string The string you want to escape.

  @return string The escaped string.
  */
  function escape_text(string $s_string) : string {
```

```
    return str_replace(
      [
        '&',
        '<',
      ],
      [
        '&amp;',
        '&lt;',
      ],
      $s_string,
    );
  }//end escape_text()



  /**
  When you want that the string appears as is in an XHTML tag attribute
  that starts and ends with double quotes: my_attribute="something".
  Example: a title attribute.
  /!\ in double quotes /!\ my_attribute='something' is not supported.

  @param string $s_string The string you want to escape.

  @return string The escaped string.
  */
  function escape_attribute(string $s_string) : string {
    return str_replace(
      [
        '&',
        '<',
        '"',
        "\n",
      ],
      [
        '&amp;',
        '&lt;',
        '&quot;',
        '&#10;',
      ],
      $s_string,
    );
  }//end escape_attribute()



  /**
```

```
  When you want that the string appears verbatim inside XHTML <pre> tags.

  @param string $s_string The string you want to escape.

  @return string The escaped string.
  */
  function escape_pre(string $s_string) : string {
    return str_replace(
      [
        '&',
        '<',
      ],
      [
        '&amp;',
        '&lt;',
      ],
      $s_string,
    );
  }//end escape_pre()
}//end namespace string_escaping\XHTML




/**
The namespace string_escaping is splitted into subnamespaces for each
language.
This subnamespace is for JS (you're hopeless?)...

@package DevOrSysAdminScripts
@subpackage string_escaping\JS
*/
namespace string_escaping\JS {

  use function string_escaping\XHTML\escape_text as XHTML_escape_text;



  /**
  When you want that the string can be put in a JS string definition.

  @param string $s_string The string you want to escape.

  @return string The escaped string.
  */
  function escape(string $s_string) : string {
```

```
    return str_replace(
      [
        '\\',
        "\n",
        "'",
        '"',
      ],
      [
        '\\\\',
        '\n',
        '\\\'',
        '\\"',
      ],
      $s_string,
    );
  }//end escape()



  /**
  When you want that the string can be put in a JS string definition
  to be included inside an XHTML file <script> tag.

  @param string $s_string The string you want to escape.

  @return string The escaped string.
  */
  function escape_for_string_definition_inside_XHTML(string $s_string)
  : string {
    return XHTML_escape_text(escape($s_string));
  }//end escape_for_string_definition_inside_XHTML()
}//end namespace string_escaping\JS




/**
The namespace string_escaping is splitted into subnamespaces for each
language.
This subnamespace is for TeX.

@package DevOrSysAdminScripts
@subpackage string_escaping\TeX
*/
namespace string_escaping\TeX {
  /**
```

```
  When you want that the string can be written to a TeX file inside a
  part where TeX is in text mode.

  @param string $s_string The string you want to escape.

  @return string The escaped string.
  */
  function escape_text(string $s_string) : string {
    // https://tex.stackexchange.com/a/34586/270015
    return str_replace(
      [
        '\\',
        '&',
        '%',
        '$',
        '#',
        '_',
        '{',
        '}',
        '~',
        '^',
      ],
      [
        '\\textbackslash',
        '\\&',
        '\\%',
        '\\$',
        '\\#',
        '\\_',
        '\\{',
        '\\}',
        '\\textasciitilde',
        '\\textasciicircum',
      ],
      $s_string,
    );
  }//end escape_text()
}//end namespace string_escaping\TeX

/*
$ php
<?php
require_once('string_escaping.libr.php');
var_dump(string_escaping\HTML\escape_text('<&"\'truc\\'));
var_dump(string_escaping\HTML\escape_attribute("<&\"\n'truc\\"));
var_dump(string_escaping\HTML\escape_pre("<&\"\n'truc\\"));
```

```
var_dump(string_escaping\XHTML\escape_text('<&"\'truc\\'));
var_dump(string_escaping\XHTML\escape_attribute("<&\"\n'truc\\"));
var_dump(string_escaping\XHTML\escape_pre("<&\"\n'truc\\"));
var_dump(string_escaping\JS\escape("abcd & < > \\ \n  \"'"));
var_dump(string_escaping\JS\escape_for_string_definition_inside_XHTML(
  "abcd & < > \\ \n  \"'"
));
*/
?>
```

3.67 unicode.libr.php

```
<?php

/**
This file is part of DevOrSysAdminScripts library.

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of the GNU Lesser General Public License
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either version 3 of the License,
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This file was renamed from "unicode.php" to "unicode.libr.php".

@category Library
@package DevOrSysAdminScripts
@author Laurent Lyaudet <laurent.lyaudet@gmail.com>
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@license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
*/

```

```
declare(strict_types=1);
declare(encoding='UTF-8');



/**
Converts an int to the byte-string corresponding to the UTF-8 encoding
of the Unicode code point of same integer value.
decimal_code_point is a slight abuse of language to say that you write the
code point in decimal notation in PHP code and you get the UTF-8.
If you provide the integer of the code point without using decimal
notation, it will also work.

@param int $i_code_point_in_decimal_notation The code point.

@throws Exception When the code point is outside of Unicode range.

@return string The corresponding UTF-8 character.
*/
function decimal_code_point_to_UTF8(
  int $i_code_point_in_decimal_notation
) : string {
  // var_dump($i_code_point_in_decimal_notation);
  // 0xxxxxxx ASCII
  if($i_code_point_in_decimal_notation < 128){
    return chr($i_code_point_in_decimal_notation);
  }
  $i_continuation_base_value = 128;
  // 110xxxxx 10xxxxxx
  if($i_code_point_in_decimal_notation < 2048){  // 2**11
    $i_first_byte_base_value = 192;
    $i_last_byte_significant_bits = (
      $i_code_point_in_decimal_notation % 64  // 2**6
    );
    $i_first_byte_significant_bits = intdiv(
      (
        $i_code_point_in_decimal_notation
        - $i_last_byte_significant_bits
      ),
      64,  // 2**6
    );
    $s_result = chr(
      $i_first_byte_base_value
      + $i_first_byte_significant_bits
    );
    $s_result .= chr(
```

```
      $i_continuation_base_value
      + $i_last_byte_significant_bits
    );
    return $s_result;
  }//end if($i_code_point_in_decimal_notation < 2048)
  // 1110xxxx 10xxxxxx 10xxxxxx
  if($i_code_point_in_decimal_notation < 65536){  // 2**16
    $i_first_byte_base_value = 224;
    $arr_arr_data_per_byte_reverse = [
      ['i_bits' => 6, 'i_base_value' => $i_continuation_base_value],
      ['i_bits' => 6, 'i_base_value' => $i_continuation_base_value],
      ['i_bits' => 4, 'i_base_value' => $i_first_byte_base_value],
    ];
    $i_rest = $i_code_point_in_decimal_notation;
    $s_result = '';
    foreach($arr_arr_data_per_byte_reverse as $arr_data){
      // _Pragma unroll
      // J'aurais pu "roller" les 4 cas des ifs.
      // Mais pour les perfs, il faudrait tout unroller vu le nombre
      // d'itérations. Et j'aimerais bien comparer du code unrollé
      // aux 2 niveaux à la main avec celui unrollé par le compilateur
      // sur les 2 niveaux. Je préjuge/présuppose qu'il unroll
      // l'intérieur d'abord (pour ne pas refaire le même boulot)
      // puis l'extérieur et qu'il oublie de réoptimiser l'intérieur
      // en fonction de l'extérieur unrollé. Est-ce que j'ai raison ?
      $i_significant_bits = $i_rest % (2**$arr_data['i_bits']);
      $i_rest = intdiv(
        $i_rest - $i_significant_bits,
        2**$arr_data['i_bits'],
      );
      // .= or =.
      // Below could have been a case of "=.".
      $s_result = (
        chr($i_significant_bits + $arr_data['i_base_value'])
        .$s_result
      );
    }//end foreach($arr_arr_data_per_byte_reverse as $arr_data)
    return $s_result;
  }//end if($i_code_point_in_decimal_notation < 65536)
  // 11110xxx 10xxxxxx 10xxxxxx 10xxxxxx
  if($i_code_point_in_decimal_notation < 2097152){  // 2**21
    $i_first_byte_base_value = 240;
    $arr_arr_data_per_byte_reverse = [
      ['i_bits' => 6, 'i_base_value' => $i_continuation_base_value],
      ['i_bits' => 6, 'i_base_value' => $i_continuation_base_value],
      ['i_bits' => 6, 'i_base_value' => $i_continuation_base_value],
```

```
      ['i_bits' => 3, 'i_base_value' => $i_first_byte_base_value],
    ];
    $i_rest = $i_code_point_in_decimal_notation;
    $s_result = '';
    foreach($arr_arr_data_per_byte_reverse as $arr_data){
      // _Pragma unroll
      $i_significant_bits = $i_rest % (2**$arr_data['i_bits']);
      $i_rest = intdiv(
        $i_rest - $i_significant_bits,
        2**$arr_data['i_bits'],
      );
      // .= or =.
      // Below could have been a case of "=.".
      $s_result = (
        chr($i_significant_bits + $arr_data['i_base_value'])
        .$s_result
      );
    }
    return $s_result;
  }//end if($i_code_point_in_decimal_notation < 2097152)
  // phpcs:disable Squiz.Strings.DoubleQuoteUsage.NotRequired
  throw new Exception(
    "UTF-8 avec jusqu'à 6 octets a été abandonné il y a longtemps."
    ." Autre anecdote \"débile\" : MySQL a de base UTF-8 avec jusqu'à"
    ." 3 octets au lieu de 4 :face-palm:."
    ." Du coup, ça peut donner quelques \"surprises\"."
    ." Sinon, y'a aussi ASCII dans MySQL."
    ." Et comme ça joue sur la mémoire et les performances,"
    ." pourquoi ils n'ont pas fait ASCII, UTF-8-2, UTF-8-3, UTF-8-4 ?"
    ." :)"
  );
  // phpcs:enable Squiz.Strings.DoubleQuoteUsage.NotRequired
}//end decimal_code_point_to_UTF8()



/**
Converts a string of hexadecimal digits to the byte-string corresponding
to the UTF-8 encoding of the Unicode code point of same integer value
that is denoted by the hexadecimal digits.

@param string $s_code_point_in_hexadecimal_notation The code point.

@throws Exception When the code point is outside of Unicode range.

@return string The corresponding UTF-8 character.
```

```
173 */
174 function hexa_code_point_to_UTF8(
175   string $s_code_point_in_hexadecimal_notation
176 ) : string {
177   $s_code_point_in_hexadecimal_notation = strtoupper(
178     $s_code_point_in_hexadecimal_notation
179   );
180   $i_code_point_in_decimal_notation = 0;
181   $i_ord_0 = ord('0');
182   $i_ord_A = ord('A');
183   $i_digit_offset = -$i_ord_0;
184   $i_letter_offset = 10 - $i_ord_A;
185   for(
186     $i = 0, $i_max = strlen($s_code_point_in_hexadecimal_notation);
187     $i < $i_max;
188     ++$i
189   ){
190     $s_hexa_digit = $s_code_point_in_hexadecimal_notation[$i];
191     $i_code_point_in_decimal_notation *= 16;
192     if(ctype_digit($s_hexa_digit)){
193       $i_code_point_in_decimal_notation += (
194         ord($s_hexa_digit) + $i_digit_offset
195       );
196       continue;
197     }
198     if(ctype_xdigit($s_hexa_digit)){
199       $i_code_point_in_decimal_notation += (
200         ord($s_hexa_digit) + $i_letter_offset
201       );
202       continue;
203     }
204     throw new Exception('Not an hexadecimal digit.');
205   }
206   return decimal_code_point_to_UTF8(
207     $i_code_point_in_decimal_notation
208   );
209 }//end hexa_code_point_to_UTF8()
210
211
212
213 /**
214 This function throws an exception with extended debug informations
215 if the input string is not valid ASCII.
216 Otherwise, it returns true.
217
218 @param string $s_string The input string.
```

```
@throws Exception When the input string is not valid ASCII.

@return bool
*/
function check_string_is_valid_ASCII(string $s_string) : bool {
  $i_offset_in_octets_from_string_start = 0;
  $i_current_line_number = 1;
  $i_offset_in_octets_from_line_start = -1;
  for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i){
    $i_current_octet = ord($s_string[$i]);
    $i_offset_in_octets_from_string_start = $i;
    ++$i_offset_in_octets_from_line_start;
    if($i_current_octet > 127){
      throw new Exception(
        'Non-ASCII character found on line '
        .$i_current_line_number
        .'; the octet/character '
        .($i_offset_in_octets_from_line_start + 1)
        .' has value '
        .$i_current_octet
        .' which is not ASCII.'
        .' (Sequential position without line splitting:'
        .' This is at octet/character '
        .($i_offset_in_octets_from_string_start + 1)
        .'.)',
      );
    }
    // if($s_string[$i] === "\n"){
    if($i_current_octet === 10){
      ++$i_current_line_number;
      $i_offset_in_octets_from_line_start = -1;
    }
  }//for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i)
  return true;
}//end check_string_is_valid_ASCII()



/**
This function throws an exception with extended debug informations
if the content of the file with given file_path is not valid ASCII.
Otherwise, it returns true.

@param string $s_file_path The input file_path.

```

```
@throws Exception When the input string is not valid ASCII.

@return bool
*/
function check_file_is_valid_ASCII(string $s_file_path) : bool {
  $s_string = file_get_contents($s_file_path);
  if($s_string === false){
    throw new Exception('File '.$s_file_path.' not found.');
  }
  return check_string_is_valid_ASCII($s_string);
}//end check_file_is_valid_ASCII()



/**
This function throws an exception with extended debug informations
if the input string is not valid UTF-8.
Otherwise, it returns true.

@param string $s_string The input string.

@throws Exception When the input string is not valid UTF-8.

@return bool
*/
function check_string_is_valid_UTF8(string $s_string) : bool {
  $i_continuation_octet_needed = 0;
  $i_offset_in_octets_from_string_start = 0;
  $i_offset_in_characters_from_string_start = -1;
  $i_character_start_position_from_string_start = 0;
  $i_current_line_number = 1;
  $i_offset_in_octets_from_line_start = -1;
  $i_offset_in_characters_from_line_start = -1;
  $i_character_start_position_from_line_start = -1;
  for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i){
    $i_current_octet = ord($s_string[$i]);
    $i_offset_in_octets_from_string_start = $i;
    ++$i_offset_in_octets_from_line_start;
    if($i_continuation_octet_needed > 0){
      if($i_current_octet < 128 || $i_current_octet >= 192){
        throw new Exception(
          'Non-UTF8 character found on line '
          .$i_current_line_number
          .'; the octet '
          .($i_offset_in_octets_from_line_start + 1)
          .', part of the character '
```

```
311           .($i_offset_in_characters_from_line_start + 1)
312           .', has value '
313           .$i_current_octet
314           .' which is not a continuation octet.'
315           .' This character starts at octet '
316           .($i_character_start_position_from_line_start + 1)
317           .' of the current line.'
318           .' (Sequential positions without line splitting:'
319           .' This is at character '
320           .($i_offset_in_characters_from_string_start + 1)
321           .' and octet '
322           .($i_offset_in_octets_from_string_start + 1)
323           .'.'
324           .' This character starts at octet '
325           .($i_character_start_position_from_string_start + 1)
326           .'.)',
327         );
328       }//end if($i_current_octet < 128 || $i_current_octet >= 192)
329       --$i_continuation_octet_needed;
330       continue;
331     }//end if($i_continuation_octet_needed > 0)
332
333     ++$i_offset_in_characters_from_string_start;
334     ++$i_offset_in_characters_from_line_start;
335     $i_character_start_position_from_string_start = (
336       $i_offset_in_octets_from_string_start
337     );
338     $i_character_start_position_from_line_start = (
339       $i_offset_in_octets_from_line_start
340     );
341     if($i_current_octet < 128){ // 0xxxxxxx ASCII
342       // if($s_string[$i] === "\n"){
343       if($i_current_octet === 10){
344         ++$i_current_line_number;
345         $i_offset_in_octets_from_line_start = -1;
346         $i_offset_in_characters_from_line_start = -1;
347       }
348       continue;
349     }
350     if($i_current_octet >= 128 && $i_current_octet < 192){
351       throw new Exception(
352         'Non-UTF8 character found on line '
353         .$i_current_line_number
354         .'; the octet '
355         .($i_offset_in_octets_from_line_start + 1)
356         .', part of the character '
```

```
357         .($i_offset_in_characters_from_line_start + 1)
358         .', has value '
359         .$i_current_octet
360         .' which is a continuation octet.'
361         .' This character starts at octet '
362         .($i_character_start_position_from_line_start + 1)
363         .' of the current line.'
364         .' (Sequential positions without line splitting:'
365         .' This is at character '
366         .($i_offset_in_characters_from_string_start + 1)
367         .' and octet '
368         .($i_offset_in_octets_from_string_start + 1)
369         .'.'
370         .' This character starts at octet '
371         .($i_character_start_position_from_string_start + 1)
372         .'.)',
373       );
374     }//end if($i_current_octet >= 128 && $i_current_octet < 192)
375     if($i_current_octet >= 192 && $i_current_octet < 224){
376       // 110xxxxx 10xxxxxx
377       $i_continuation_octet_needed = 1;
378       continue;
379     }
380     if($i_current_octet >= 224 && $i_current_octet < 240){
381       // 1110xxxx 10xxxxxx 10xxxxxx
382       $i_continuation_octet_needed = 2;
383       continue;
384     }
385     if($i_current_octet >= 240 && $i_current_octet < 248){
386       // 11110xxx 10xxxxxx 10xxxxxx 10xxxxxx
387       $i_continuation_octet_needed = 3;
388       continue;
389     }
390     throw new Exception(
391       'Non-UTF8 character found on line '
392       .$i_current_line_number
393       .'; the octet '
394       .($i_offset_in_octets_from_line_start + 1)
395       .', part of the character '
396       .($i_offset_in_characters_from_line_start + 1)
397       .', has value '
398       .$i_current_octet
399       .' which is invalid.'
400       .' This character starts at octet '
401       .($i_character_start_position_from_line_start + 1)
402       .' of the current line.'
```

```
      .' (Sequential positions without line splitting:'
      .' This is at character '
      .($i_offset_in_characters_from_string_start + 1)
      .' and octet '
      .($i_offset_in_octets_from_string_start + 1)
      .'.'
      .' This character starts at octet '
      .($i_character_start_position_from_string_start + 1)
      .'.)',
    );
  }//end for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i)
  return true;
}//end check_string_is_valid_UTF8()



/**
This function throws an exception with extended debug informations
if the content of the file with given file_path is not valid ASCII.
Otherwise, it returns true.

@param string $s_file_path The input file_path.

@throws Exception When the input string is not valid UTF-8.

@return bool
*/
function check_file_is_valid_UTF8(string $s_file_path) : bool {
  $s_string = file_get_contents($s_file_path);
  if($s_string === false){
    throw new Exception('File '.$s_file_path.' not found.');
  }
  return check_string_is_valid_UTF8($s_string);
}//end check_file_is_valid_UTF8()



/*
?>
<?php
require_once("./unicode.libr.php");
var_dump(hexa_code_point_to_UTF8("002B"));
var_dump(hexa_code_point_to_UTF8("00E6"));
var_dump(hexa_code_point_to_UTF8("1400"));
var_dump(hexa_code_point_to_UTF8("10111"));
// https://en.wikibooks.org/wiki/Unicode/Character_reference/2000-2FFF
```

```
// Some invisible unicode characters that are a problem:
// 2000-200F 2028-202F 205F 2060-206F
$arr = [
  '2000', '2001', '2002', '2003', '2004', '2005', '2006', '2007',
  '2008', '2009', '200A', '200B', '200C', '200D', '200E', '200F',
  '2028', '2029', '202A', '202B', '202C', '202D', '202E', '202F',
  '205F',
  '2060', '2061', '2062', '2063', '2064', '2065', '2066', '2067',
  '2068', '2069', '206A', '206B', '206C', '206D', '206E', '206F',
];
$arr_s = [];
$arr_s2 = [];
$arr_s3 = [];
foreach($arr as $s_hexa){
  $s = hexa_code_point_to_UTF8($s_hexa);
  $arr_s []= $s;
  if(strlen($s) === 2){
    $s2 = '\x'.bin2hex($s[0]).'\x'.bin2hex($s[1]);
  }
  if(strlen($s) === 3){
    $s2 = '\x'.bin2hex($s[0]).'\x'.bin2hex($s[1])
      .'\x'.bin2hex($s[2]);
  }
  $arr_s2 []= $s2;
  $s3 = "  | sed -e 's/".$s2."/ /g'\\";
  $arr_s3 []= $s3;
  echo($s3."\n");
}
//*/
?>
```

3.68 upgrade_build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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```

```
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# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "upgrade_build_and_checks.sh"
# to "upgrade_build_and_checks.exec.sh".

upgrade_build_and_checks(){
  # Options:
  #   --check-download (=keep to keep the downloaded files)
  #   --fixed-point-build
  #   --verbose
  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
  fi
  local LFBFL_fixed_point_build=""
  if [[ "$*" == *--fixed-point-build* ]]; then
    LFBFL_fixed_point_build="--fixed-point-build"
  fi
  readonly LFBFL_verbose
  local LFBFL_check_download=""
  if [[ "$*" == *--check-download=keep* ]]; then
    LFBFL_check_download="--check-download=keep"
  elif [[ "$*" == *--check-download* ]]; then
    LFBFL_check_download="--check-download"
  fi
  readonly LFBFL_check_download
  (\
    cd build_and_checks_dependencies/\
    && ./update_common_build_and_checks.exec.sh "${LFBFL_verbose}"\
  )
  ./build_and_checks.exec.sh "."\
    "${LFBFL_verbose}"\
    "${LFBFL_fixed_point_build}"\
    "${LFBFL_check_download}"
}

upgrade_build_and_checks "$@"
```

3.69 wget_sha512.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# the GNU General Public License
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# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "wget_sha512.sh"
# to "wget_sha512.libr.sh".

wget_sha512(){
  # $1=file_path
  # $2=download_URL
  # $3=correct_sha512
  # Options:
  #   --check-download (=keep to keep the downloaded file)
  #   --verbose
  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    printf "%s wget_sha512 %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
  fi
  readonly LFBFL_verbose
  local LFBFL_check_download=""
  if [[ "$*" == *--check-download=keep* ]]; then
    LFBFL_check_download="--check-download=keep"
  elif [[ "$*" == *--check-download* ]]; then
    LFBFL_check_download="--check-download"
```

```
  fi
  readonly LFBFL_check_download
  declare -i LFBFL_i_error=0

  if [[ ! -f "$1" ]]; then
    wget ${LFBFL_verbose:+"${LFBFL_verbose}"} -O "$1" "$2"
  fi
  declare -r LFBFL_present_sha512=$(
    sha512sum "$1"\
    | cut --fields=1 --delimiter=' '
  )
  if [[ "${LFBFL_present_sha512}" != "$3" ]]; then
    printf "%s does not have correct sha512:\n - wanted %s\n - found %s\n"\
      "$1" "$3" "${LFBFL_present_sha512}"
    LFBFL_i_error=1
  fi
  if [[ -n "${LFBFL_check_download}" ]]; then
    wget ${LFBFL_verbose:+"${LFBFL_verbose}"} -O "$1.wget_sha512.temp" "$2"
    declare -r LFBFL_present_sha512_2=$(
      sha512sum "$1.wget_sha512.temp"\
      | cut --fields=1 --delimiter=' '
    )
    if [[ "${LFBFL_check_download}" != "--check-download=keep" ]]; then
      rm "$1.wget_sha512.temp"
    fi
    if [[ "${LFBFL_present_sha512_2}" != "$3" ]]; then
      printf\
        "%s does not have correct sha512:\n - wanted %s\n - found %s\n"\
        "Online version of $1" "$3" "${LFBFL_present_sha512_2}"
      LFBFL_i_error=1
    fi
  fi
  # shellcheck disable=SC2248
  return ${LFBFL_i_error}
}
```

Merci Dieu ! Merci Père ! Merci Seigneur ! Merci Saint Esprit !